Smoke Aerosol Measurement Experiment Re-flight (SAMER)

by

David L. Urban[1], Gary A. Ruff[2] and William Sheredy[3], Marit Meyer,
NASA Glenn Research Center, Cleveland, Ohio 44135
Thomas Cleary[4] and Jiann Yang[5]
National Institute of Standards and Technology, Gaithersburg, Maryland 20899
George Mulholland[6]
The University of Maryland, College Park, Maryland 20742
and
Zeng-Guang Yuan[7]
National Center for Space Exploration Research, Cleveland, Ohio 44135

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1 Branch Chief, Combustion and Reacting Systems Branch, 21000 Brookpark Road, MS 77-5, Senior Member.
2 Advanced Capabilities Project Office, 21000 Brookpark Road, MS 77-7, Associate Fellow.
3 ISS Projects Office, 21000 Brookpark Road, MS 77-5.
4 Engineered Fire Safety Group, 100 Bureau Drive, Stop 8664.
5 Fire Measurements Group Leader, 100 Bureau Drive, Stop 8662.
6 Visiting Professor of Mechanical Engineering, Glenn L. Martin Hall.
7 Staff Scientist, 21000 Brookpark Road, MS 110-3.

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Abstract

Smoke Aerosol Measurement Experiment Re-flight (SAMER) is an extension or continuation of Smoke Aerosol Measurement Experiment (SAME), conducted in space from September 9 to October 5, 2007. On the success of SAME, SAMER included the following modifications: 1) a new test sample material Pyrell®, an intumescent fire-retardant foam by Foamex was introduced to replace DBP samples, 2) test condition ranges was extended to include higher heating temperature and no-flow conditions, and 3) a new instrument, M-pass, developed in-house GRC, was used to replace one of the two Commercial Diagnostic Enclosures used in SAME. SAMER flight tests were conducted from July 16, to September 9, 2010, successfully completing 66 test points. As a part of SAMER effort, three major post–flight ground tests were performed: the SAMER apparatus flow tests to obtain accurate dilution ratio data, the TEM tests to obtain the 1-g counterpart of TEM samples and a batch of SMPS tests to examine the validity of the lognormal distribution assumption of aerosol particle size for the tested materials. Based on the results of both SAMER flight and ground-based tests, it is concluded that the moment method in conjunction the lognormal distribution assumption is a valid and effective approach to characterization of the aerosol encountered in space vehicles. Particle size distributions of aerosol produced from 5 tested materials commonly found in space crafts were identified.

Introduction

Smoke Aerosol Measurement Experiment Re-flight (SAMER) is one of the awarded Rapid Turn Around (RTA) projects, solicited by NASA Headquarters in June 2009. SAMER is an extension or continuation of Smoke Aerosol Measurement Experiment (SAME), which was conducted in the Microgravity Science Glovebox (MSG) onboard the International Space Station (ISS) from September 9 to October 5, 2007.

Generally speaking, SAME experiments were completed successfully in that the apparatus developed for the experiments functioned well and desired data from instruments were obtained. However, there was a major problem of the SAME experiments. After the thermal precipitators returned to the ground, it was found that all the TEM grids were severely contaminated with some unidentified foreign particles, making these grids unusable although some of them showed shapes of certain types of smoke particles.

Based on the experience gained from SAME, SAMER included the following modifications: 1) a new test material, Pyrell®, an intumescent fire-retardant foam by Foamex, was introduced to replace DBP samples, 2) test conditions were broadened including higher heating temperatures and no-flow during sample heating time, and 3) a new instrument, M-pass, developed in-house GRC, was used to replace one of the two Commercial Diagnostic Enclosures that includes a DustTrak and an ionization detector used in SAME. SAMER flight tests were conducted from July 16, to September 9, 2010, successfully completing 66 test points.

After SAMER flight tests, three major post–flight ground tests were performed to improve data reduction process and to validate the lognormal distribution assumption, the key to the moment method employed by SAME and SAMER. The first ground-based test is the SAMER apparatus flow tests. The objective of the test is to obtain accurate dilution ratio data of the P-trak and Dusttrak so that the accuracy of aerosol concentration can be improved. The second batch of ground tests was the TEM tests which were conducted under exactly the same conditions as those for the SAMER flight tests so that the gravity effects on the TEM samples can be identified. The third batch of ground tests was to a SMPS apparatus to obtain the actual particle size distribution under the same condition as the SAMER flight tests. The objective of these tests is to examine the validity of lognormal particle size distribution assumption.

Besides these ground tests, there were two improvements in the data reduction procedures. One is to derive sample heating temperatures from the logged engineering data so that each test can be characterized by the sample heating temperature instead of R_t/R_0 as has always been used. It was realized that even two tests use the same R_t/R_0, the sample heating temperatures can be quite different. Characterizing a test with sample heating temperature makes it

easier to understand the particle generation from the view point of material pyrolysis. The second improvement is to device a method of calculating the percentage of cumulative moment detected by an instrument that has a limited particle size detection range. This development allows us to improve uncertainty analysis of the results we obtained.

Since the apparatus used in SAMER is mostly unchanged from that of SAME and the basic approach – the moment method – used by the two experiments are the same, in this report, these two parts – the apparatus and the moment methods – will not be repeated. What will be included in this final report are the newly added part, the improved data reduction methods and the new results of particle distributions.

Flight Operation

SAME-R utilized the SAME hardware already on orbit with reloaded sample carousels, 30 extra loose samples, rebuilt thermal precipitators and refilled alcohol cartridge for P-trak use. Table below lists sample locations in carousels.

	Position 1	Position 2	Position 3	Position 4	Position 5	Position 6
Carousel #1	Silicone	Lamp wick	Teflon	Kapton	Pyrell	Silicone
Carousel #2	Silicone	Silicone	Lamp wick	Lamp wick	Teflon	Teflon
Carousel #3	Kapton	Kapton	Pyrell	Pyrell	Silicone	Silicone
Carousel #4	Lamp wick	Lamp wick	Lamp wick	Teflon	Teflon	Teflon
Carousel #5	Kapton	Kapton	Kapton	Pyrell	Pyrell	Pyrell (bad)
Carousel #6	Silicone	Lamp wick	Teflon	Kapton	Pyrell	Silicone

The Pyrell sample installed in position 6 of carousel 5 was found unusable during the test, which was probably shacked loose during the launch.

In addition to those preloaded carousels, thirty loose samples were also uploaded to ISS. These loose samples included 6 Pyrell, 6 silicone rubber, 9 Kapton, 5 lamp wick and 4 Teflon samples. These samples were later installed in carousel #4, six a group, to replace used ones for more runs. Although physically only carousel #4 is repeatedly used to load these loose samples, in data logging sheet, nominated carousel numbers, i.e. carousel # 7(4) to carousel # 11(4) were used in order to distinguish those additional runs from actual carousel # 4. Sample number of each additional carousel loading is listed in Table below.

	Pyrell	Silicone	Kapton	Lamp wick	Teflon
Carousel #7 (4)	2	1	1	1	1
Carousel #8 (4)	1	1	3	1	
Carousel #9 (4)	1	1	2	1	1
Carousel #10(4)	1	2	1	1	1
Carousel #11(4)	1	1	2	1	1
Total	**6**	**6**	**9**	**5**	**4**

Six re-assembled Thermal Precipitators, each having 12 TEM grids in it, were uploaded to ISS for SAMER.

Since the SAMER project is a re-flight of SAME, the hardware, the project approach, the principle of moment method, post flight data analysis and uncertainty analysis will not be repeated in this report. Information in these areas can be found in the final report of the original SAME project. This report emphasizes information unique for SAMER only and new development made in SAMER project.

The SAME-R flight operation was carried out from July 16, 2010 to September 9, 2010 in 7 time periods: July 16, July 19-22, July 29-30, August 2, August 23-27, August 30 and September 7-9. Totally 67 runs were conducted including two runs using reused samples. The test conditions of each run can be found in Appendix A, where the secondary aging runs are listed as separate entities, making the total run number to 109. In the SAMER flight tests, 70 TEM samples were collected totally.

During the tests, SAMER engineer team and science team were in the Tele-Science Center in GRC, where they could communicate with the astronaut who is conducting the test and have a real time video of the apparatus when

space to ground signal transmission is available. The engineering housekeeping data files generated during the tests were downlinked shortly after tests were completed each day. The entire flight test operation was quite smooth and as planned. The hardware worked fine without any unexpected events.

After entire SAMER tests were completed, the six carousels, all tested samples and the six thermal precipitators were shipped to the GRC for measuring sample weight losses and TEM grid analysis.

SAMER results

The preliminary results of SAMER experiments are best summarized in a paper titled "Particle Morphology and Size results from the Smoke Aerosol Measurement Experiment-2" [ref. xx]. Below are excerpts from the relevant part of the paper. The referenced materials that are beyond the text of excerpts can be found in the original paper.

Over 100 sample materials were tested. These were comprised of samples of 6 materials: Teflon™, Kapton™, Pyrell™, silicone rubber, cellulose (lamp wick), and dibutyl-phthalate deposited on a porous wick. The test conditions included multiple sample temperatures and air flow rates. The baseline air flow rate was 8 cm/s with runs conducted at rates down to 2 cm/s with a limited set conducted with no airflow while the sample was heated. The sample temperatures were based on the pyrolysis properties of the material and were thus material specific. The baseline temperature was selected to produce 1 to 2 mg of weight loss in 60 seconds and have reasonable signal on each of the moment instruments. To examine the effect of sample temperature, additional higher temperature conditions were also selected that would not saturate the moment instruments. The TEM girds recovered from the thermal precipitators on SAME-1 were unfortunately contaminated with extraneous particles whose origin is currently under investigation. For SAME-2 more rigorous assembly controls were followed and numerous good particle images were obtained. For typical runs, the particle concentrations were a few million particles / cc and the mass concentrations were 2 to 15 mg/m^3 however there were runs with fewer particles. For these conditions, the typical numbers of particles sampled by the instruments were in the billions although for the test with the lowest particle concentrations the sample sizes were of the order of several million particles.

A. Moment Instrument Results

Geometric mean, count mean, and average mass diameter results from the moment instruments for baseline runs are presented in Table 1. As described above, the count mean diameter and the diameter of average mass measurements are independent of the nature of the size distribution however, the calculation of the geometric mean diameter and the geometric standard deviation (σ_g) depend on the assumption of a log-normal distribution. This assumption cannot be directly tested with the available data since this would require a device that can separate the particles into size bins. However, there are two indications that the distribution can be approximated by a log-normal distribution. In the cases listed below, the count mean diameter is less than the diameter of average mass which is a necessary condition for a log-normal distribution (See Fig. 1). The second indication is that the geometric standard deviations are all physically reasonable (i.e. greater than 1 (see Eq. 3) and less than approximately 3.5 which is the practical limit on the magnitude of aerosol geometric standard deviation). The aging results show a substantial increase in the observed diameters. As expected the increase in the count mean diameter is greater than the corresponding increase in the diameter of average mass. This can be understood since coagulation consists of particle collisions joining two particles. For a polydisperse aerosol, these collisions usually involve the collision of one of the smallest (high mobility) particles with the largest (high cross-section) particles. This collision has little effect on the size of the large particle but it removes the smaller particles from the size distribution. This has a larger effect on the count mean diameter since it is linear in diameter while the diameter of average mass varies with the cube root of the diameter. Theoretical predictions suggest that this process will produce a near lognormal distribution and that the standard deviation will tend toward a self-preserving distribution with a value of 1.32.[14]

		Geometric Mean Diameter (Dg) (µm)	Count Mean Diameter (M_1/M_0) (µm)	Diameter of Average Mass $(M_3/M_0)^{1/3}$ (µm)	σ_g
Kapton	Unaged	0.047	0.062	0.106	2.072
	Aged 720 s	0.102	0.121	0.169	1.790

Lampwick	Unaged	0.103	0.142	0.271	2.232
	Aged 720 s	0.269	0.308	0.401	1.673
Silicone	Unaged	0.152	0.221	0.468	2.382
	Aged 720 s	0.320	0.400	0.628	1.956
Teflon	Unaged	0.093	0.111	0.173	2.081
	Aged 720 s	0.080	0.115	0.237	2.342
Pyrell	Unaged	0.17**7**	0.227	0.377	2.032
	Aged 720 s	0.348	0.400	0.529	1.697

Table 1: Diameter results for baseline runs for each material (Diameters are in micrometers).

Overall, the Teflon and Kapton particles were very small. The lamp wick and silicone results exhibited substantially larger diameters of average mass, particularly for the aged cases. As reported previously[10], the Dust Trak results for lamp wick and silicone demonstrated substantial portions of the particle distribution possessed aerodynamic diameters larger than 1 micrometer. The comparatively much smaller diameters for the Kapton and Teflon will make detection of this smoke challenging for light scattering devices, on the other hand the large sizes seen with the lamp wick and silicone would generate very large signals on a light scattering system. As discussed by Urban et al.[13], to have a system that would have the signal gain sensitive enough to trigger an alarm for the lower signal from small particles such as Teflon and Kapton without excessive nuisance alarms from large dust particles is difficult without some level of particle size discrimination. This suggests that detection of these particles against the background environment will require a detection system capable of measuring more than one moment of the particle size distribution.

The results for Teflon, Kapton and lampwick are plotted versus duct velocity in figures 6, 7 and 8. These figures show a strong trend of increased particle size at low flow rates. This indicates that smoke trapped in an avionics enclosure can be expected to be substantially larger than smoke released in ventilated areas.

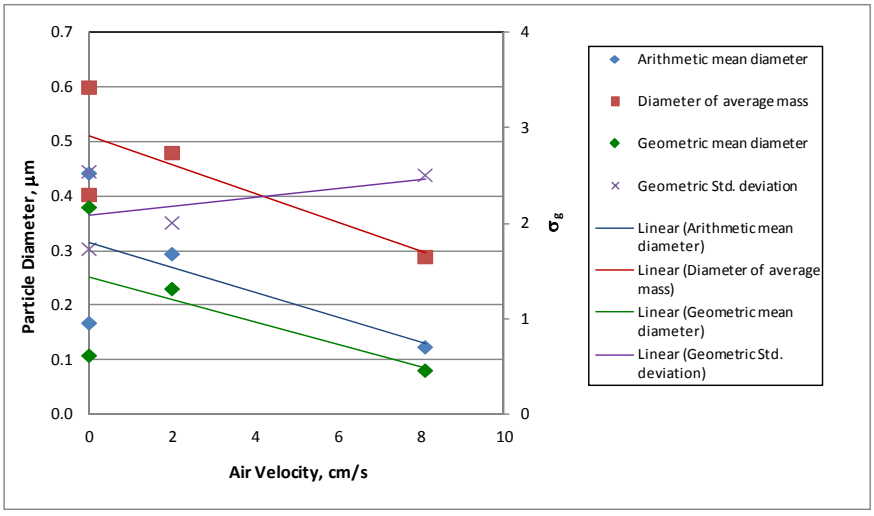


Figure 6: Particle diameters as a function of air velocity for lampwick. The lines are linear (least square) fits. The repeatability of the diameter measurements is estimated to be +/- 15%.

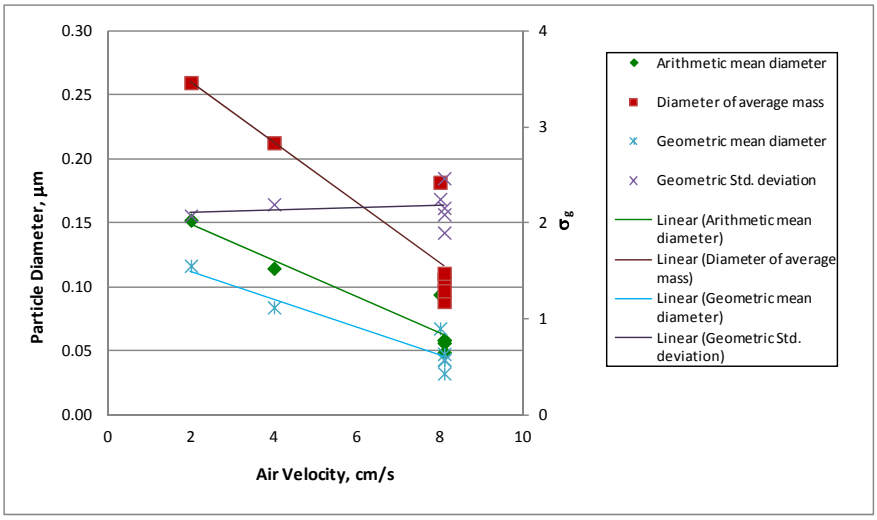


Figure 7: Particle diameters as a function of air velocity for Kapton. The lines are linear (least square) fits. The repeatability of the diameter measurements is estimated to be +/- 15%.

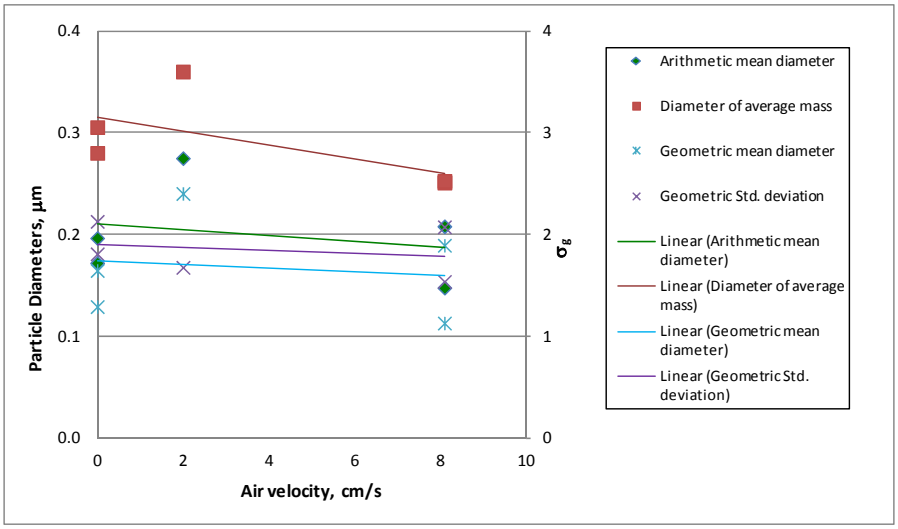


Figure 8: Particle diameters as a function of air velocity for Teflon. The lines are linear (least square) fits. The repeatability of the diameter measurements is estimated to be +/- 15%.

B. MPASS Results

The MPASS was included in SAME-2 to demonstrate the effectiveness of a two-moment smoke detector. Because of how they are used, a smoke detector is not calibrated for specific materials and instead has a single calibration for all conditions. The objective is to maintain a flat sensor response, regardless of the type of smoke or the particle size so the signal is proportional only to a selected moment (e.g. mass or number concentration). The approach used was to derive correlations to predict the smoke number concentration and mass concentration from the MPASS output. This correlation was made using the other SAME measurements (number and mass concentration) as a reference. A least-squares, third-order, correlation was computed on all test points with a signal to noise ratio greater than 2 independent of all other parameters (e.g. material, heating rate, and aging time).

The results are shown in figures 9 and 10, for prediction of the mass concentration and the number concentration, respectively. Ideally these response graphs should be horizontal lines. The normalized output is plotted along with the normalized correlation result as a function of geometric mean diameter. Ideally the

normalized result should be horizontal lines. The uniformity of response (scatter) is expressed by the least squares variance relative to the mean (Δ) and is shown in the legend. Both figures show that the correlation decreases the relative variance by roughly a factor of at least 2 relative to that provided by a conventional single-output sensor. The resulting variance for the mass response is substantially lower than for the number concentration. This is consistent with the fact that the original optimization of the MPASS was to mass concentration not to number concentration. Thus with appropriate processing, the MPASS output can be proportional to either the number concentration or the mass concentration over a wide range of particle sizes and material compositions. This independence demonstrates that by combining 2 moments the MPASS is able to scale its response to the particle size being sampled giving a more accurate indication of the amount of smoke present than a single moment detector. The MPASS therefore has the ability to discern the difference between a strong signal from a high smoke concentration and a strong signal from a lower concentration of very large dust particles, a feature which can minimize false smoke detector alarms.

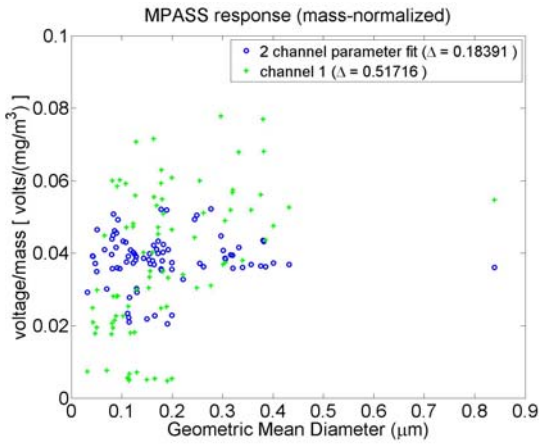


Figure 9: Mass normalized signal channel output voltage and correlated MPASS output voltage versus geometric mean diameter.

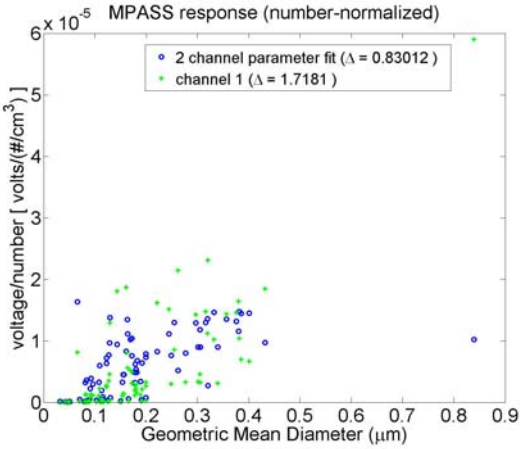


Figure 10: Count normalized output voltage and correlated MPASS output voltage versus geometric mean diameter.

C. TEM Results

Overall, 72 particle samples were taken on Transmission Electron Microscope (TEM) grids using the thermal precipitator. These samples were imaged with a TEM to evaluate the particle morphology as a function of flow conditions, material and heating level. Overall, silicone rubber presented no particles that were visible via TEM, presumably because the particles were condensed liquid that evaporated before they could be imaged in the TEM. Figure 11 presents typical images for the major materials.

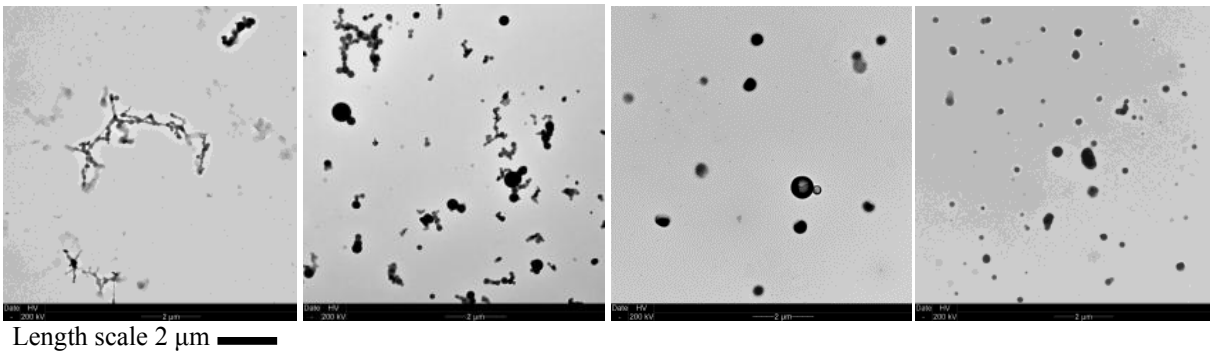
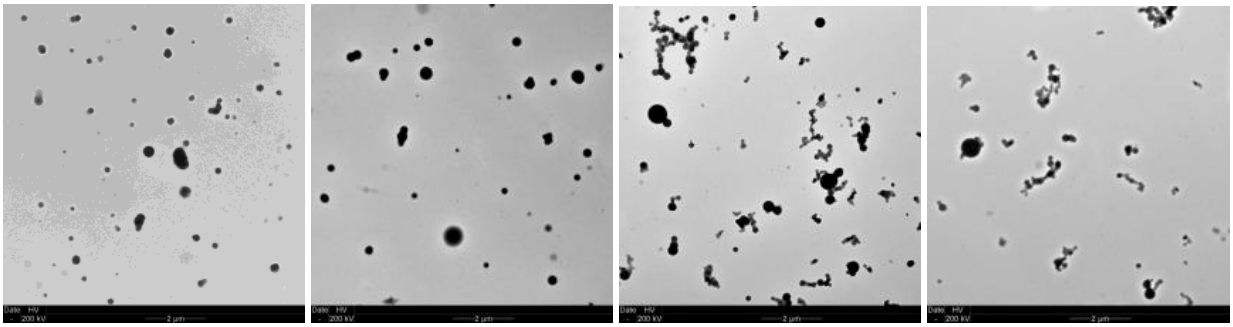


Figure 11: TEM images for Teflon (Run 56), Pyrell (Run 63), Lampwick (Run 54) and Kapton (Run 62).

As seen in Table 1, aging did increase the average particle size, however this level of change is not readily visible in the TEM images, instead the most visible change is the disappearance of the smallest particles. (Fig 12)

As described in the discussion of Table 1, this is consistent with expectations since the smaller particles have the highest mobility and consequently are readily incorporated into the larger particles. Particularly for the semi-liquid particles, this absorption of the smallest particles into larger particles is expected to have little or no apparent effect on the size of the larger particles because the diameter is the cube root of the volume. In the case of non-liquid particles, the smallest particles will nevertheless be unlikely to be visible via TEM when they are attached to larger particles.



Length scale 2 μm

Figure 12: TEM images showing the effect of aging for Kapton (Run 62) and Pyrell (Run 63). From Left to right, unaged Kapton, aged Kapton, unaged Pyrell, aged Pyrell. The notable change is the elimination of the smallest particles.

D. Impact of flow on particle morphology

As the samples are heated by the wire, chemical bonds are broken and gaseous products are released via pyrolysis. These products may include partially oxidized species resulting from the reaction with oxygen. As the gases move from the wire, the temperature and species concentration will decrease as a result of conduction/diffusion and convection. With the decrease in temperature, the super saturation ratio of the pyrolysis gas will pass a critical stage and nucleation of smoke particles will occur. These smoke nuclei grow rapidly from the condensation of the pyrolysis gases. The time scale for the nucleation and condensation is on the order of miliseconds or less. Over a longer time scale, the growing smoke particles undergoing Brownian motion collide and stick together in a process known as coagulation. This process may continue for time scales of up to thousand seconds. If the particles are liquid, the colliding particles will coalesce and form larger droplets. This is the case for the lampwick, silicone rubber, and Kapton smokes (Fig. 11). If the particles are solid, they will retain their structure as they stick resulting in an agglomerate structure composed of many primary particles connected in a low density fractal-like structure. This is the case for the Teflon and Pyrell smokes (Fig. 11).

Key parameters affecting the smoke properties of the smoke produced here are the low-gravity condition, the convective flow past the heated wire, the wire temperature, and the aging time. As the wire temperature increases, more of the sample will be pyrolyzed and these gases will subsequently condense on the nuclei leading to a larger mass concentration of the aerosol and a larger diameter of average mass.

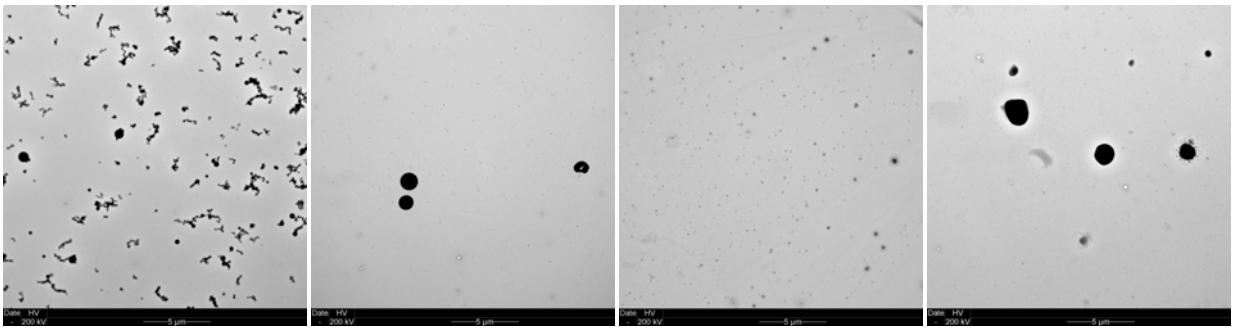
An increase in the convective flow for a fixed wire temperature is expected to result in an increase in the pyrolysis rate and thus an increase in the mass concentration of the aerosol. Some results follow this trend with silicone rubber, lampwick, and Pyrell increasing slightly with flow while Teflon and Kapton decrease. A possible explanation is that holding the wire temperature fixed does not necessarily mean that the sample surface temperature distribution is precisely the same at different air flow speeds. Although these are expected to be small variations, given the exponential dependence of the pyrolysis kinetics on temperature, changes in the heat transport environment could change the sample surface temperature between the wire coils.

A case of special interest is the case of zero-flow. The transport for the zero-flow case in μ-g is controlled by molecular diffusion and thermal expansion. Over a period of 60 seconds, the gases would diffuse a distance on the order of 4 cm. Assuming a cubical geometry this would correspond to the smoke being confined to a volume of about 64 cm^3 rather than being diluted in the 6000 cm^3 piston. If the same number of particles is generated as for the flow case, the initial concentration will be on the order of 50 times higher for the zero-flow case. This is

expected to lead to more coagulation and a larger particle size. It is found for all of the materials the diameter of average mass is 30 % to 40 % larger for the zero-flow tests compared to the flow tests.

The zero flow condition was not in the original design of the SAME hardware but was examined for 11 tests by modifying the valve operation in the SAME hardware. This resulted in a very high concentration of particulate where the aging process was accelerated compared to that in the aging chamber at dilute conditions. An unfortunate side effect was that approximately half the smoke was lost from the system because it diffused out of the inlet into the MSG volume. As a result the total particle counts were lower than normal and the TEM grids had few particles. It is important to recognize that this condition is unique to low-gravity scenarios. If one performed the same experiment at 1-g, the buoyancy of the rising gases would greatly increase the mixing zone. So the larger particle diameters observed in these zero-flow conditions are important to consider in the development of improved fire detection equipment for space vehicles.

For the zero-flow case, the heat is transported primarily by conduction rather than convection. This is likely to result in a significant increase in the gas temperature relative to the case with 8 cm/s air flow. A striking effect of this temperature change is the change in the morphology of the Pyrell smoke from fractal-like clusters with up to 30 primary spheres for the case with a flow of 8 cm/s to spherical particles for the zero-flow condition (Fig 13).



Pyrell 8 cm/s air flow (Run 63) Pyrell no air flow (Run 95) Lampwick 8 cm/s air flow (Run 80) Lampwick no air flow (Run 80)

Length scale: 5 μm

Figure 13: TEM images showing the effect of flow for Pyrell and lampwick under 8 cm/s air flow and quiescent conditions.

SAME Smoke-in-Drums Ground-Based Experiment

In order to assess whether the size distribution of the smoke is lognormal, detailed particle size distributions were measured with a reference instrument. This cannot be accomplished in low gravity, so this investigation was performed with the ground-based engineering SAME hardware which is identical to the setup on the ISS, incorporating the flight aerosol instruments which had been returned to earth. A Scanning Mobility Particle Sizer (SMPS) Spectrometer (3936, TSI, Shoreview, MN, USA) was used as the reference instrument in validation experiment. The SMPS requires a two minute scan through a range of voltages to acquire a high resolution particle size distribution; however, the duration of smoke supplied from the SAME aging chamber is at most 30 seconds. Therefore, the smoke was collected in an intermediate container which served two purposes. The first purpose was to sufficiently dilute the smoke from the SAME chamber in order to effectively stop coagulation (aging) of the smoke particles during the SMPS scans. The second purpose was to have a large enough quantity of dilute smoke for multiple SMPS scans. A 55 gallon drum was chosen for this purpose and the SAME smoke-in-drums setup was developed to enable SMPS measurements on a portion of the smoke output from the SAME piston chamber. The configuration is shown in Figure 14, which shows the original SAME hardware inside the dashed outline. One DustTrak™ was removed from the original SAME configuration and its portion of the smoke sample was diverted from the setup to one of two drums which hold the diluted smoke during multiple 2-minute SMPS measurements. One drum collected fresh smoke from the heated sample material and the other drum was filled after a controlled aging period in the piston chamber. Thus both aged and unaged smoke could be measured with the SMPS.

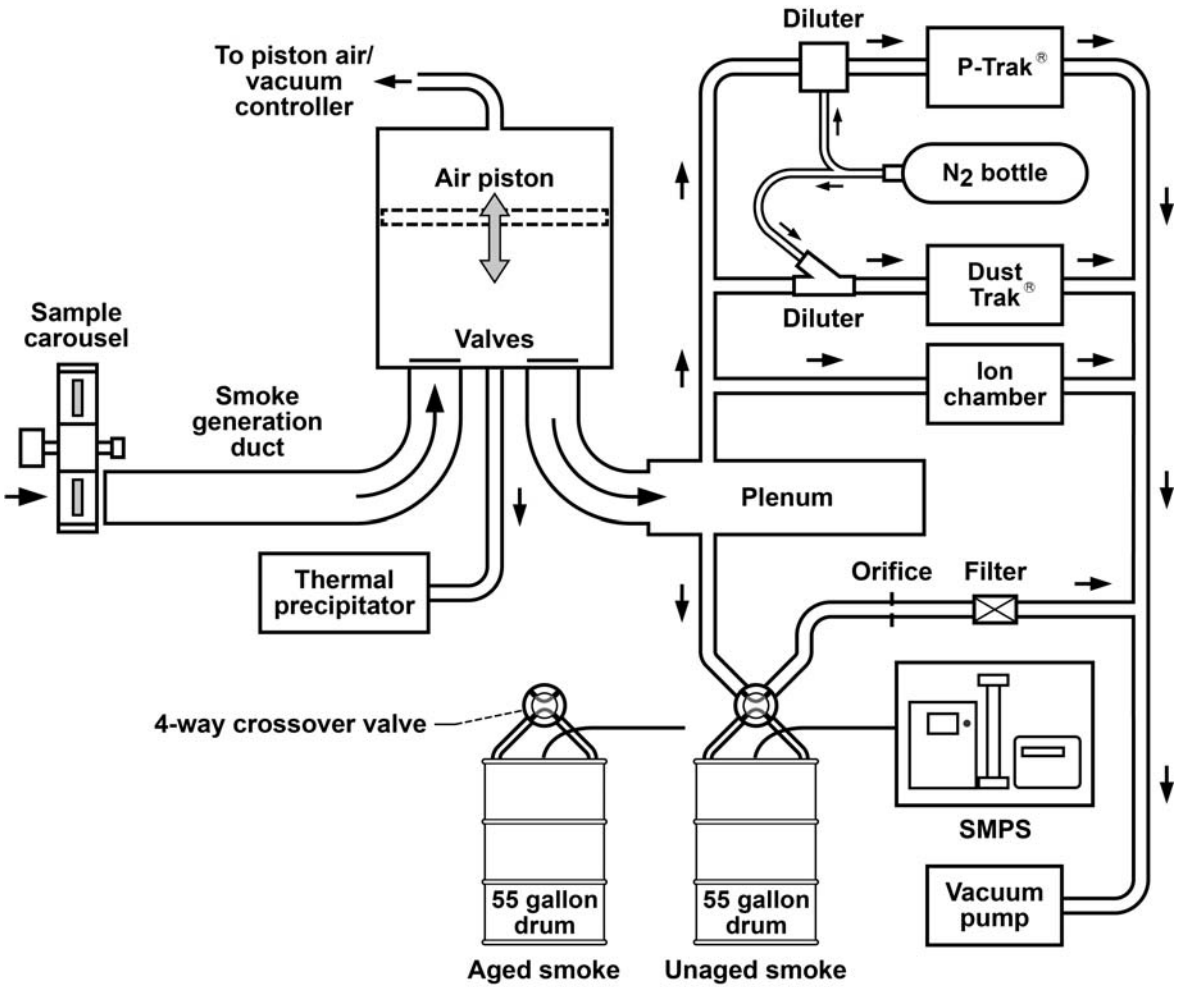


Figure 14: SAMER drum testing schematic. Some smoke is diverted from the setup to fill one of two drums which hold the diluted smoke during three 2-minute SMPS measurements. Two drums were needed to contain and measure both un-aged and aged smoke.

Drum Hardware and Experimental Process

The two 55 gallon drums were minimally modified for use in the experiment as follows: A muffin fan was bolted to the circular drum floor opposite the removable lid. All other modifications were made to the lid, including feed-through adapters for tubing and the electrical fan cord, and mounted HEPA filters through which clean air was provided for purging. A four-way crossover valve enabled the drum to be either filled with smoke via a vacuum pump preceded by a flow-controlling orifice, or bypassed by the smoke, which effectively isolated the SAME hardware from high pressure during the drum purging process. Transfer of un-aged smoke into the drum began when the SAME software commanded the valve opening and lowered the piston in the aging chamber causing half of the chamber volume to be expelled (3 liters), with the remainder of the smoke retained in the chamber and allowed to age for 12 minutes. The flow rate into the drum was 1.7 L/min, matching the flow rate of the DustTrak™ instrument that the drums replaced. It took less than 2 minutes to transfer the smoke sample into the drum. Tubing from the SAME hardware to the drums was less than one meter in order to minimize transport losses of smoke particles. Initial calculations from previous SAME data indicated that for all materials, the aerosol concentration in the drum would be less than 2×10^4 particles/cm^3, which is below the rule-of-thumb threshold concentration of 10^6 particles/cm^3. Below this level, coagulation can be neglected since it will occur at a very slow rate.[15] Thus it could be assumed that very little aging took place in the drum during the SMPS measurements of un-aged smoke. The drum concentration was well within the counting range of the SMPS so no additional dilution was required for

the measurements. During each SMPS sample, the volume of air removed was between 0.5 % and 1.5 % of the drum volume and a correspondingly small number of particles were removed so the drum aerosol concentration was relatively constant during the measurements. Any make-up air needed during the sampling was drawn in through the HEPA filters on the drum lid. The fan in the drum was used to mix the smoke and prevent stratification which could skew the SMPS measurements. First, a 10:1 sheath-to-aerosol flow ratio was used for three consecutive measurements with high fidelity giving particle sizes up to 660 nm. Three subsequent samples were taken with a flow rate ratio of 5:1, in order to reach the largest measurable SMPS particle size of 1000 nm. Considering that aging increases particle size, the higher flow rate ratio (smaller range) gives results for only a portion of the true aged size distribution for most materials in this experiment. Therefore the larger range is important for measurements in spite of the low resolution. After six SMPS scans at two different flow rates, the drum was purged with a continuous flow of HEPA-filtered house air at approximately 309 kPa-g (45 psig), which was exhausted into a fume hood. An additional P-Trak™ was used to indicate when the drum concentration was below 30 particles/cm^3, which was considered 'clean'. One set of SMPS measurements for one smoke sample (both aged and unaged) was accomplished in approximately 30 minutes.

SMPS Results for Spherical Smoke Aerosols

Five SAME materials were tested in the drum configuration at two temperature levels, baseline and high temperature. Typical particle size distributions from the ground testing validation experiments of the un-aged and aged smoke for the more spherical aerosols are plotted in Figs. 6 through 8. Pyrell® and Teflon® are not spherical aerosols and thus are not relevant to the moment method approach. The mean particle size distributions are shown in the upper plots which have open plot marker symbols for unaged smoke and solid symbols for aged smoke. The plotted lines represent a lognormal curve-fit with the MATLAB Statistics Toolbox function 'nlinfit' which performs non-linear least squares regression with the Levenberg-Marquardt algorithm (MATLAB version R2012a, The MathWorks, Inc.). Residual plots, shown also in the lower portion of Figs. 15 through 17 are aligned with the corresponding size distribution plots above, showing the deviations from the lognormal fits. This visual test for 'goodness-of-fit' would result in randomly scattered residual points both above and below zero, for a good lognormal fit. It is common to observe a wedge-shaped spread of residuals, as in Kapton (Fig. 15), where the tails of the distribution have mostly small residual values, with a wider spread of residuals around the peak diameter. In general, the spread of the residuals is more compact for Kapton and lamp wick, indicating a better lognormal fit for these materials. Note that the Kapton residuals are mostly positive up to 70 nm, which indicates that the data are less steep than the lognormal fit. The lamp wick residual plot (Fig. 16) also shows a small but systematic deviation from the lognormal fit, which is evident by the change in sign of the unaged data residuals between 500 and 600 nm. Silicone residual plots (Fig. 17) show the least randomness, which indicates that the lognormal fit is less valid. The residuals change signs on both sides of the peak, indicating a shoulder in the small sizes (residuals go from positive to negative), the peak is offset from the fit (positive residuals around 400 nm) and the data is steeper than the lognormal fit in the large sizes (negative residuals). Some samples have an initial uptick in the small diameter tail, which is believed to be a sampling anomaly in the SMPS, possibly an artifact from the previous sample, as the scans were performed in rapid succession. This anomaly did not have a significant effect on the parameters obtained in the fitting of the SMPS data.

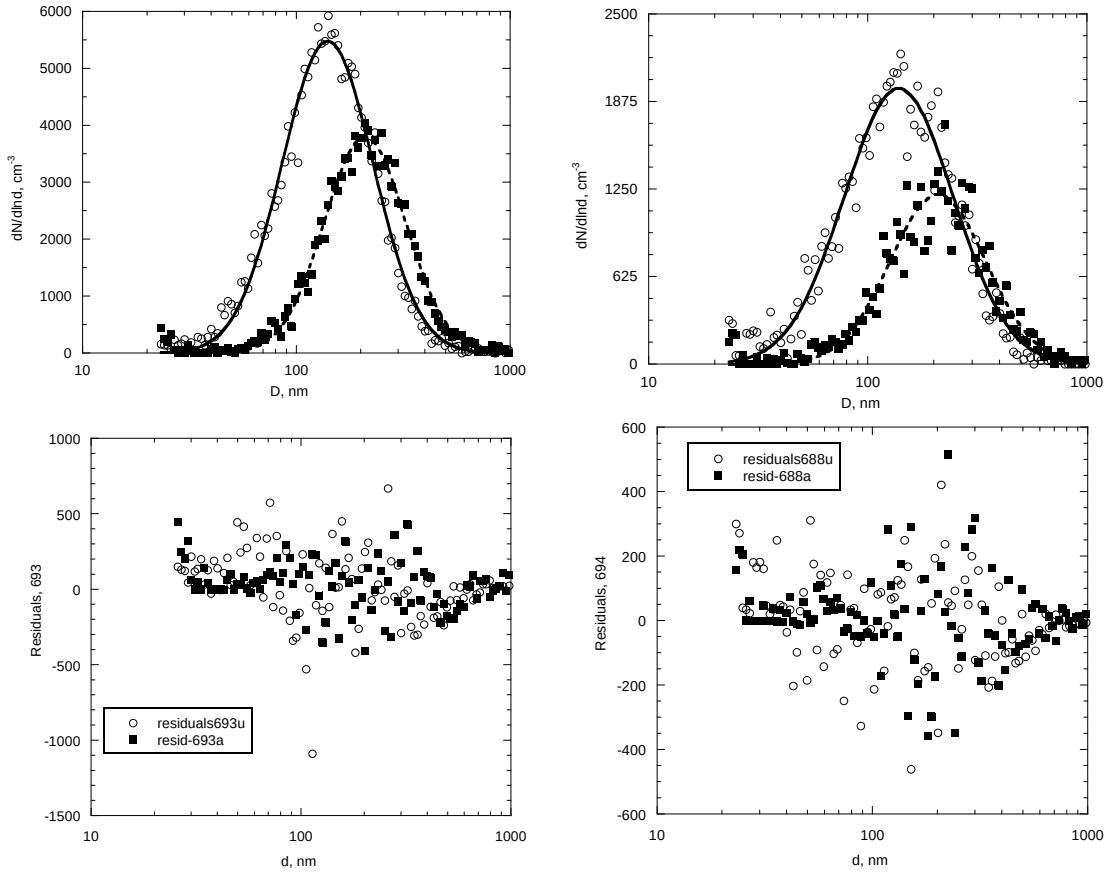


FIG. 15: Smoke particle size distributions from SMPS measurements for Kapton® baseline temperature (left) and high temperature (right) with open plot markers for unaged smoke and solid symbols for aged smoke. The solid curves represent the non-linear least square fits. Residual plots showing the deviations from the lognormal fits are aligned below the respective size distributions.

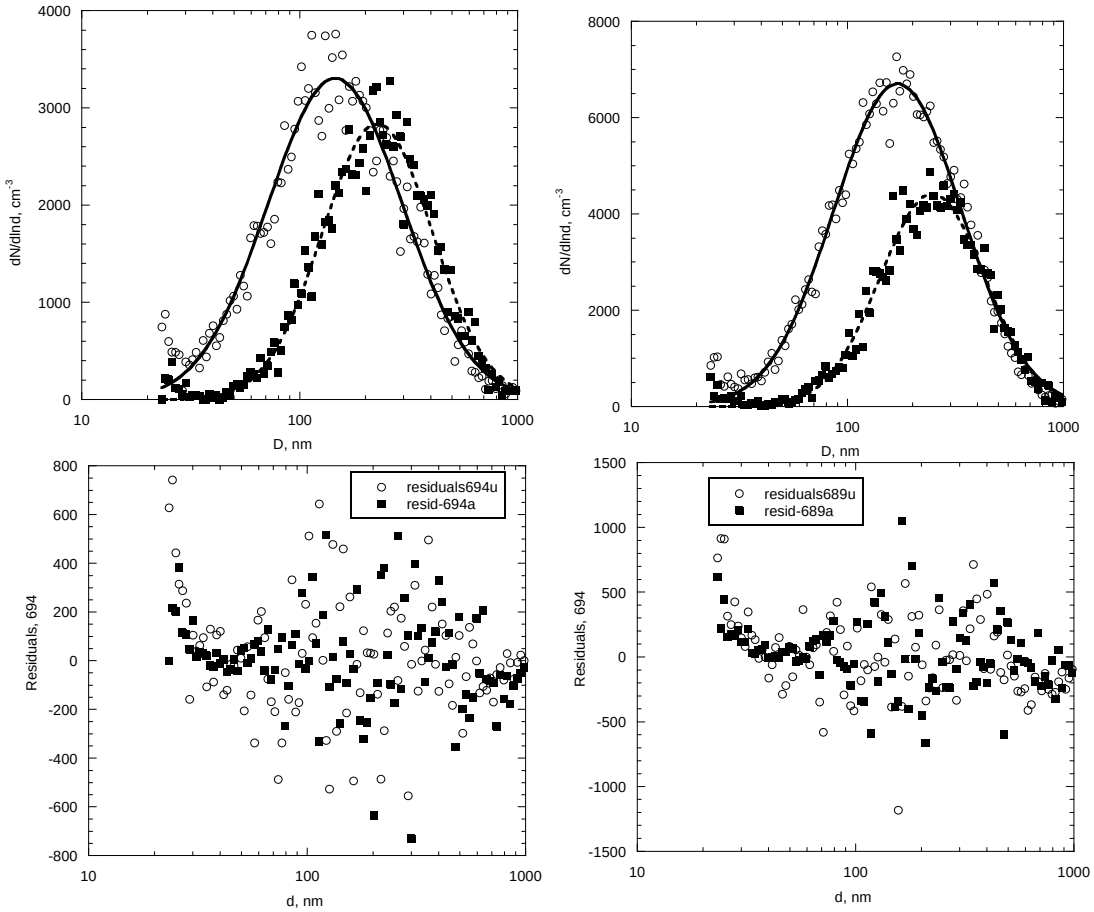


Figure 16: Un-aged smoke particle size distributions from SMPS measurements for lamp wick baseline temperature (left) and high temperature (right) with open plot markers for unaged smoke and solid symbols for aged smoke. The solid curves represent the non-linear least square fits. Residual plots showing the deviations from the lognormal fits are aligned below the respective size distributions.

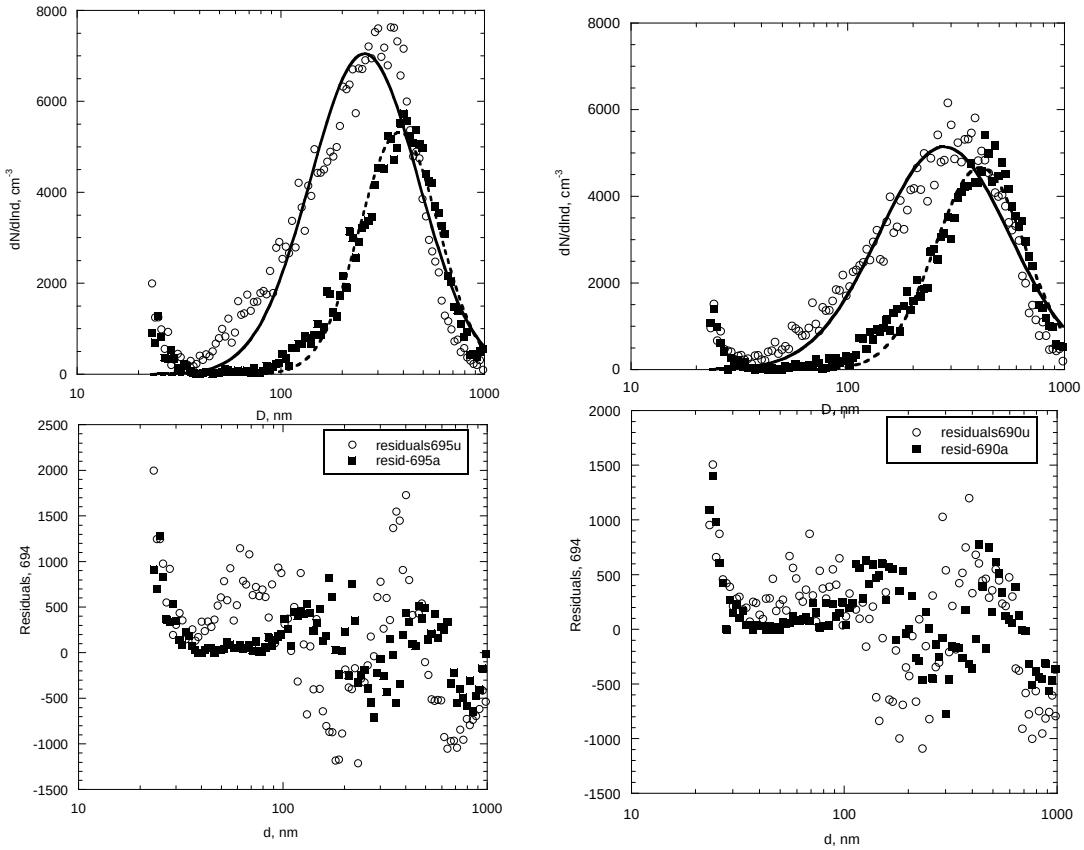


FIG. 17: Un-aged smoke particle size distributions from SMPS measurements for Silicone baseline temperature (left) and high temperature (right) with open plot markers for unaged smoke and solid symbols for aged smoke. The solid curves represent the non-linear least square fits. Residual plots showing the deviations from the lognormal fits are aligned below the respective size distributions.

Ground TEM test: Comparison with drum test results

A TEM lamp wick particle size distribution of particles collected during ground testing is compared with SMPS results in Fig. 18. Two different smoke tests are compared but the heating temperatures are within 0.6 °C so the pyrolysis can be considered to be similar. For the TEM analysis, 17 images containing 4668 particles were processed. TEM images displayed two types of particles: dark, high contrast particles and smaller lower contrast particles. The faint particles could be an artifact on the TEM grid or be the result of poor image quality. Another explanation could be a secondary particle formation event from the pyrolysis products. For this size distribution analysis, all particles were counted without thresholding, so the small faint particles were included in the totals, which reduced the geometric mean diameter and caused σ_{gn} to be quite a bit larger than the SMPS results. The SMPS and TEM volume distributions have a similar spread but do not agree well in the peak location. The largest size bin of this TEM analysis had twelve particles which is considered a statistically significant quantity for this type of analysis (Hinds 1999). The SMPS size distribution ends at 1000 nm, but there is no upper size limit for particles in the TEM size distribution. Results of the ISS flight TEM size distribution analyses and moment method calculations appear in Table 2.

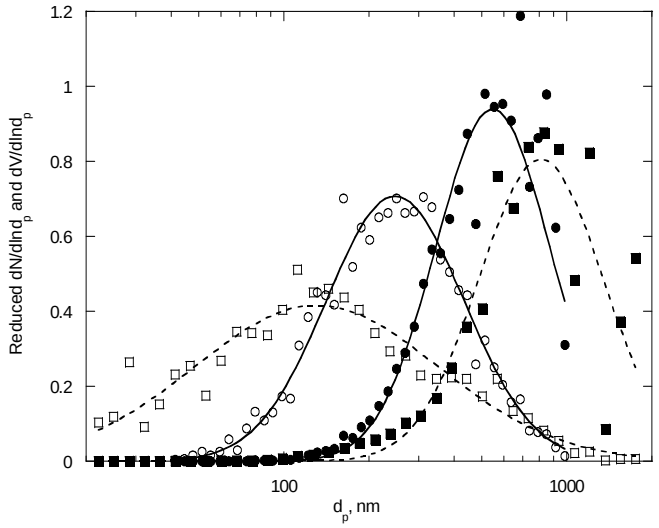


Fig. 18: Number and volume distribution for LW aged test (285.8 °C) 689 from SMPS (circle symbols) and aged test 669 (286.4 °C) from TEM (square symbols). The open symbols correspond to the number distribution and the closed to the volume distribution. The best fit parameters for SMPS are d_{gn} = 248 nm, d_{gv}=548 nm , σ_{gn} = 1.75 and σ_{gv}=1.59 and for TEM are d_{gn} = 131 nm, d_{gv}=812 nm , σ_{gn} = 2.70 and σ_{gv}=1.65.

Table 2: Flight Data Comparison of TEM and Moment Method Distribution Parameters

Sample	Temperature, °C	# particles counted	D_m µm		D_{ave} µm		D_g µm		Sigma g		Hatch-Choate Diameters	
			TEM	Moment	TEM	Moment M1/M0	TEM D_{gn}	Moment	TEM σ_{gn}	Moment	Dav Hatch Choate from TEM sigma and Dg	Dm Hatch Choate from TEM sigma and Dg
Lampwick #54 (ISS) Unaged	Low Temp 264.7	511	0.403	0.253	0.304	0.11	0.255	0.072	1.84	2.493	0.307	0.445
Lampwick #55 (ISS) Aged	Low Temp 262.9	196	0.499	0.320	0.359	0.149	0.284	0.102	2.070	2.400	0.370	0.628
Kapton #94 (ISS) Aged	Low Temp 527.3	1353	0.159	0.176	0.120	0.136	0.103	0.119	1.720	1.669	0.119	0.160
Kapton #62 (ISS) Unaged	High Temp 574.3	1320	0.226	0.251	0.180	0.148	0.158	0.114	1.680	2.066	0.181	0.237
Kapton #62 (ISS) Aged	High Temp 574.3	796	0.289	0.383	0.235	0.243	0.210	0.194	1.630	1.962	0.237	0.300
Pyrell #63 (ISS) Unaged	Low Temp 225.3	3436	0.394	0.392	0.290	0.207	0.244	0.151	1.810	2.222	0.291	0.414
Pyrell #63 (ISS) Aged	Low Temp 225.3	2658	0.468	0.539	0.354	0.359	0.292	0.293	1.920	1.892	0.361	0.553
Pyrell #64 (ISS) Unaged	High Temp 242.1	1888	0.411	0.445	0.298	0.254	0.249	0.192	1.840	2.115	0.300	0.435
Pyrell #64 (ISS) Aged	High Temp 242.1	1560	0.540	0.616	0.403	0.450	0.328	0.385	1.980	1.750	0.414	0.660
Silicone #60 (ISS) Unaged	High Temp 379.9	557	0.153	0.533	0.120	0.241	0.107	0.162	1.590	2.440	0.119	0.148

TEM Particle Morphology Results from Flight Tests

The TEM grids were examined with a Philips™ CM20 Transmission Electron Microscope at 200KV and images were collected using an Olympus™ Veleta camera. The digital camera was calibrated using a MagiCal® calibration reference which is NIST certified standard.

Kapton particles are smaller but rarely agglomerated. The spherical shape and uniform density indicate growth by condensation in the saturated vapor of the pyrolysis products. Figure 19 shows the effect of aging, with the unaged (left) having a much higher population of very small particles, and the aged particles (right) appearing only slightly larger.

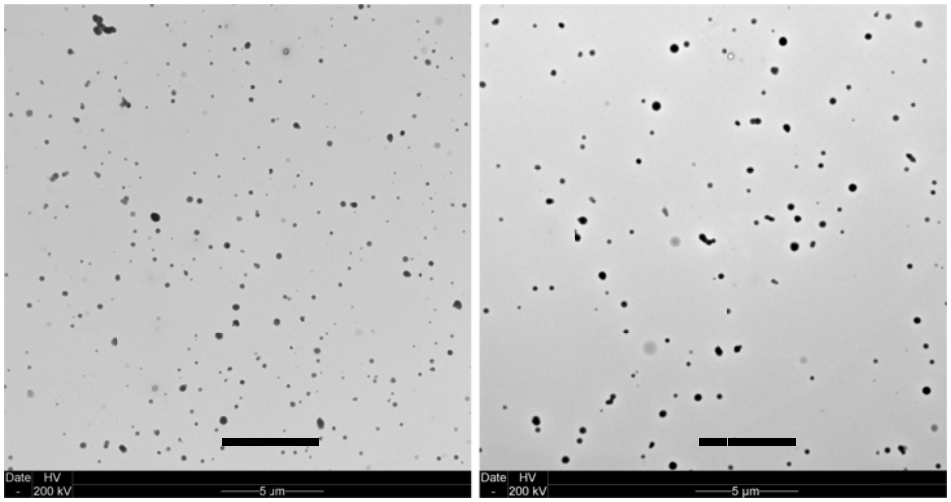


FIG. 19: TEM images of Kapton particle morphology from ISS high temperature testing, unaged on the left, aged on the right. Sample heating temperature was 574 °C, bar is 5 µm.

Lamp wick smoke aerosols (Fig. 20) are known to be primarily spherical droplet type particles that grow by condensation of pyrolysis gases (Mulholland et al, 2005). Occasional doublets are seen but most are unagglomerated. Two distinct large particle types are observed: uniformly dense or lighter in the center suggesting they arrive at the carbon film as a liquid. Some TEM images display additional faint particles that covered only one or two pixels.

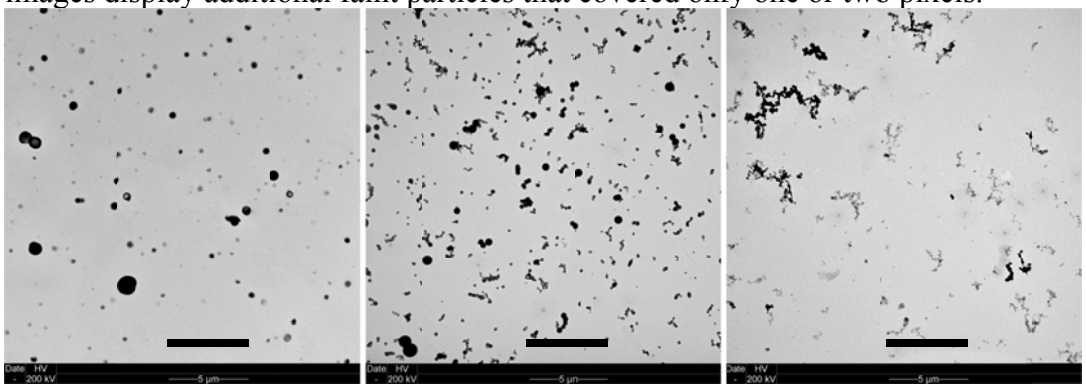


FIG. 20: TEM images of unaged smoke from flight tests-- Lamp wick, 264.7 °C (left) and Pyrell, 242.1 °C (middle) and Teflon, 516.3 °C (right), bars are 5 µm.

Pyrell smoke particles consist of agglomerates made up of primary particles ranging from 30 nm to 100 nm. The non-uniform primary particle size Teflon primary particles are much smaller and are agglomerates with a fractal structure. The darker agglomerates are more electron-dense and indicate that the fainter particles may have partially evaporated in the electron beam. Additionally, some particles were not completely adhered to the TEM grid and movement could be observed as the force of the electron beam influenced the loose ends of the agglomerates.

Silicone particles were not wholly preserved on the TEM grids owing to the volatile nature of the pyrolysis products. Only very small and faint particles remained after the return flight to earth, as seen in Figure 21. Note that the magnification in this figure is nearly double that of the other particle images shown. SMPS ground test data indicates that fresh silicone particles are much

larger, as seen in Fig. 4. Previous to this work, it was thought that pyrolysis smoke was made up of spherical droplets (Mulholland, 2005). In the companion paper (Mulholland et al. 2014) an explanation for the formation of the agglomerates is presented.

Morphology results show that only Kapton and lamp wick are spherical aerosols so they are better suited to the moment method. Although the TEM images of Silicone do not reflect spherical morphology, it is considered a spherical smoke aerosol as it consists of liquid droplets. Meaningful values of d_{av} can be calculated regardless of shape, and the material-specific calibration of the DustTrak with fundamental aerosol mass measurements provide moment method values of d_m which are valid for the nonspherical materials Pyrell and Teflon (by equations (3) and (4)).

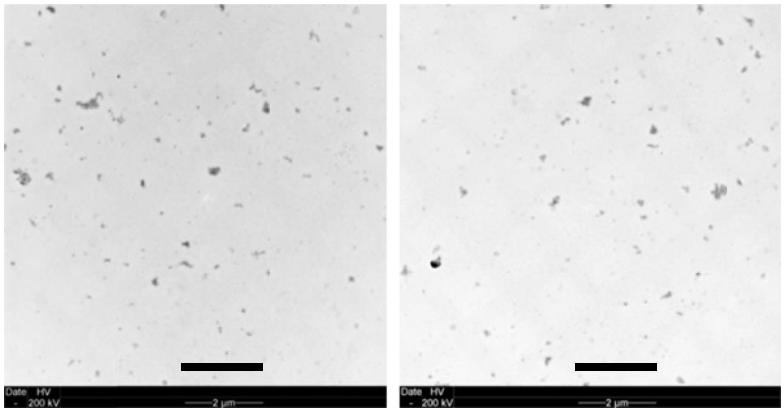


FIG. 21: TEM images of residual Silicone smoke particles from SAME pyrolysis at approximately twice the magnification. Sample heating temperature was 380 °C, bar = 2 µm.

TEM Particle Size Distribution Results

Particle size distributions were created by an image analysis of the particles captured on TEM grids as an independent verification of particle measurements. The size distributions were constructed by first locating the downstream edge of the particle deposition. Sequential images were then taken by traversing the grid in a line moving from one edge of the aerosol deposition to the other. This protocol was repeated until a large number of particles were captured. Images were processed to remove any background and adjusted to high contrast before setting a threshold. Particles were measured as the projected area, even when in elongated agglomerates.

The TEM particle size analysis method used ImageJ, an open source, public domain image processing progam (Rasband, W.S., ImageJ, U. S. National Institutes of Health, Bethesda, Maryland, USA, http://imagej.nih.gov/ij/, 1997-2012). The diameter of the area equivalent circle was computed for each particle and the volume distribution was based on the assumption of spherical particles.

The limitations of Silicone TEM images outlined above and the fractal nature of Teflon particles preclude the creation of a reliable size distribution by microscopy. Therefore, Kapton, lamp wick and Pyrell were analyzed. Representative size distributions of a high temperature Kapton test are shown in Figure 22. The unaged smoke has d_g = 158 nm and σ_g =1.68 and after 12 minutes of aging, d_g increases to 210 nm and σ_g shrinks to 1.63. This supports observations on Figure 15, which shows aged particles with a narrower size range and uniformly larger diameters. TEM size distributions for nonspherical Pyrell appear in Figure 22, with the unaged

smoke having d_g = 249 nm and σ_g =1.84 and after 14 minutes of aging, d_g increases to 328 nm and σ_g increases slightly to 1.98. Considering that Pyrell is not a spherical material, yet a spherical assumption was made in this image analysis, these results are quite reasonable.

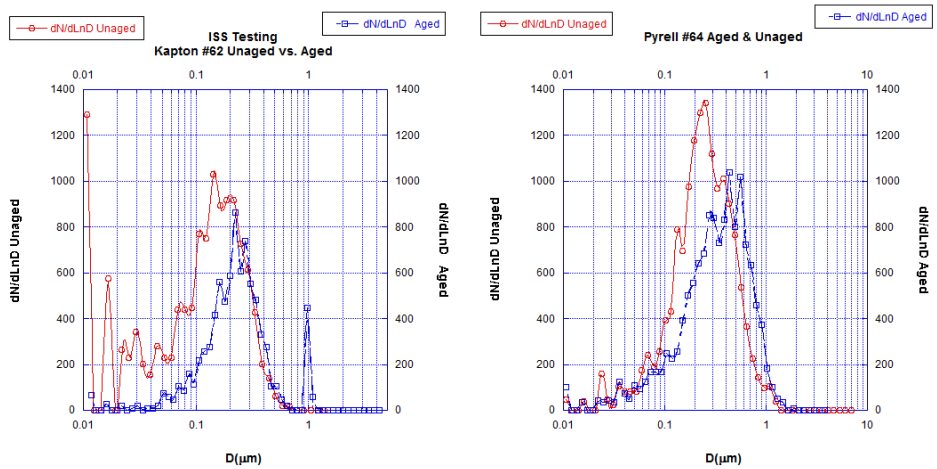


FIG. 22: Kapton particle size distribution (left) from TEM image analysis of ISS high temperature testing. Sample heating temperature was 574 °C. Pyrell particle size distribution (right) from TEM image analysis of ISS high temperature testing. Sample heating temperature was 242 °C.

Limitations of Aerosol Instrument Measurement Ranges

Calibration of SAME instruments was intended to empirically account for differences in the ranges of the instruments. In the ground validation tests, however, the truncated distribution formula (equation 13) can shed light on the limitations and uncertainty of the SMPS measurement range for the fuels tested. Particularly when the particle size distributions are converted to surface area or volume distributions, the percentage of the distribution that is lacking can be significant. Furthermore, the upper end of the SMPS size distribution measurement may not be as reliable because it can be affected by poor counting statistics and these bins are more susceptible to multiple charge correction errors. Therefore it is prudent to compare a more conservative upper SMPS limit of 700 nm along with the full recorded range to 1000 nm, to see the effect of the SMPS measurement range. Thus, if we were to consider the SMPS data most reliable (least uncertainty) in the range from 23 to 700 nm, the truncated distribution formula for $M_{p,rel}$ can indicate what percentage of the distribution would be captured with this limitation. Figure 10 shows the percentage of the distributions captured by the SMPS for two example materials. Kapton is the best candidate for SMPS validation, as the highest percentages of each type of distribution are within the SMPS measurement range. For example, considering both the conservative 700 nm limit and the 1000 nm limit, the bar graph indicates that 98% to 100% of the distribution has been measured. Silicone smoke is not a good candidate for SMPS validation because of the lack of measurement range, particularly when converting to surface and volume size distributions. Particularly in the volume distribution in Fig. 23 (right

graph), the black bar representing a 1000 nm upper size limit is only at 40%, indicating that 60% of the distribution is missed by the instrument, but when losses are considered and a range of 700 nm is relied upon, approximately 20% of the distribution is captured (the white bar).

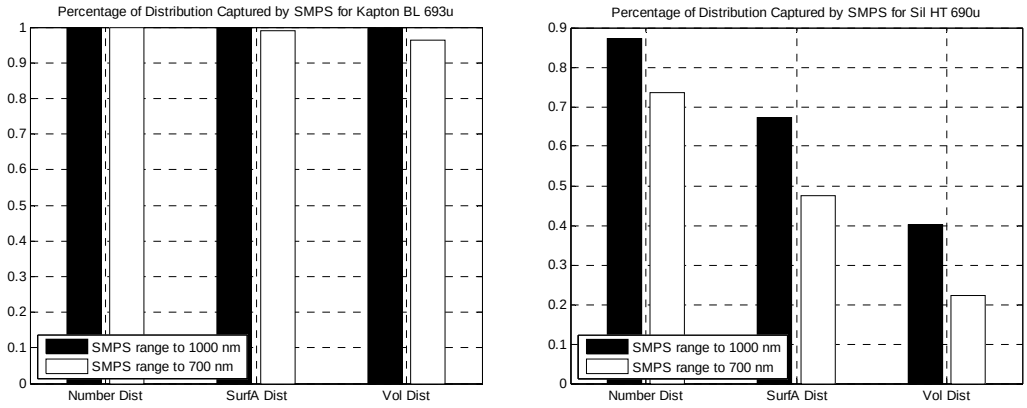


FIG. 23: The percentage of the particle size distributions captured by the SMPS for low temperature (510.5 °C) unaged Kapton smoke (left) and high temperature (369.2 °C) unaged Silicone smoke (right). Black bars represent an SMPS upper limit of 1000 nm (as measured) and white bars represent a more conservative upper limit of 700 nm

Thus, the truncated distribution formula for $M_{p,rel}$ can be a useful indicator of the suitability of an aerosol reference instrument and the level of uncertainty in measurments. If enough of the size distribution is known to get parameters of a lognormal fit from curve-fitting software, one can determine how comprehensive the size distribution measurement is and whether conversion of the distribution will produce reliable results.

Log-probability Plot

The log-probability plot is a graphical technique used to assess lognormality of an aerosol. Discrete data from the SMPS size bins and TEM size distributions are plotted on the probability scale. Lognormal distributions will be a straight line, and volume and number size distributions should be parallel, indicating that σ_g is the same for both distributions but with different means. Extremes on the graphs can be neglected in the log probability plots for number distributions when they deviate from the straight line fit for 5% or less of the extremes of the probability scale (Hinds, 1999). For the SMPS data, these deviations are the result of poor counting statistics in the large diameter tail and lack of measurement range (for lamp wick and silicone), which particularly affects the volume distribution plots. Thus the volume distribution plot will display a line that is slightly curved at the large sizes, which corresponds to a lack of linearity of the number distribution line beyond 95% of the distribution. Figures 24 and 25 show log-probability plots for Kapton®, silicone and lamp wick at high temperature. The plotted cumulative frequencies are quite linear for Kapton® and the number and volume distribution lines are nearly parallel. The log probability plot for silicone is not linear and thus fits a lognormal distribution poorly. The lognormal fit and the residual plot in Fig. 17 reflect the same result, where the data are steeper than the fit curve for large diameters and less steep than the fit results for smaller diameters, thus the log probability plot on the right side in Fig. 24 has deviations from the straight line fits at both the smallest and largest percentages of the distributions.

The lamp wick volume distribution has a smaller σ_g which is evident by the shallower slope. This is caused by the limited SMPS measurement range, which misses a portion of the high

temperature lamp wick large diameter tail, a deficiency which is emphasized in the conversion to volume concentration. Both aged and unaged log probability plots are shown for lamp wick (left and right graphs in Fig. 25) to demonstrate the flattening of the slope with aging, which corresponds to the narrowing of the distribution as coagulation takes place.

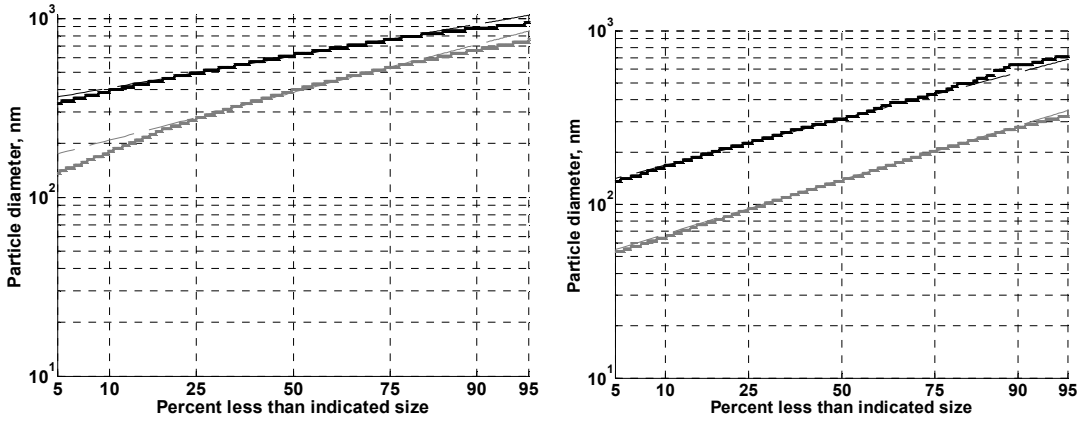


FIG. 24: Log-probability graphs for Kapton® high temperature sample, unaged (left) and high temperature Silicone aged (right), based on discrete SMPS bin data. Number distributions are grey and volume distributions are black.

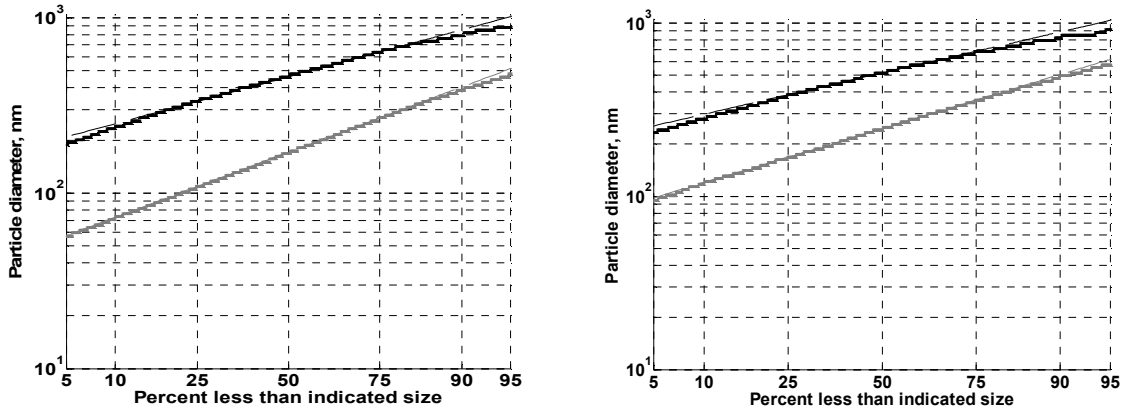


FIG. 25: Log-probability graphs for high temperature lamp wick unaged (left) and aged (right), based on discrete SMPS bin data. Number distributions are grey and volume distributions are black.

Comparison of SMPS Data, Discretely Calculated Moment Diameters vs. Hatch-Choate Diameters based on Lognormal Fit Parameters

The above graphs provide an overall qualitative measure whether the size distributions can be assumed lognormal. A useful quantitative measure of the validity of the lognormal assumption is the comparison between the SMPS measured moment, particle size, or σ_g and the corresponding value obtained by the moment method d_g and σ_g converted with the Hatch-Choate equations.

The SMPS bin data (based on a 64-channel histogram) can be used in equation (14) formulas to calculate diameters of average properties, which can then be compared with diameters calculated from equation (9) using the geometric mean diameter and σ_g from the lognormal fit of the SMPS particle size distribution. This is equivalent to calculating N_{tot}, L_{tot} and M_{tot} from SMPS binned data to obtain d_{av}, d_m, by equations (3) and (4). Thus, the continuous distribution parameters used in the conversion equations will be compared to the grouped data, and the expectation is that these diameters will be equal if the lognormal assumption is valid. Two examples of these comparisons are shown in Figure 26, which compares the count mean (number average, or d_{50} of the number distribution) diameter, surface area diameter, diameter of average mass, mass median diameter (d_{50} of the volume distribution), and the count median diameter (which is the geometric mean diameter if the distribution is lognormal). As can be seen, there is good agreement in all diameters for Kapton but not as good agreement for Silicone. The diameters with the largest deviations are those having to do with the mass. This is not surprising, since these are heavily influenced by the large diameter particles and the often the SMPS raw counts in the upper bins of the tail have fewer than 10 particles so there is the potential for discrepancies in the discrete bin calculations due to insufficient statistics.

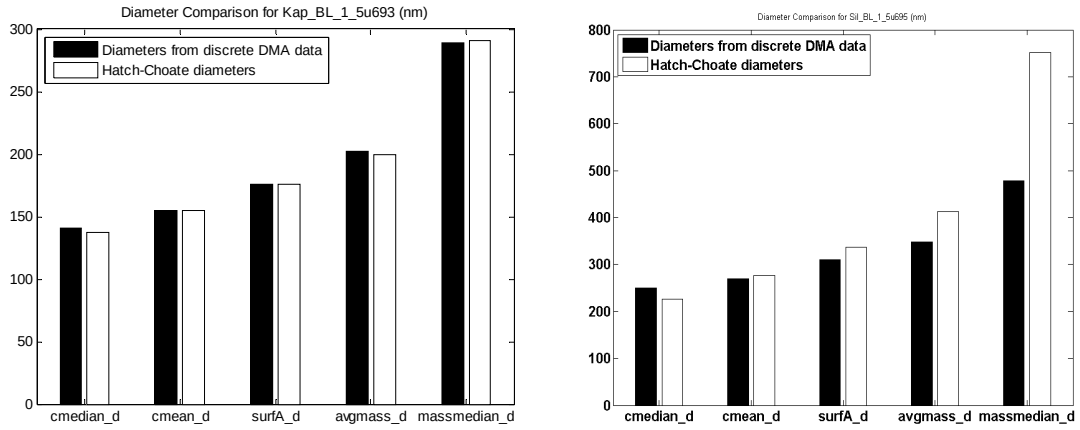


FIG. 26: Comparison of diameters calculated from SMPS data in two ways: 1) Converted by Hatch-Choate conversion equations using SMPS d_g and $\int_g$ and 2) Calculated using discrete SMPS bin data. Low temperature unaged Kapton smoke (left) and low temperature unaged silicone smoke (right).

Figure 27 shows the results of these bar graphs from all three spherical aerosols on one plot (including unaged and aged diameters, at all temperatures tested). Data falling on the 1:1 reference line meet the lognormal assumption, whereas those that differ significantly do not. While some information is lost in this scatter plot vs. the bar graphs (eg. which data marker represents which moment diameter), the graph shows that for Kapton and lamp wick, the diameters calculated by both methods coincide and thus can be considered lognormal. It is evident that diameters above 500 nm, which are the higher moment diameters, and mostly Silicone, do not coincide. Overall, the qualitative comparison of these diameters strongly suggests that the smoke particle size distributions for Kapton® and lamp wick can be considered lognormal and silicone should not.

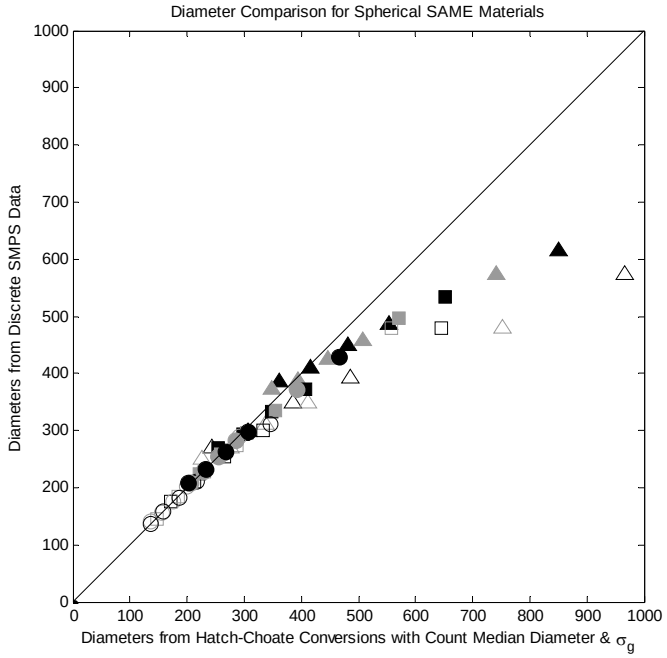


FIG. 27: Comparison of diameters calculated from SMPS data in two ways: 1) Converted by Hatch-Choate conversion equations using SMPS d_g and $\int_g$ and 2) Calculated using discrete SMPS bin data. All diameters as represented in Fig. 14 are combined here for all materials and test conditions: open markers are unaged, solid are aged; grey represents baseline temperature and black represents high temperature tests, and marker shapes are Kapton – circle, lamp wick – square, Silicone – triangle.

Comparison Between Size Distribution Parameters from TEM and Moment Method Measurements from Flight Data

The TEM particle size analysis involved determination of d_g and σ_g of the distribution as well as d_m and d_{av}. These values can be compared with the moment method results measured simultaneously during the smoke particle collection on the TEM grids. As has been discussed, Kapton and lamp wick are the most lognormal smoke aerosols tested. Unfortunately, the flight lamp wick TEM data was not comprehensive, so complete comparisons cannot be made. The Kapton TEM data was much more complete and thus can be compared systematically against the moment method data. Figure 28 shows the comparisons for high and low temperature Kapton tests, with the TEM values in the patterned bars and moment method values in the solid colored bars.

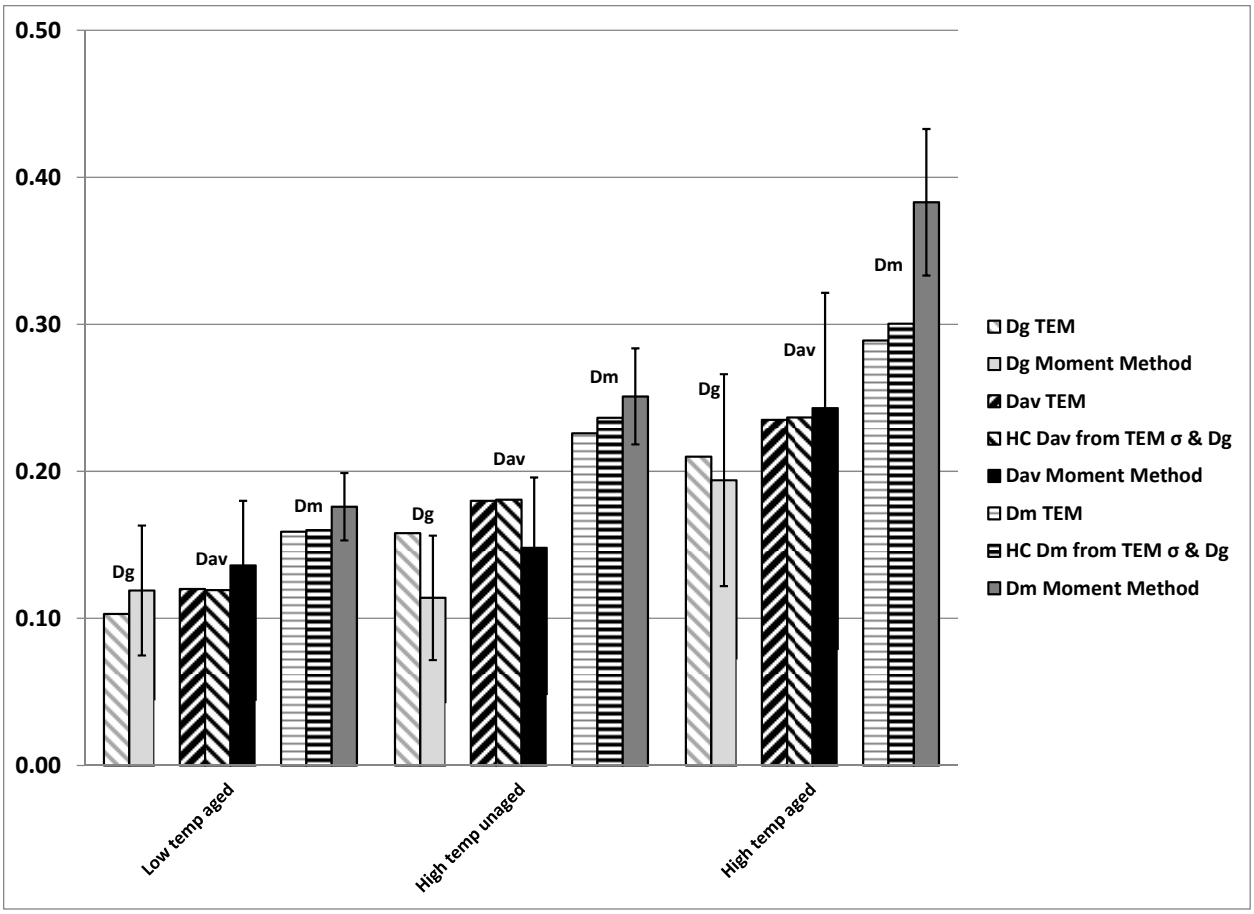


FIG. 28: Diameter Comparison for Kapton smoke from flight data. TEM particle size distribution diameters are compared to converted diameters calculated from the Hatch-Choate conversion equations using the TEM analysis d_g and $\int_g$ and also compared with diameters from the moment method.

Error bars on Figure 28 apply to moment method quantities and are based on the error analysis using the uncertainties from Cleary, et al. (2009), which quantify the first moment instrument error at ± 12 %, the third moment instrument error at ± 25 %. The zeroth moment instrument error was estimated to be ± 30 %.

As expected, Kapton has good agreement with most moment method diameters, as it has a lognormal size distribution. The TEM diameters from the size analysis and the converted values are very consistent as well. Thus, it can be concluded that Kapton smoke at these temperatures is very well suited to measurement by the moment method.

Although meticulous calibrations were performed, many smoke aerosols exceeded the calibration range for the ionization detector which measures the first moment. As evident in equations (12) and (13), error in the first moment measurement has the largest influence on the resulting calculations. Kapton consists of smaller particles and was not affected by this shortcoming, so these tests provided the most reliable moment method results. Since the first

moment measurement is not used in the calculation of d_m, this quantity can be compared for the other materials.

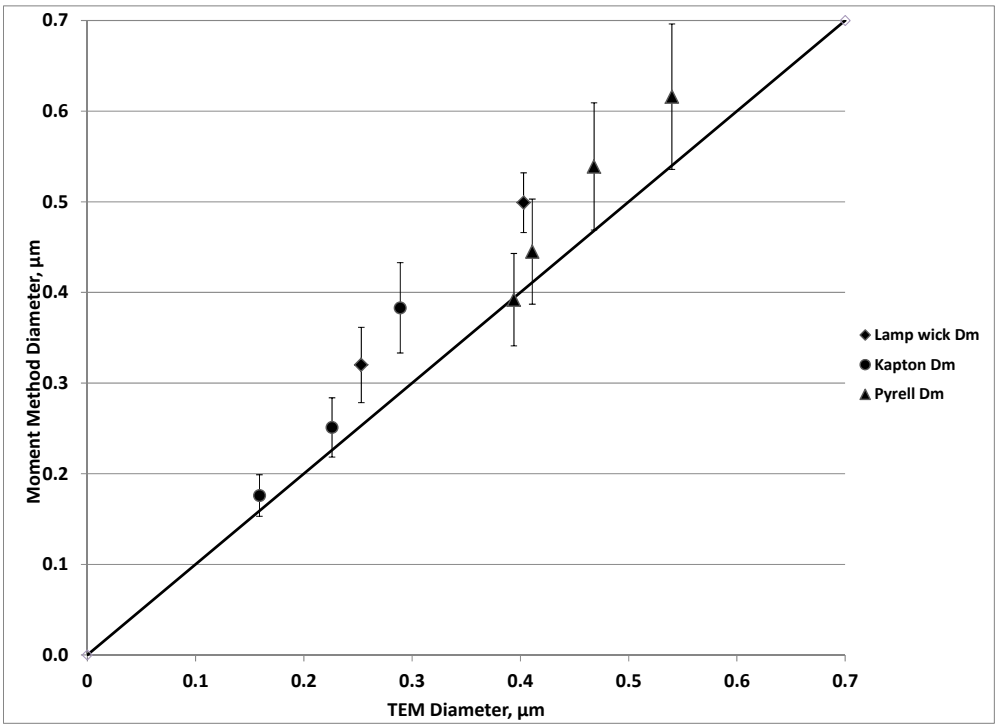


FIG. 29: Comparison of TEM analysis and moment method diameter of average mass for smoke from flight data.

Figure 29 shows the comparison of the flight TEM diameter of average mass with the value calculated from the moment instrument data using equation (4). Error bars for the moment method values of diameter of average mass were calculated as in Figure 28. Since the third moment instrument was calibrated for each smoke type with a direct-reading reference instrument, the measurements are assumed to be relatively shape-independent. For the three materials in this graph, the moment method provides a reasonably good measurement for the diameter of average mass. Thus it can be concluded that the zeroth and third moment instruments maintained their calibration sufficiently to quantify one of the moment average diameters from flight data.

Flow test –verification of dilution ratio

In SAME/SAMER apparatus, to ensure the concentration of aerosol stream entering the P-trak and Dusttrak were within the ranges of the instruments, the aerosol stream was diluted with the N2 gas provided by the ISS. For P-trak, this was done through a custom-designed device called dynamic diluter in which a valve controls the diluting N2 in such manner that the ration between the N2 flow and that of the incoming aerosol stream meets the specified value while the total flow rate is maintained as a constant. When performing data reduction, the commanded dilution ratio is always used. For DustTraks, the dilution flow can be controlled as on/off two statuses. For each run, whether the dilution needs to be on or not is specified in the commend sheet, i.e., predetermined. Once the dilution is turned on, the N2 flow is controlled by an orifice, meant to be operated under critical flow regime. The total flow through a DustTrak is controlled by an orifice operating under critical flow regime. Thus, in normal condition, the flow rate of incoming stream is the difference between these two constant flow rates and the dilution ratio can be calculated accordingly.

In order to make sure the dilution ratio used in the data reduction procedures are accurate, a series flow tests were conducted by inserting mass flow meters in different location to measure the actual flow rates.

The test results about P-Trak dilution show that

a) The correlation between actual sample flow and commanded flow to LFE for the Flight Dynamic diluter is:
   actual flow (ccm) = 0.9474*command value (ccm) -1.2395 (ccm).
b) There are no valid test results to give such a correlation for the Engineering Dynamic diluter. Thus the above correlation will be used for the engineering Dynamic diluter if needed.
c) The total air flow the Engineering P-trak is, in average, 703.96 ccm.
d) The total air flow the Flight P-trak is, in average, 705.64 ccm.
e) The column "Aerosol Concentration to P-Trak" in the SAMER Combined .xlsx file is the sample gas flow to LFE over the total air flow to the P-Trak.

The results about DustTrak dilution are more complicated. First of all, it was realized that the orifice that controls the diluting N2 flow was not working under the critical flow regime as original designed. From the logged data in the engineering data file, the N2 pressure upstream the orifice, which is a regulated pressure, is not high enough to maintain a critical flow. This pressure varied much more than it was supposed to, indicating that the regulator is undersized. Thus, the actual diluting N2 flow rate depends on both upstream and downstream pressure of the orifice and the size of the orifice and the temperature as well, although the temperature dependency is weak. The total flow rate going through a DustTrak is controlled by the orifice in each CDE, which is truly working under critical flow regime. Thus for the two DustTraks in the SAME apparatus, there are totally 4 orifices, two for the diluting N2 flows installed in the fluid control unit and two for the total flows installed with each DustTrak in its CDE. These 4 orifices are of the same model from the same manufacture. Unfortunately, these for orifices were not calibrated at procurement. Thus we only have the generic characteristic curve for that model of orifice provided by the manufacture, which is as follows

N2 flow (SLPM) = 0.308*(Regulated N2 pressure)^0.5283

This equation is used to calculate the actual N2 diluting flow rate based on logged upstream and downstream N2 pressures.

After the flight CDE units were shipped to the ground, the actual total flow rate though the Dusttrak and the MPASS were measured.

The total flow rate of Flight unit DustTrak B (used in SAMER) is later found to be 1.69076 LPM.

The total flow rate of Flight unit Mpass (used in SAMER) is later found to be 1.68576 LPM.

Sample heating temperature derivation

Sample heating temperature is one of the key parameters that dictate the properties of generated smoke particles. When studying effects of gravity on the smoke properties, we want to compare smokes generated at the same temperature. For SAME and SAMER, during a test, the software controls the temperature via the resistance of heating wire. It is true that for the same wire coil, constant wire resistance means constant wire temperature. Traditionally, test conditions are specified by Rt/R0, the ratio between the resistance of heated wire over the resistance of the wire at room temperature. Thus, before each run, the sample wire resistance is measured, then multiplied by this parameter to get the target resistance for the control circuitry. Although, Rt/R0 can uniquely define the temperature, it is not clear what actual sample temperature is. Since the wire resistance is one of the parameters that are logged in the engineering data sheet which makes it possible to derive actual temperature from the logged resistances. However, when deriving wire temperatures, one has to take the fact into account that the wire length varies from sample to sample and so called room temperature varies too. After the heating temperatures were obtained, it is surprising that for the same R_t/R_0, the actual heating temperature difference can be as large as 30 degrees Celsius. The following is how this is carried out.

SAME heater temperature modeling.

Objective:

It was identified that each sample has a significant length of leading section of the resistive wire. This section does not touch the sample material and is exposed in air flow. Thus the temperature of this leading section is different from the portion of the wire that contacts the sample material. The control circuitry only controls the total resistance of the heating wire, not individual portion of the wire.

Our goal is to find the temperature of the wire that is in contact with the sample material.

Formulation:
Hypothesis: the temperature difference between the two portions of the wire is caused by different convection heat transfer from the two. The wire that is in contact with the sample has less heat loss from the convection heat transfer because about half of its surface area is covered by the sample material.

Assumptions and simplifications:

1) *Assume the entire wire consists of two distinct portions, one is in air (portion 1) and the other (portion 2) is in touch with the sample so that only half of the wire area is exposed to air.*
2) *Uniform cross section area of the wire, so the resistance is proportional to the length and the resistivity of the wire.*
3) *Resistivity is a known function of temperature.*
4) *Temperature distribution inside wire is uniform.*
5) *Newton convective heat transfer equation applies to both sections but with different coefficients. Although the port 2 is half exposed in air where convective equation works and half in contact with sample where transient conduction heat transfer should be used, for the sake of simplicity, we lump the two mechanisms together and use convection only with a modified heat transfer coefficient.*

Notation:
R_t *– total resistance, Ohm*
R_1 *– resistance of part 1, Ohm*
R_2 *– resistance of part 2, Ohm*
h *– heat transfer coefficient of air,* $w/^oC/mm^2$
T_1 *– temperature of part 1,* ^oC
T_2 *– temperature of part 2,* ^oC
L_1 *– length of part 1, mm*
L_2 *– length of part 2, mm*
R *– radius of the heating wire, mm*
I *-- electric current, amp*
m *– a parameter within 0.5 and 1.*
$\sigma(T)$ *– resistivity as a known function of temperature, in a form of polynomial, Ohm*mm*

Equations:
Total measured resistance equals the sum of the resistances of part 1 and part 2 of the wire

(1) $$R_t = (L_1\sigma(T_1) + L_2\sigma(T_2))/(\pi r^2)$$

This R_t *can be obtained from the engineering data file.*

In steady state, the heat generated in the wire and heat loss from the wire should be the same.
For the part 1, the heat loss is via convective heat transfer to the air.

(2) $$I^2\sigma(T_1)L_1/(\pi r^2) = 2\pi r L_1 h(T_1 - T_0)$$

For part 2, only about the half of the wire surface is exposed to air. The other half is in contact with the solid sample.
Thus,

(3) $I^2\sigma(T_2)L_2/(\pi r^2) = h\pi r L_2(T_2 - T_0)$ *+ conduction to the sample*

Since the conduction part is also proportional to the surface area and (T_2-T_0)*, Eq. 3 can be rewritten as*

(4) $$I^2\sigma(T_2)L_2/(\pi r^2) = (h+k)\pi r L_2(T_2 - T_0)$$

where k is a coefficient for conduction part. This "handwavy" because the conduction part is much more complicated and the coefficient k won't be a constant. It should vary with time. It is probably safe to assume k, as such defined, is smaller than h, i.e., 0<k<h. Then Eq. 4 can further be reduced to

(5) $I^2\sigma(T_2)L_2/(\pi r^2) = 2\mu\pi r L_2 h(T_2 - T_0)$

where m is an unknown, but bounded coefficient: $0.5 < \mu < 1$.

Dividing Eq. (5) by Eq. (2), we got

(6) $\sigma(T_2)/\sigma(T_1) = \mu(T_2-T_0)/(T_1-T_0)$

Eq. 1 can be rewritten as

(7) $R_t = [L_1 + L_2\sigma(T_2)/\sigma(T_1)]\,\sigma(T_1)/(\pi r^2)$

Substituting (6) into (7), we got

(8) $R_t = [L_1 + L_2\,\mu(T_2-T_0)/(T_1-T_0)]\,\sigma(T_1)/(\pi r^2)$

Now Eq. 6 and Eq.8 are two independent equations for two unknowns T_1 *and* T_2*, with m being a parameter within 0.5 and 1.0. For each test run, we can get* R_t, T_0, L_1, L_2 *and r from engineering data files, then assume a number of* μ*s to solve for* T_1 *and* T_2 *to see how the temperatures are sensitive to* μ.

Cumulative moments for limited particle size range

Theoretically, a lognormal distribution is defined in the particle size from zero to infinity, whereas any instrument used to measure certain moment of an aerosol inevitably has a limited particle size range. In some cases, an impacter is inserted in the inlet of an instrument to eliminate particles larger than certain size. In these cases, the instrument cannot sense the entire range of the distribution as the theory describes. Thus, one would like to know how much signal is lost due to the limited effective particle size range, or what the percentage of the total signal the instrument receives. This piece of information is useful to assess errors induced by omitting particles in a certain size range. It was found this question can be answered conveniently using the error function calculations, which is available in the Excel spread sheet or some other computational software or even in the tables of most mathematical handbooks. The following is the detailed discussion of how to do this.

Derivation of an equation to calculate the cumulative moment function in a limited range for the lognormal distribution function via the error function.

We will mostly use the same notation that is used in the Hinds' book except using d instead of d_p for the particle size for the sake of convenience. Thus d can be either particle size or the differentiation operator depending on the context.

The p moment cumulative function of a lognormal distribution with d_g and σ_g is defined as:

$$M_p(D) = \frac{1}{\sqrt{2\pi}ln\sigma_g}\int_0^D d^{p-1}exp\left(-\frac{(lnd-lnd_g)^2}{2(ln\sigma_g)^2}\right)dd \qquad (1)$$

Using the following new variables, we can simplify the derivation.

$$A = \frac{1}{\sqrt{2\pi}ln\sigma_g},\ \ B = 2(ln\sigma_g)^2,\ \ C = lnd_g \text{ and } L = lnd. \qquad (2)$$

Thus we have a simplified form of (1):

$$M_p(D) = A\int_{-\infty}^{lnD} exp\left(pL - \frac{(L-C)^2}{B}\right)dL \qquad (3)$$

It can be shown that

$$pL - \frac{(L-C)^2}{B} = pC + 0.25p^2B - \frac{1}{B}(L - C - 0.5pB)^2 \qquad (4)$$

Let's define two new variables to simplify the derivation.

$$\xi = pC + 0.25p^2B \text{ and } \eta = \frac{L-C-0.5pB}{\sqrt{B}} \quad (5)$$

From (5), we have that $dL = \sqrt{B}d\eta$

Now, (3) can be rewritten as

$$M_p(D) = A\sqrt{B}\int_{-\infty}^{\eta_D} \exp(\xi - \eta^2)\, d\eta = A\sqrt{B}\exp(\xi)\int_{-\infty}^{\eta_D} \exp(-\eta^2)\, d\eta \quad (6)$$

or, in terms of the error function, we have

$$M_p(D) = \frac{\sqrt{\pi}}{2}A\sqrt{B}exp(\xi)[1 + erf(\eta_D)] \quad (7)$$

where $\eta_D = \frac{lnD-C-0.5pB}{\sqrt{B}} = \frac{ln\left(\frac{D}{d_g}\right)}{\sqrt{2}ln\sigma_g} - \frac{1}{\sqrt{2}}pln\sigma_g$ (8)

Now let's eliminate all intermediate variables to express $M_p(D)$ in terms of d_g, σ_g and p. Equation (7) can be rewritten as:

$$M_p(D) = \frac{1}{2}d_g^p exp\left(0.5\left(pln\sigma_g\right)^2\right)[1 + erf(\eta_D)] \quad (9)$$

It is interesting to see when D approaches infinity, what the total p moment is. As D approaches infinity, η_D approaches infinity too and error function approaches 1. Thus, the total p moment of the lognormal distribution is

$$M_p = M_p(\infty) = d_g^p exp\left[0.5\left(pln\sigma_g\right)^2\right] \quad (10)$$

Both equations (9) and (10) are very useful, if we haven't seen them in some other places. These two equations give absolute value of the *p-th* moment, partial or total. The dimension of the *p-th* moment is $[L^p]$. However, in many cases, we are interested in the relative value to the total moment or the moment value normalized by the total *p-th* moment value. There relative values can tell us how much present signal is lost because of the limited detection range of an instrument. Now let's move further to see what this relative cumulative *p-th* moment is. Dividing Eq. (9) by Eq. (10), we have

$$M_{pr} = \frac{1}{2}[1 + erf(\eta_D)] \quad (11)$$

We can see (11) is amazingly simple and useful. Note that in equation (8) D is normalized by d_g. Thus M_{pr}, the relative cumulative value only depends on 3 parameters, normalized D, σ_g and p.

Implementation

If we want to know how much percentage of the signal an instrument for the *p-th* moment receives, which cannot detect particles larger than diameter D, assuming a lognormal distribution with d_g and σ_g.

This can be done as follows.

1) Calculate η_D using (8) for given d_g, σ_g, p and D.
2) Find $erf(\eta_D)$ from any source that is available.
3) Using (11) to calculate $M_{pr}(D)$, which corresponds to the percentage of the signal received.

Sometimes an instrument has a lower and upper limits of its particle size detection range. We can make a miner change of Eq. (9) and (11) to find out how much signal the instrument actually detects.

Let's introduce the following notations.

D_l and D_u - the lower and upper limits, respectively, of particle diameters the instrument can detect.

η_{dl} and $\eta_{du} - \eta_d$ corresponding to D_l and D_u, respectively, using equation (8).

M_{pi} – the pth moment signal the instrument detects.

M_{pri} – the pth moment signal the instrument detects in percentage.

From Eq. (9), we have

$$M_{pi} = M_p(D_u) - M_p(D_l) = \frac{1}{2} d_g^p exp\left(0.5\left(pln\sigma_g\right)^2\right)[erf(\eta_{du}) - erf(\eta_{dl})] \quad (12)$$

Dividing Eq (12) by Eq (10), we get

$$M_{pri} = \frac{1}{2}[erf(\eta_{du}) - erf(\eta_{dl})] \quad (13)$$

We can use eqs. (12) and (13) to find how much an instrument can see if its lower and upper detection limit are D_l and D_u.

Appendix A: Test matrix

Smoke Generation	Carousel	1	1	1	1	1	1	1
	Sample Type	Silicone	Lampwick	Teflon	Kapton	Kapton	Pyrell	Silicone
	Sample ID	S03	L13	T01	K01	K02	P02	S04
	Test ID Number	1S	2L	3T	4K	4K	5P	6S
	Position in Carousel	1	2	3	4	4	5	6
	SAME Test #	48	49	50	51	51a	52	53
	Description	Low Temp	B.L. High RT/R0	B.L. High RT/R0	B.L. new formula	B.L. new formula	Baseline	HighTemp
	Date (mm/dd/2010)	7/16	7/19	7/20	7/21	7/22	7/22	7/22
	Inlet Velocity (cm/sec)	8	8	8.1	8.1	8.1	8.1	8.1
	Rt/R0	1.304	1.251	1.452	1.380	1.380	1.220	1.350
	Duration (sec)	60	60	60	60	60	60	60
	Primary Aging time (sec)	11	11	11	11		11	1
	Primary Mixing time (sec)	8	8	8	8		8	8
	Net Aging time (sec)	62	51	33	38	261	n/a	52
Primary Diagnostics	Prim. Aging Therm Precip	y	y	y	y		y	y
	Prim. Aging Therm Precip time	60	90	75	?		60	?
	Use STS Detector	n	n	n	n		n	n
	Use Comm Diag A (MPASS)	y	y	y	y		y	y
	Use Comm Diag B	y	y	y	y		y	y
	Diluting A (MPASS)	y	y	n	n	n	y	y
	Diluting B	y	y	n	n	n	y	y
	Cyclone Inlet on CDA (MPASS)	y	y	y	y		y	y
	Impactor Inlet on CDB	10	10	10	10		10	10
	Use P-Trak	y	y	y	y		y	y
	Aerosol Concentration To P-Trak	0.0318	0.0385	0.0520	0.0385	0.0385	0.0385	0.0318
Secondary Diagnostics Volume	Secondary Diagnostics Volume %							
	Secondary Aging Duration (sec)				180			
	STS Detector				n			
	Comm Diag A (MPASS)				y			
	Comm Diag B				y			
	Diluting A (MPASS)				n			
	Diluting B				n			
	Ues PTrak				y			
	Aerosol Concentration To P-Trak				0.0385			
	2nd agign therml precip duration				0			

Smoke Generation	Carousel	2	2	2	2	2	2
	Sample Type	Lampwick	Lampwick	Lampwick	Teflon	Lampwick	Teflon
	Sample ID	L14	L15	L15	T02	L14	T03
	Test ID Number	7L	8L	8La	9T	10L	11T
	Position in Carousel	3	4	4	5	3	6
	SAME Test #	54	55	55a	56	57	58
	Description	Baseline	High RT/R0 test aging	aged	Baseline	High RT/R0 reburn sample	Baseline aging
	Date (mm/dd/2010)	7/29	7/29	7/29	7/29	7/29	7/30
	Inlet Velocity (cm/sec)	8.1	8.1	8.1	8.1	8.1	8.1
	Rt/R0	1.245	1.245	1.245	1.445	1.253	1.445
	Duration (sec)	60	60		60	60	60
	Primary Aging time (sec)	11	11		11	11	11
	Primary Mixing time (sec)	8	8		8	8	8
	Net Aging time (sec)	57	47	218	41	n/a	34
Primary Diagnostics	Prim. Aging Therm Precip	y	y		y		y
	Prim. Aging Therm Precip time	120	60		75		75
	Use STS Detector	n	n		n	n	n
	Use Comm Diag A (MPASS)	n	n		y	n	y
	Use Comm Diag B	y	y		y	y	y
	Diluting A (MPASS)	n	n	n	n	n	n
	Diluting B	y	y	y	n	y	n
	Cyclone Inlet on CDA (MPASS)	y	y		y	y	y
	Impactor Inlet on CDB	10	10		10	10	10
	Use P-Trak	y	y		y	y	y
	Aerosol Concentration To P-Trak	0.0385	0.0385	0.0385	0.0520	0.0385	0.0520
Secondary Diagnostics Volume	Secondary Diagnostics Volume %						
	Secondary Aging Duration (sec)						
	STS Detector						
	Comm Diag A (MPASS)						
	Comm Diag B						
	Diluting A (MPASS)						
	Diluting B						
	Ues PTrak						
	Aerosol Concentration To P-Trak						
	secondary agign therml precip duration						

Smoke Generation	Carousel	2	2	2	2	2
	Sample Type	Teflon	Silicone	Silicone	Silicone	Silicone
	Sample ID	T03	S07	S07	S09	S09
	Test ID Number	11Ta	12S	12Sa	13S	13Sa
	Position in Carousel	6	1	1	2	2
	SAME Test #	58a	59	59a	60	60a
	Description	aged	Baseline aging	aged	Htemp aging	aged
	Date (mm/dd/2010)	7/30	7/30	7/30	7/30	7/30
	Inlet Velocity (cm/sec)	8.1	8.1	8.1	8.1	8.1
	Rt/R0	1.445	1.327	1.327	1.350	1.350
	Duration (sec)		60		60	
	Primary Aging time (sec)		11		11	
	Primary Mixing time (sec)		8		8	
	Net Aging time (sec)	805	24	856	49	856
Primary Diagnostics	Prim. Aging Therm Precip				y	
	Prim. Aging Therm Precip time				720	
	Use STS Detector		n		n	
	Use Comm Diag A (MPASS)		y		y	
	Use Comm Diag B		y		y	
	Diluting A (MPASS)	n	y	y	y	y
	Diluting B	n	y	y	y	y
	Cyclone Inlet on CDA (MPASS)		y		y	
	Impactor Inlet on CDB		10		10	
	Use P-Trak		y		y	
	Aerosol Concentration To P-Trak	0.0520	0.0318	0.0318	0.0318	0.0318
Secondary Diagnostics Volume	Secondary Diagnostics Volume %	50		70		70
	Secondary Aging Duration (sec)	720		720		720
	STS Detector	n		n		n
	Comm Diag A (MPASS)	y		y		y
	Comm Diag B	y		y		y
	Diluting A (MPASS)	n		y		y
	Diluting B	n		y		y
	Ues PTrak	y		y		y
	Aerosol Concentration To P-Trak	0.0520		0.0318		0.0318
	secondary agign therml precip duration	??				

Smoke Generation	Carousel	3	3	3	3	3	3
	Sample Type	Kapton	Kapton	Kapton	Kapton	Pyrell	Pyrell
	Sample ID	K02	K02	K03	K03	P03	P03
	Test ID Number	14K	14Ka	15K	15Ka	16P	16Pa
	Position in Carousel	1	1	2	2	3	3
	SAME Test #	61	61a	62	62a	63	63a
	Description	Baseline Aging	aged	High Temp aging	aged	Baseline Aging	aged
	Date (mm/dd/2010)	8/2	8/2	8/2	8/2	8/23	8/23
	Inlet Velocity (cm/sec)	8.1	8.1	8.1	8.1	8.1	8.1
	Rt/R0	1.380	1.380	1.480	1.480	1.220	1.220
	Duration (sec)	60		60		60	
	Primary Aging time (sec)	11		11		11	
	Primary Mixing time (sec)	8		8		8	
	Net Aging time (sec)	38	807	39	808	56	840
Primary Diagnostics	Prim. Aging Therm Precip	n		Y		y	
	Prim. Aging Therm Precip time			60		60	
	Use STS Detector	n		n		n	
	Use Comm Diag A (MPASS)	y		y		y	
	Use Comm Diag B	y		y		y	
	Diluting A (MPASS)	n	n	n	n	y	y
	Diluting B	n	n	n	n	y	y
	Cyclone Inlet on CDA (MPASS)	y		y		y	
	Impactor Inlet on CDB	10		10		10	
	Use P-Trak	y		y		y	
	Aerosol Concentration To P-Trak	0.0385	0.0385	0.0385	0.0385	0.0385	0.0385
Secondary Diagnostics Volume	Secondary Diagnostics Volume %				50		70
	Secondary Aging Duration (sec)	720			720		720
	STS Detector				n		n
	Comm Diag A (MPASS)				y		y
	Comm Diag B				y		y
	Diluting A (MPASS)				n		y
	Diluting B				n		y
	Ues PTrak				y		y
	Aerosol Concentration To P-Trak				0.0385		0.0385
	secondary agign therml precip duration				60		60

Smoke Generation	Carousel	3	3	3	3	3	3
	Sample Type	Pyrell	Pyrell	Silicone	Silicone	Silicone	Silicone
	Sample ID	P04	P04	S12	S12	S13	S13
	Test ID Number	17P	17Pa	18S	18Sa	19S	19Sa
	Position in Carousel	4	4	5	5	6	6
	SAME Test #	64	64a	65	65a	66	66a
	Description	High Temp aging	aged	Baseline long Aging	aged	High Temp aging	aged
	Date (mm/dd/2010)	8/23	8/23	8/23	8/23	8/23	8/23
	Inlet Velocity (cm/sec)	8.1	8.1	8.1	8.1	8.1	8.1
	Rt/R0	1.230	1.230	1.327	1.327	1.360	1.360
	Duration (sec)	60		60		60	
	Primary Aging time (sec)	11		11		11	
	Primary Mixing time (sec)	8		8		8	
	Net Aging time (sec)	47	842	50	1923	42	837
Primary Diagnostics	Prim. Aging Therm Precip	y					
	Prim. Aging Therm Precip time	45					
	Use STS Detector	n		n		n	
	Use Comm Diag A (MPASS)	y		y		y	
	Use Comm Diag B	y		y		y	
	Diluting A (MPASS)	y	y	y	y	y	y
	Diluting B	y	y	y	y	y	y
	Cyclone Inlet on CDA (MPASS)	y		y		y	
	Impactor Inlet on CDB	10		10		10	
	Use P-Trak	y		y		y	
	Aerosol Concentration To P-Trak	0.0385	0.0385	0.0318	0.0318	0.0318	0.0318
Secondary Diagnostics Volume	Secondary Diagnostics Volume %		70		70		70
	Secondary Aging Duration (sec)		720		1800		720
	STS Detector		n		n		n
	Comm Diag A (MPASS)		y		y		y
	Comm Diag B		y		y		y
	Diluting A (MPASS)		y		y		y
	Diluting B		y		y		y
	Ues PTrak		y		y		y
	Aerosol Concentration To P-Trak		0.0385		0.0318		0.0318
	secondary agign therml precip duration		45				

Smoke Generation	Carousel	6	6	6	6	6	6
	Sample Type	Silicone	Lampwick	Teflon	Kapton	Pyrell	Silicone
	Sample ID	S02	L25	T07	K07	P08	S05
	Test ID Number	20S	21L	22T	23K	24P	25S
	Position in Carousel	1	2	3	4	5	6
	SAME Test #	67	68	69	70	71	72
	Description	High temp aged STS det	High temp aged STS det	High temp aged STS det	High temp aged STS det	High temp aged STS det	Baseline STS det
	Date (mm/dd/2010)	8/24	8/24	8/24	8/24	8/24	8/25
	Inlet Velocity (cm/sec)	8.1	8.1	8.1	8.1	8.1	8.1
	Rt/R0	1.350	1.280	1.450	1.480	1.230	1.327
	Duration (sec)	60	60	60	60	60	60
	Primary Aging time (sec)	720	720	720	720	720	11
	Primary Mixing time (sec)	8	8	8	8	8	8
	Net Aging time (sec)	751	747	745	747	750	42
Primary Diagnostics	Prim. Aging Therm Precip	y	y	y	y	y	y
	Prim. Aging Therm Precip time	300	90	45	60	45	60
	Use STS Detector	y	y	y	y	y	y
	Use Comm Diag A (MPASS)	y	y	y	y	y	n
	Use Comm Diag B	y	y	y	y	y	y
	Diluting A (MPASS)	y	n	n	n	y	n
	Diluting B	y	y	n	n	y	y
	Cyclone Inlet on CDA (MPASS)	y	y	y	y	y	y
	Impactor Inlet on CDB	10	10	10	10	10	10
	Use P-Trak	y	y	y	y	y	y
	Aerosol Concentration To P-Trak	0.0318	0.0385	0.0520	0.0385	0.0385	0.0318
Secondary Diagnostics Volume	Secondary Diagnostics Volume %						
	Secondary Aging Duration (sec)						
	STS Detector						
	Comm Diag A (MPASS)						
	Comm Diag B						
	Diluting A (MPASS)						
	Diluting B						
	Ues PTrak						
	Aerosol Concentration To P-Trak						
	secondary agign therml precip duration						

Smoke Generation	Carousel	5	5	5	5	5
	Sample Type	Kapton	Kapton	Pyrell	Pyrell	Kapton
	Sample ID	K04	K04	P05	P05	K05
	Test ID Number	26K	26Ka	27P	27Pa	28K
	Position in Carousel	1	1	4	4	3
	SAME Test #	73	73a	74	74a	75
	Description	Baseline Aging	aged	Baseline Aging	aged	Mid Temp aging (no Ptrak)
	Date (mm/dd/2010)	8/26	8/26	8/26	8/26	8/26
	Inlet Velocity (cm/sec)	8.1	8.1	8.1	8.1	8.1
	Rt/R0	1.380	1.380	1.220	1.220	1.430
	Duration (sec)	60		60		60
	Primary Aging time (sec)	11		11		11
	Primary Mixing time (sec)	8		8		8
	Net Aging time (sec)	33	560	48	623	32
Primary Diagnostics	Prim. Aging Therm Precip	y		y		y
	Prim. Aging Therm Precip time	60		60		60
	Use STS Detector	n		n		n
	Use Comm Diag A (MPASS)	y		y		y
	Use Comm Diag B	y		y		y
	Diluting A (MPASS)	n	n	y	y	n
	Diluting B	n	n	y	y	n
	Cyclone Inlet on CDA (MPASS)	y		y		y
	Impactor Inlet on CDB	10		10		10
	Use P-Trak	y		y		y
	Aerosol Concentration To P-Trak	0.0385	0.0385	0.0385	0.0385	0.0385
Secondary Diagnostics Volume	Secondary Diagnostics Volume %	58		58		58
	Secondary Aging Duration (sec)	480		480		480
	STS Detector	n		n		n
	Comm Diag A (MPASS)	y		y		y
	Comm Diag B	y		y		y
	Diluting A (MPASS)	n		y		n
	Diluting B	n		y		n
	Ues PTrak	y		y		y
	Aerosol Concentration To P-Trak	0.0385		0.0385		0.0385
	secondary agign therml precip duration		60		60	

Smoke Generation	Carousel	5	5	5	5	5
	Sample Type	Kapton	Pyrell	Pyrell	Kapton	Kapton
	Sample ID	K05	P06	P06	K06	K06
	Test ID Number	28Ka	29P	29Pa	30K	30Ka
	Position in Carousel	3	5	5	2	2
	SAME Test #	75a	76	76a	77	77a
	Description	aged	High Temp aging	aged	High Temp aging	aged
	Date (mm/dd/2010)	8/26	8/26	8/26	8/26	8/26
	Inlet Velocity (cm/sec)	8.1	8.1	8.1	8.1	8.1
	Rt/R0	1.430	1.230	1.230	1.480	1.480
	Duration (sec)		60		60	
	Primary Aging time (sec)		11		11	
	Primary Mixing time (sec)		8		8	
	Net Aging time (sec)	563	51	625	31	559
Primary Diagnostics	Prim. Aging Therm Precip		y		y	
	Prim. Aging Therm Precip time		45		60	
	Use STS Detector		n		n	
	Use Comm Diag A (MPASS)		y		y	
	Use Comm Diag B		y		y	
	Diluting A (MPASS)	n	y	y	n	n
	Diluting B	n	y	y	n	n
	Cyclone Inlet on CDA (MPASS)		y		y	
	Impactor Inlet on CDB		10		10	
	Use P-Trak		y		y	
	Aerosol Concentration To P-Trak	0.0385	0.0385	0.0385	0.0385	0.0385
Secondary Diagnostics Volume	Secondary Diagnostics Volume %		58		58	
	Secondary Aging Duration (sec)		480		480	
	STS Detector		n		n	
	Comm Diag A (MPASS)		y		y	
	Comm Diag B		y		y	
	Diluting A (MPASS)		y		n	
	Diluting B		y		n	
	Ues PTrak		y		y	
	Aerosol Concentration To P-Trak		0.0385		0.0385	
	secondary agign therml precip duration	60		45		60

Smoke Generation	Carousel	4	4	4	4	4	4
	Sample Type	Lampwick	Lampwick	Teflon	Teflon	Lampwick	Lampwick
	Sample ID	L16	L16	T04	T04	L18	L18
	Test ID Number	32L	32La	33T	33Ta	34L	34La
	Position in Carousel	1	1	4	4	3	3
	SAME Test #	78	78a	79	79a	80	80a
	Description	Baseline aging	aged	Baseline aging	aged	High temp aging	aged
	Date (mm/dd/2010)	8/26	8/26	8/26	8/26	8/26	8/26
	Inlet Velocity (cm/sec)	8.1	8.1	8.1	8.1	8.1	8.1
	Rt/R0	1.245	1.245	1.445	1.445	1.280	1.280
	Duration (sec)	60		60		60	
	Primary Aging time (sec)	11		11		11	
	Primary Mixing time (sec)	8		8		8	
	Net Aging time (sec)	37	581	33	560	40	580
Primary Diagnostics	Prim. Aging Therm Precip	y		y		y	
	Prim. Aging Therm Precip time	90		75		90	
	Use STS Detector	n		n		n	
	Use Comm Diag A (MPASS)	y		y		y	
	Use Comm Diag B	y		y		y	
	Diluting A (MPASS)	n	n	n	n	n	n
	Diluting B	y	y	n	n	y	y
	Cyclone Inlet on CDA (MPASS)	y		y		y	
	Impactor Inlet on CDB	10		10		10	
	Use P-Trak	y		y		y	
	Aerosol Concentration To P-Trak	0.0385	0.0385	0.0520	0.0520	0.0385	0.0385
Secondary Diagnostics Volume	Secondary Diagnostics Volume %	58		58		58	
	Secondary Aging Duration (sec)	480		480		480	
	STS Detector	n		n		n	
	Comm Diag A (MPASS)	y		y		y	
	Comm Diag B	y		y		y	
	Diluting A (MPASS)	n		n		n	
	Diluting B	y		n		y	
	Ues PTrak	y		y		y	
	Aerosol Concentration To P-Trak	0.0385		0.0520		0.0385	
	secondary agign therml precip duration		90		75		90

Smoke Generation	Carousel	4	4	4	4	4	4
	Sample Type	Teflon	Teflon	Lampwick	Lampwick	Teflon	Teflon
	Sample ID	T06	T06	L17	L17	T05	T05
	Test ID Number	35T	35Ta	36L	36La	37T	37Ta
	Position in Carousel	6	6	2	2	5	5
	SAME Test #	81	81a	82	82a	83	83a
	Description	High temp aging	aged	Mid temp aging	aged	Mid temp aging	aged
	Date (mm/dd/2010)	8/26	8/27	8/26	8/27	8/26	8/27
	Inlet Velocity (cm/sec)	8.1	8.1	8.1	8.1	8.1	8.1
	Rt/R0	1.450	1.450	1.265	1.265	1.448	1.448
	Duration (sec)	60		60		60	
	Primary Aging time (sec)	11		11		11	
	Primary Mixing time (sec)	8		8		8	
	Net Aging time (sec)	30	559	39	579	30	557
Primary Diagnostics	Prim. Aging Therm Precip	y		y		y	
	Prim. Aging Therm Precip time	45		90		60	
	Use STS Detector	n		n		n	
	Use Comm Diag A (MPASS)	y		y		y	
	Use Comm Diag B	y		y		y	
	Diluting A (MPASS)	n	n	n	n	n	n
	Diluting B	n	n	y	y	n	n
	Cyclone Inlet on CDA (MPASS)	y		y		y	
	Impactor Inlet on CDB	10		10		10	
	Use P-Trak	y		y		y	
	Aerosol Concentration To P-Trak	0.0520	0.0520	0.0385	0.0385	0.0520	0.0520
Secondary Diagnostics Volume	Secondary Diagnostics Volume %	58		58		58	
	Secondary Aging Duration (sec)	480		480		480	
	STS Detector	n		n		n	
	Comm Diag A (MPASS)	y		y		y	
	Comm Diag B	y		y		y	
	Diluting A (MPASS)	n		n		n	
	Diluting B	n		y		n	
	Ues PTrak	y		y		y	
	Aerosol Concentration To P-Trak	0.0520		0.0385		0.0520	
	secondary agign therml precip duration		45		90		60

Smoke Generation	Carousel	7(4)	7(4)	7(4)	7(4)	7(4)	7(4)
	Sample Type	Pyrell	Pyrell	Kapton	Kapton	Silicone	Silicone
	Sample ID	P	P	K	K	S	S
	Test ID Number	37P	37Pa	38K	38Ka	39S	39Sa
	Position in Carousel						
	SAME Test #	84	84a	85	85a	86	86a
	Description	Mid Temp aging retry	aged	Mid Temp aging run. Repeat 75.	aged	Mid temp aging	aged
	Date (mm/dd/2010)	8/30	8/30	8/30	8/30	8/30	8/30
	Inlet Velocity (cm/sec)	8.1	8.1	8.1	8.1	8.1	8.1
	Rt/R0	1.225	1.225	1.430	1.430	1.345	1.345
	Duration (sec)	60		60		60	
	Primary Aging time (sec)	11		11		11	
	Primary Mixing time (sec)	8		8		8	
	Net Aging time (sec)	53	632	33	563	50	628
Primary Diagnostics	Prim. Aging Therm Precip	y		y			
	Prim. Aging Therm Precip time	60		60			
	Use STS Detector	n		n		n	
	Use Comm Diag A (MPASS)	y		y		y	
	Use Comm Diag B	y		y		y	
	Diluting A (MPASS)	y	y	n	n	y	y
	Diluting B	y	y	n	n	y	y
	Cyclone Inlet on CDA (MPASS)	y		y		y	
	Impactor Inlet on CDB	10		10		10	
	Use P-Trak	y		y		y	
	Aerosol Concentration To P-Trak	0.0385	0.0385	0.0385	0.0385	0.0318	0.0318
Secondary Diagnostics Volume	Secondary Diagnostics Volume %	58		58		58	
	Secondary Aging Duration (sec)	480		480		480	
	STS Detector	n		n		n	
	Comm Diag A (MPASS)	y		y		y	
	Comm Diag B	y		y		y	
	Diluting A (MPASS)	y		n		y	
	Diluting B	y		n		y	
	Ues PTrak	y		y		y	
	Aerosol Concentration To P-Trak	0.0385	0.0385	0.0385	0.0385	0.0318	
	secondary agign therml precip duration		60				

Smoke Generation	Carousel	7(4)	7(4)	7(4)	7(4)	7(4)
	Sample Type	Pyrell	Lampwick	Teflon	Kapton	Kapton
	Sample ID	P	L	T	K	K
	Test ID Number	40P	41L	42T	43K	43Ka
	Position in Carousel					
	SAME Test #	87	88	89	90	90a
	Description	Baseline STS det	Baseline STS det	Baseline STS det	Mid Temp aging reburn 85	aged
	Date (mm/dd/2010)	8/30	8/30	8/30	8/30	8/30
	Inlet Velocity (cm/sec)	8.1	8.1	8.1	8.1	8.1
	Rt/R0	1.220	1.245	1.445	1.430	1.430
	Duration (sec)	60	60	60	60	
	Primary Aging time (sec)	11	11	11	11	
	Primary Mixing time (sec)	8	8	8	8	
	Net Aging time (sec)	45	n/a	31	38	569
Primary Diagnostics	Prim. Aging Therm Precip	y		y		
	Prim. Aging Therm Precip time	60	90	75	60	
	Use STS Detector	y	y	y	n	
	Use Comm Diag A (MPASS)	n	n	n	y	
	Use Comm Diag B	y	y	y	y	
	Diluting A (MPASS)	y	n	n	n	n
	Diluting B	y	y	n	n	n
	Cyclone Inlet on CDA (MPASS)	y	y	y	y	
	Impactor Inlet on CDB	10	10	10	10	
	Use P-Trak	y	y	y	y	
	Aerosol Concentration To P-Trak	0.0385	0.0385	0.0520	0.0385	0.0385
Secondary Diagnostics Volume	Secondary Diagnostics Volume %	0	0	0	58	
	Secondary Aging Duration (sec)	0	0	0	480	
	STS Detector				n	
	Comm Diag A (MPASS)				y	
	Comm Diag B				y	
	Diluting A (MPASS)				n	
	Diluting B				n	
	Ues PTrak				y	
	Aerosol Concentration To P-Trak				0.0385	0.0385
	secondary agign therml precip duration					

Smoke Generation	Carousel	8(4)	8(4)	8(4)	8(4)	8(4)
	Sample Type	Silicone	Kapton	Kapton	Kapton	Kapton
	Sample ID	S	K	K	K	K
	Test ID Number	44S	45K	46K	46Ka	47K
	Position in Carousel					
	SAME Test #	91	92	93	93a	94
	Description	Baseline STS det	Baseline STS det	Mid Temp aging run 75 repeat	aged	Mid Temp aging run mid high
	Date (mm/dd/2010)	9/7	9/7	9/7	9/7	9/7
	Inlet Velocity (cm/sec)	8.1	8.1	8.1	8.1	8.1
	Rt/R0	1.327	1.380	1.430	1.430	1.450
	Duration (sec)	60	60	60		60
	Primary Aging time (sec)	11	11	11		11
	Primary Mixing time (sec)	8	8	8		8
	Net Aging time (sec)	46	35	36	568	42
Primary Diagnostics	Prim. Aging Therm Precip			y		y
	Prim. Aging Therm Precip time			60		60
	Use STS Detector	y	y	n		n
	Use Comm Diag A (MPASS)	n	n	y		y
	Use Comm Diag B	y	y	y		y
	Diluting A (MPASS)	n	n	n	n	n
	Diluting B	y	n	n	n	n
	Cyclone Inlet on CDA (MPASS)	y	y	y		y
	Impactor Inlet on CDB	10	10	10		10
	Use P-Trak	y	y	y		y
	Aerosol Concentration To P-Trak	0.0318	0.0385	0.0385	0.0385	0.0385
Secondary Diagnostics Volume	Secondary Diagnostics Volume %	0	0	58		58
	Secondary Aging Duration (sec)	0	0	480		480
	STS Detector			n		n
	Comm Diag A (MPASS)			y		y
	Comm Diag B			y		y
	Diluting A (MPASS)			n		n
	Diluting B			n		n
	Ues PTrak			y		y
	Aerosol Concentration To P-Trak			0.0385		0.0385
	secondary agign therml precip duration				60	

Smoke Generation	Carousel	8(4)	8(4)	8(4)	8(4)	8(4)
	Sample Type	Kapton	Pyrell	Pyrell	Lampwick	Lampwick
	Sample ID	K	P	P	L	L
	Test ID Number	47Ka	48P	48Pa	49L	49La
	Position in Carousel					
	SAME Test #	94a	95	95a	96	96a
	Description	aged	No flow run w/ aging	aged	No flow run w/aging	aged
	Date (mm/dd/2010)	9/7	9/7	9/7	9/7	9/7
	Inlet Velocity (cm/sec)	8.1	0	0	0	0
	Rt/R0	1.450	1.220	1.220	1.245	1.245
	Duration (sec)		60		60	
	Primary Aging time (sec)		11		11	
	Primary Mixing time (sec)		8		8	
	Net Aging time (sec)	567	38	n/a	34	582
Primary Diagnostics	Prim. Aging Therm Precip		y		y	
	Prim. Aging Therm Precip time		60		90	
	Use STS Detector		n		n	
	Use Comm Diag A (MPASS)		y		y	
	Use Comm Diag B		y		y	
	Diluting A (MPASS)	n	y	y	n	n
	Diluting B	n	y	y	y	y
	Cyclone Inlet on CDA (MPASS)		y		y	
	Impactor Inlet on CDB		10		10	
	Use P-Trak		y		y	
	Aerosol Concentration To P-Trak	0.0385	0.0385	0.0385	0.0385	0.0385
Secondary Diagnostics Volume	Secondary Diagnostics Volume %		58		58	
	Secondary Aging Duration (sec)		480		480	
	STS Detector		n		n	
	Comm Diag A (MPASS)		y		y	
	Comm Diag B		y		y	
	Diluting A (MPASS)		y		n	
	Diluting B		y		y	
	Ues PTrak		y		y	
	Aerosol Concentration To P-Trak		0.0385		0.0385	
	secondary agign therml precip duration	60				

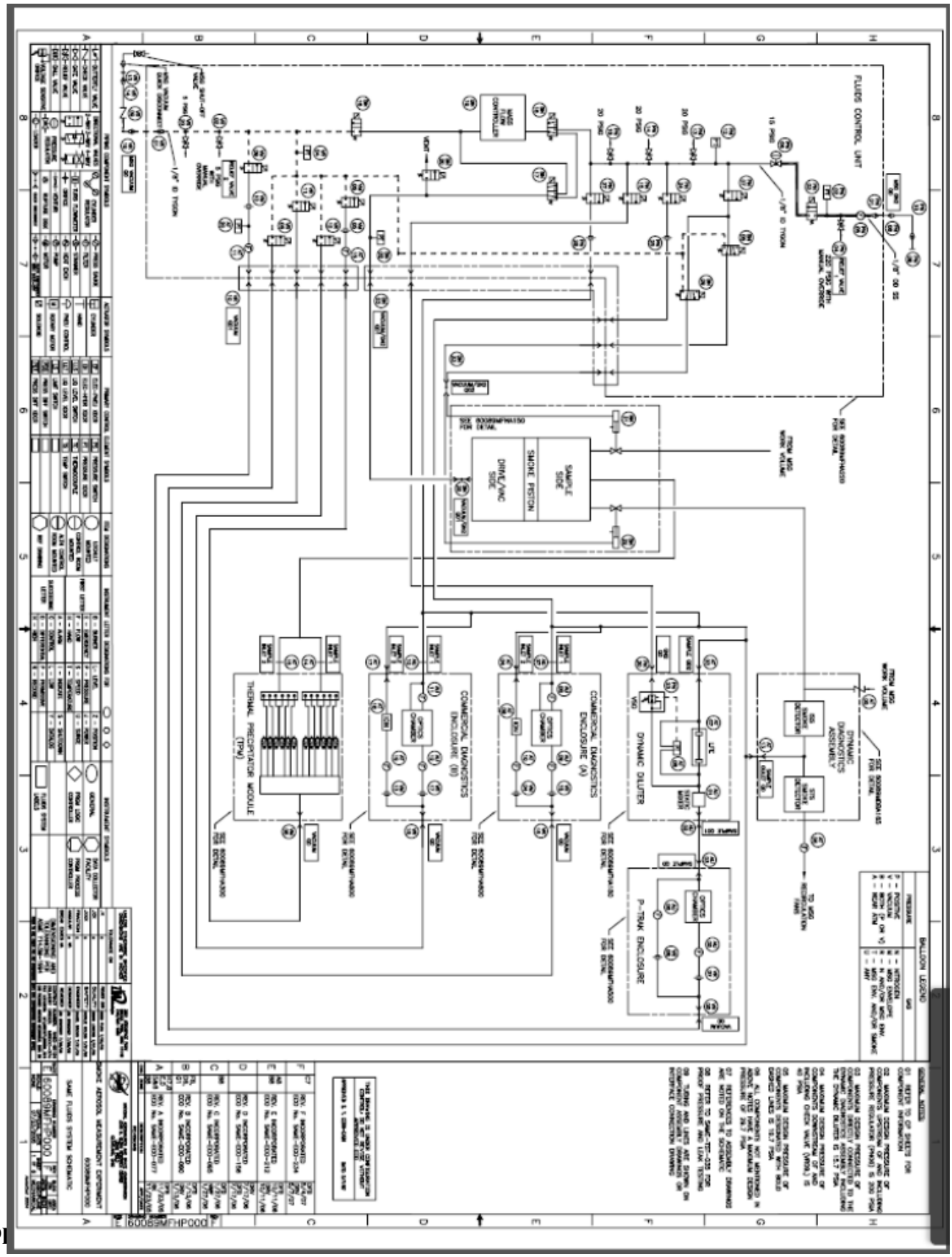
Smoke Generation	Carousel	9(4)	9(4)	9(4)	9(4)	9(4)	9(4)
	Sample Type	Pyrell	Pyrell	Lampwick	Lampwick	Silicone	Silicone
	Sample ID	P	P	L	L	S	S
	Test ID Number	50P	50Pa	51L	51La	52S	52Sa
	Position in Carousel						
	SAME Test #	97	97a	98	98a	99	99a
	Description	No flow run w/ aging	aged	No flow run w/ aging	aged	No flow run w/ aging	aged
	Date (mm/dd/2010)	9/8	9/8	9/8	9/8	9/8	9/8
	Inlet Velocity (cm/sec)	0	0	0	0	0	0
	Rt/R0	1.230	1.230	1.280	1.280	1.350	1.350
	Duration (sec)	60		60		60	
	Primary Aging time (sec)	11		11		11	
	Primary Mixing time (sec)	8		8		8	
	Net Aging time (sec)	48	624	37	585	48	627
Primary Diagnostics	Prim. Aging Therm Precip	Y		Y			
	Prim. Aging Therm Precip time	60		90			
	Use STS Detector	n		n		n	
	Use Comm Diag A (MPASS)	y		y		y	
	Use Comm Diag B	y		y		y	
	Diluting A (MPASS)	y	y	n	n	y	y
	Diluting B	y	y	y	y	y	y
	Cyclone Inlet on CDA (MPASS)	y		y		y	
	Impactor Inlet on CDB	10		10		10	
	Use P-Trak	y		y		y	
	Aerosol Concentration To P-Trak	0.0385	0.0385	0.0385	0.0385	0.0318	0.0318
Secondary Diagnostics Volume	Secondary Diagnostics Volume %	58		58		58	
	Secondary Aging Duration (sec)	480		480		480	
	STS Detector	n		n		n	
	Comm Diag A (MPASS)	y		y		y	
	Comm Diag B	y		y		y	
	Diluting A (MPASS)	y		n		y	
	Diluting B	y		y		y	
	Ues PTrak	y		y		y	
	Aerosol Concentration To P-Trak	0.0385		0.0385		0.0318	
	secondary agign therml precip duration				90		

Smoke Generation	Carousel	9(4)	9(4)	9(4)	9(4)	9(4)	9(4)
	Sample Type	Teflon	Teflon	Kapton	Kapton	Kapton	Kapton
	Sample ID	T	T	K	K	K	K
	Test ID Number	53T	53Ta	54K	54Ka	55K	55Ka
	Position in Carousel						
	SAME Test #	100	100a	101	101a	102	102a
	Description	No flow run w/ aging	aged	No flow run w/ aging	aged	No flow run w/ aging	aged
	Date (mm/dd/2010)	9/8	9/8	9/8	9/8	9/8	9/8
	Inlet Velocity (cm/sec)	0	0	0	0	0	0
	Rt/R0	1.450	1.450	1.480	1.480	1.480	1.480
	Duration (sec)	60		60		120	
	Primary Aging time (sec)	11		11		11	
	Primary Mixing time (sec)	8		8		8	
	Net Aging time (sec)	29	560	29	561	28	563
Primary Diagnostics	Prim. Aging Therm Precip	Y		Y			
	Prim. Aging Therm Precip time	75		90			
	Use STS Detector	n		n		n	
	Use Comm Diag A (MPASS)	y		y		y	
	Use Comm Diag B	y		y		y	
	Diluting A (MPASS)	n	n	n	n	n	n
	Diluting B	n	n	n	n	n	n
	Cyclone Inlet on CDA (MPASS)	y		y		y	
	Impactor Inlet on CDB	10		10		10	
	Use P-Trak	y		y		y	
	Aerosol Concentration To P-Trak	0.0520	0.0520	0.0385	0.0385	0.0385	0.0385
Secondary Diagnostics Volume	Secondary Diagnostics Volume %	58		58		58	
	Secondary Aging Duration (sec)	480		480		480	
	STS Detector	n		n		n	
	Comm Diag A (MPASS)	y		y		y	
	Comm Diag B	y		y		y	
	Diluting A (MPASS)	n		n		n	
	Diluting B	n		n		n	
	Ues PTrak	y		y		y	
	Aerosol Concentration To P-Trak	0.0520		0.0385		0.0385	
	secondary agign therml precip duration				90		

Smoke Generation	Carousel	10(4)	10(4)	10(4)	10(4)	10(4)	10(4)
	Sample Type	Silicone	Silicone	Silicone	Silicone	Teflon	Teflon
	Sample ID	S	S	S	S	T	T
	Test ID Number	56S	56Sa	57S	57Sa	58T	58Ta
	Position in Carousel						
	SAME Test #	103	103a	104	104a	105	105a
	Description	low flow run w/ aging	aged	No flow run w/ aging	aged	No flow run w/ aging	aged
	Date (mm/dd/2010)	9/9	9/9	9/9	9/9	9/9	9/9
	Inlet Velocity (cm/sec)	1	1	0	0	0	0
	Rt/R0	1.350	1.350	1.350	1.350	1.450	1.450
	Duration (sec)	60		60		60	
	Primary Aging time (sec)	11		11		11	
	Primary Mixing time (sec)	8		8		8	
	Net Aging time (sec)	48	619	51	636	24	559
Primary Diagnostics	Prim. Aging Therm Precip					Y	
	Prim. Aging Therm Precip time					75	
	Use STS Detector	n		n		n	
	Use Comm Diag A (MPASS)	y		y		y	
	Use Comm Diag B	y		y		y	
	Diluting A (MPASS)	y	y	y	y	n	n
	Diluting B	y	y	y	y	n	n
	Cyclone Inlet on CDA (MPASS)	y		y		y	
	Impactor Inlet on CDB	10		10		10	
	Use P-Trak	y		y		y	
	Aerosol Concentration To P-Trak	0.0318	0.0318	0.0788	0.0788	0.0788	0.0788
Secondary Diagnostics Volume	Secondary Diagnostics Volume %	58		58		58	
	Secondary Aging Duration (sec)	480		480		480	
	STS Detector	n		n		n	
	Comm Diag A (MPASS)	y		y		y	
	Comm Diag B	y		y		y	
	Diluting A (MPASS)	y		y		n	
	Diluting B	y		y		n	
	Ues PTrak	y		y		y	
	Aerosol Concentration To P-Trak	0.0318		0.0788		0.0788	
	secondary agign therml precip duration						75

Smoke Generation	Carousel	10(4)	10(4)	10(4)	10(4)	10(4)	10(4)
	Sample Type	Lampwick	Lampwick	Pyrell	Pyrell	Kapton	Kapton
	Sample ID	L	L	P	P	K	K
	Test ID Number	59L	59La	60P	60Pa	61K	61Ka
	Position in Carousel						
	SAME Test #	106	106a	107	107a	108	108a
	Description	No flow run w/ aging	aged	Low flow run w/ aging	aged	Low flow run w/ aging	aged
	Date (mm/dd/2010)	9/9	9/9	9/9	9/9	9/9	9/9
	Inlet Velocity (cm/sec)	0	0	1	1	1	1
	Rt/R0	1.280	1.280	1.230	1.230	1.480	1.480
	Duration (sec)	60		60		60	
	Primary Aging time (sec)	11		11		11	
	Primary Mixing time (sec)	8		8		8	
	Net Aging time (sec)	30	558	32	559	30	560
Primary Diagnostics	Prim. Aging Therm Precip	y		Y		Y	
	Prim. Aging Therm Precip time	90		60		90	
	Use STS Detector	n		n		n	
	Use Comm Diag A (MPASS)	y		y		y	
	Use Comm Diag B	y		y		y	
	Diluting A (MPASS)	n	n	y	y	n	n
	Diluting B	n	n	y	y	n	n
	Cyclone Inlet on CDA (MPASS)	y		y		y	
	Impactor Inlet on CDB	10		10		10	
	Use P-Trak	y		y		y	
	Aerosol Concentration To P-Trak	0.0788	0.0788	0.0385	0.0385	0.0385	0.0385
Secondary Diagnostics Volume	Secondary Diagnostics Volume %	58		58		58	
	Secondary Aging Duration (sec)	480		480		480	
	STS Detector	n		n		n	
	Comm Diag A (MPASS)	y		y		y	
	Comm Diag B	y		y		y	
	Diluting A (MPASS)	n		y		n	
	Diluting B	n		y		n	
	Ues PTrak	y		y		y	
	Aerosol Concentration To P-Trak	0.0788		0.0385		0.0385	
	secondary agign therml precip duration				60		90

Smoke Generation	Carousel	11(4)	11(4)	11(4)	11(4)	11(4)	11(4)
	Sample Type	Silicone	Lampwick	Teflon	Kapton	Pyrell	Kapton
	Sample ID	S	L	T	K	P	K
	Test ID Number	62S	63L	64T	65K	66P	67K
	Position in Carousel						
	SAME Test #	109	110	111	112	113	114
	Description	low flow run w STS	low flow run w STS	low flow run w STS	low flow run w STS	low flow run w STS	low flow run w STS no Ptrak
	Date (mm/dd/2010)	9/9	9/9	9/9	9/9	9/9	9/9
	Inlet Velocity (cm/sec)	2	2	2	2	2	2
	Rt/R0	1.350	1.280	1.450	1.480	1.230	1.430
	Duration (sec)	60	60	60	60	60	60
	Primary Aging time (sec)	11	11	11	11	11	11
	Primary Mixing time (sec)	8	8	8	8	8	8
	Net Aging time (sec)	28	21	43	22	31	28
Primary Diagnostics	Prim. Aging Therm Precip						
	Prim. Aging Therm Precip time						
	Use STS Detector	y	y	y	y	y	y
	Use Comm Diag A (MPASS)	n	n	n	n	n	n
	Use Comm Diag B	y	y	y	y	y	y
	Diluting A (MPASS)	n	n	n	n	n	n
	Diluting B	y	y	n	n	y	n
	Cyclone Inlet on CDA (MPASS)	y	y	y	y	y	y
	Impactor Inlet on CDB	10	10	10	10	10	10
	Use P-Trak	y	y	y	y	y	y
	Aerosol Concentration To P-Trak	0.0318	0.0385	0.0520	0.0385	0.0385	0.0385
Secondary Diagnostics Volume	Secondary Diagnostics Volume %						
	Secondary Aging Duration (sec)						
	STS Detector						
	Comm Diag A (MPASS)						
	Comm Diag B						
	Diluting A (MPASS)						
	Diluting B						
	Ues PTrak						
	Aerosol Concentration To P-Trak						
	secondary agign therml precip duration						


FLUIDS CONTROL UNIT
DYNAMIC DILUTER
COMMERCIAL DILUTION ENCLOSURE (A)
COMMERCIAL DILUTION ENCLOSURE (B)
THERMAL PRECIPITATOR MODULE (TPM)
P-TRAK ENCLOSURE
DYNAMIC DIAGNOSTICS ASSEMBLY
SAMPLE SIDE
DRAW/VAC SIDE
SMOKE PISTON
OPTICS CHAMBER
SAME FLUIDS SYSTEM SCHEMATIC

App

Appendix C: Sample mass loss of each test run

Sample Number	Preflight Average	Postflight Average	Mass Loss (mg)	Test Number	GMT
S01	0.69639	0.69391	2.48	SAMERFLT 39S	242
S02	0.68742	Carousel 6 Not Tested		SAMERFLT20S	
S03	0.72567	0.72157	4.10	SAMERFLT 1S SAMERFLT20S	197 236
S04	0.74688	0.74241	4.47	SAMERFLT 6S SAMERFLT 25S	203 238
S05	0.71125	Carousel 6 Not Tested		SAMERFLT 25S	
S06	0.6882	0.68603	2.17	SAMERFLT 44S	250
S07	0.74098	0.73976	1.22	SAMERFLT 12S	211
S08	0.7221	0.71877	3.33	SAMERFLT 52S	251
S09	0.72037	0.71691	3.46	SAMERFLT 13S	211
S10	0.72187	0.71790	3.97	SAMERFLT 57S	252
S11	0.72313	0.72027	2.86	SAMERFLT 56S	252
S12	0.72895	0.72744	1.51	SAMERFLT 18S	235
S13	0.72085	0.71612	4.73	SAMERFLT 19S	235
S14	0.7353	0.73242	2.88	SAMERFLT 62S	253
T01	1.01266	0.98870	23.96	SAMERFLT 3T SAMERFLT 22T	201 236
T02	0.99826	0.99326	5.00	SAMERFLT 9T	210
T03	1.00397	1.00306	0.91	SAMERFLT 11T	211
T04	1.10712	1.01097	96.15	SAMERFLT 32T	239
T05	1.02093	1.01122	9.71	SAMERFLT 34T	239
T06	1.00887	1.00080	8.07	SAMERFLT 36T	239
T07	1.01455	Carousel 6 Not Tested		SAMERFLT 22T	
T08	0.99927	0.99794	1.33	SAMERFLT 42T	242
T09	1.01454	1.01009	4.45	SAMERFLT 53T	251
T10	0.98925	0.98850	0.75	SAMERFLT 58T	252
T11	1.00898	0.99903	9.95	SAMERFLT 64T	253
K01	0.54711	0.54605	1.06	SAMERFLT 4K SAMERFLT 23K	202 237
K02	0.55177	0.55089	0.88	SAMERFLT 14K	214
K03	0.55777	0.55402	3.75	SAMERFLT 15K	214
K04	0.55083	0.54975	1.08	SAMERFLT 26K	238
K05	0.54795	0.54754	0.41	SAMERFLT 28K	238
K06	0.54909	0.54850	0.59	SAMERFLT 30K	238
K07	0.54642	Carousel 6 Not Tested		SAMERFLT 23K	
K08	0.54119	0.54084	0.35	SAMERFLT 38K SAMERFLT 43K	242
K09	0.55698	0.55692	0.06	SAMERFLT 45K	250

Sample Number	Preflight Average	Postflight Average	Mass Loss (mg)	Test Number	GMT
K10	0.54	0.53964	0.36	SAMERFLT 46K	250
K11	0.55059	0.55002	0.57	SAMERFLT 47K	250
K12	0.53874	0.53744	1.30	SAMERFLT 54K	251
K13	0.53809	0.53665	1.44	SAMERFLT 55K	251
K14	0.5393	0.53791	1.39	SAMERFLT 61K	252
K15	0.53997	0.53755	2.42	SAMERFLT 65K	253
K16	0.55493	0.55443	0.50	SAMERFLT 67K	253
P01	0.46979	0.46929	0.50	SAMERFLT 37P	242
P02	0.4849	0.48422	0.68	SAMERFLT 5P SAMERFLT 24P	203 237
P03	0.48571	0.48515	0.56	SAMERFLT 16P	235
P04	0.4843	0.48363	0.67	SAMERFLT 17P	235
P05	0.47617	0.47560	0.57	SAMERFLT 27P	238
P06	0.46931	Not run - bad sample		SAMERFLT 31P	
P07	0.47816	0.47750	0.66	SAMERFLT 29P	238
P08	0.48425	Carousel 6 Not Tested		SAMERFLT 24P	
P09	0.47746	0.47697	0.49	SAMERFLT 40P	242
P10	0.48293	0.48251	0.42	SAMERFLT 48P	250
P11	0.47584	0.47541	0.43	SAMERFLT 50P	251
P12	0.4659	0.46538	0.52	SAMERFLT 60P	252
P13	0.48357	0.48270	0.87	SAMERFLT 66P	253
L02	0.58364	0.58264	1.00	SAMERFLT 41L	242
L03	0.61397	0.61259	1.38	SAMERFLT 49L	250
L04	0.60396	0.60019	3.77	SAMERFLT 51L	251
L05	0.55782	0.58989	2.93	SAMERFLT 59L	252
L06	0.59582	0.59011	5.71	SAMERFLT 63L	253
L13	0.60437	0.60135	3.02	SAMERFLT 2L SAMERFLT 21L	236
L14	0.60061	0.59915	1.46	SAMERFLT 7L SAMERFLT 10L	210
L15	0.60374	0.60251	1.23	SAMERFLT 8L	210
L16	0.60329	0.60204	1.25	SAMERFLT 31L	239
L17	0.59287	0.58944	3.43	SAMERFLT 33L	239
L18	0.60209	0.60051	1.58	SAMERFLT 35L	239
L25	0.60465	Carousel 6 Not Tested		SAMERFLT 21L	

Red text indicates a repeated test due to the astronaut's error

Red highlighting signifies tests that were not conducted

Appendix D: TEM grids map

		SAME-R Thermal Precipitator Usage	could measure	ubiquitous - tiny or some particles, possible measure
				Not measureable

Thermal Precipitator 1

	Date viewed - comments	A	Date viewed - comments	B
1	8/24 - no particles - No images	48 (SAMERFLT 1S)	2/22 - few particles, mostly debris	67 (SAMERFLT 20S)
2	8/24 - no particles Ubiquitous particles	49 (SAMERFLT 2L)	9/12 - no particles	68 (SAMERFLT 21L)
3	9/1 no particles	50 (SAMERFLT 3T)	2/22 - few particles, some background particles	69 (SAMERFLT 22T)
4	8/31 - no particles - tinies/ chains	51 (SAMERFLT 4K)	9/13 - no particles	70 (SAMERFLT 23K)
5	9/12 no particles	52 (SAMERFLT 5P)	2/22 - few particles, mostly debris	71 (SAMERFLT 24P)
6	2/22 - few particles, mostly debris	53 (SAMERFLT 6S)	2/22 - few particles, mostly debris	72 (SAMERFLT 25S)

Thermal Precipitator 2

	Date viewed - comments	A	Date viewed - comments	B
1	9/16 & 4/23 - particles present	54 (SAMERFLT 7L)	9/16 - particles present	62 preage (SAMERFLT 15K)
2	9/28 - particles present	55 (SAMERFLT 8L)	2/17 - particles present	62 postage (SAMERFLT 15K)
3	9/21 - particles present	56 (SAMERFLT 9T)	2/17 - particles present	63 preage (SAMERFLT 16P)
4	9/21 - no measureable particles	#58 Preage (SAMERFLT 11T)	2/17 - particles present	63 postage (SAMERFLT 16P)
5	2/17 - particles present	#58 Postage (SAMERFLT 11T)	9/21 - particles present	64 preage (SAMERFLT 17P)
6	9/28 & 4/23 particles present	60 (SAMERFLT 13S)	2/22 - Particles present	64 postage (SAMERFLT 17P)

Thermal Precipitator 5

	Date viewed/comments	A	Date viewed - comments	B
1	10/28 - no particles	73 preage (SAMERFLT 26K)	11/9 - no particles extremely clean	73 postage (SAMERFLT 26K)
2	11/3 - no particles	74 preage (SAMERFLT 27P)	11/9 - particles found upstream at 3y.	74 postage (SAMERFLT 27P)
3	10/28 - no particles	75 preage (SAMERFLT 28K)	11/9 - possible population of very tiny particles	75 postage (SAMERFLT 28K)
4	11/3 - many agglomerated particles	76 preage (SAMERFLT 29P)	11/3 - particles found not as far upstream	76 postage (SAMERFLT 29P)
5	11/3 - very tiny particles	77 preage (SAMERFLT 30K)	11/3 - more very tiny particles	77 postage (SAMERFLT 30K)
6		empty		empty

Thermal Precipitator 4

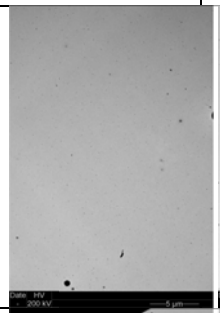
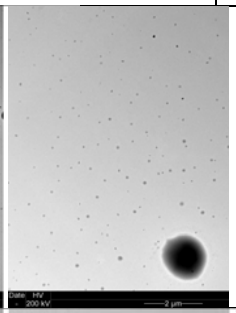
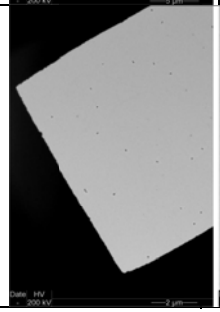
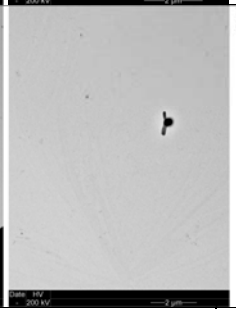
	Date viewed - comments	A	Date viewed - comments	B
1	12/9 & 4/23 - no particles	78 preage (SAMERFLT 31L)	12/9 - no particles	78 postage (SAMERFLT 31L)
2	10/5 - no particles	79 preage (SAMERFLT 32T)	12/9 - no particles	79 postage (SAMERFLT 32T)
3	12/9 - No particles	80 preage (SAMERFLT 33L)	12/9 - no particles	80 postage (SAMERFLT 33L)
4	10/5 - no particles	81 preage (SAMERFLT 34T)	10/6 - no particles	81 postage (SAMERFLT 34T)
5	12/9 - No particles	82 preage (SAMERFLT 35L)	12/9 & 4/23 - no particles	82 postage (SAMERFLT 35L)
6	10/5 - no measurable particles	83 preage (SAMERFLT 36T)	10/6 - no particles	83 postage (SAMERFLT 36T)

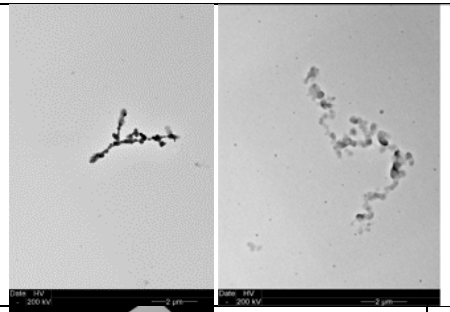
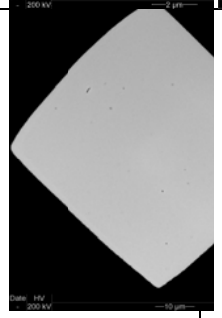
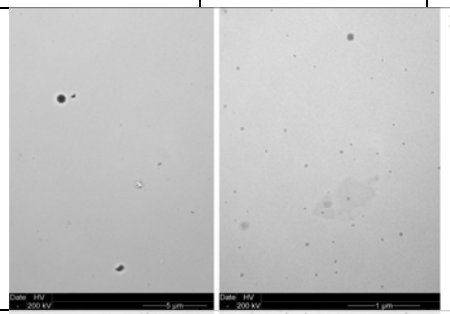
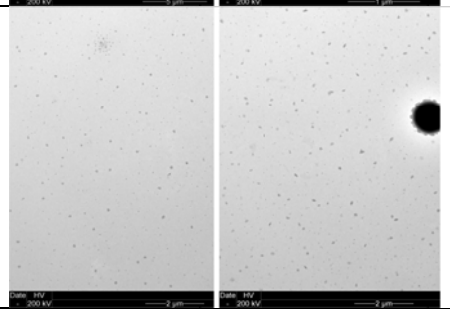
Thermal Precipitator 3

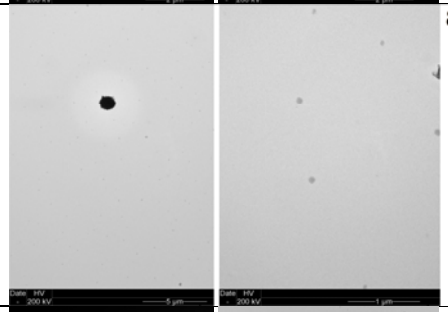
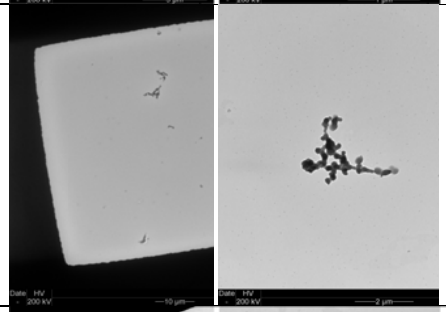
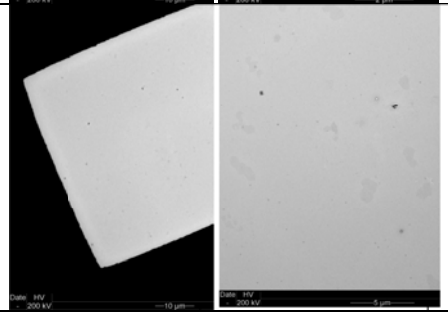
	Date viewed - comments	A	Date viewed - comments	B
1	10/5 - particles noted	84 preage (SAMERFLT 37P)	10/5 - particles noted	84 postage (SAMERFLT 37P)
2	2/6/12 - no measureable particles	85 (SAMERFLT 38K)	2/6/12 - particles present	87 (SAMERFLT 40P)
3	3/19 - particles mixed in with extraneous bkg.	106 (SAMERFLT 59L)	3/19 - particles present	89 (SAMERFLT 42T)
4	2/6/12 - Particles but not enough to measure - no images	105 preage (SAMERFLT 58T)	2/6/12 - very small particles - not measureable.	105 postage (SAMERFLT 58T)
5	3/29/12 - particles present	107 preage (SAMERFLT 60P)	2/22 - particles present	107 postage (SAMERFLT 60P)
6	3/29/12 - very small particles present	108 preage (SAMERFLT 61K)	3/29/12 - very small particles	108 postage (SAMERFLT 61K)

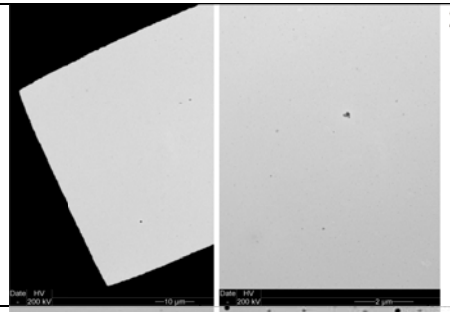
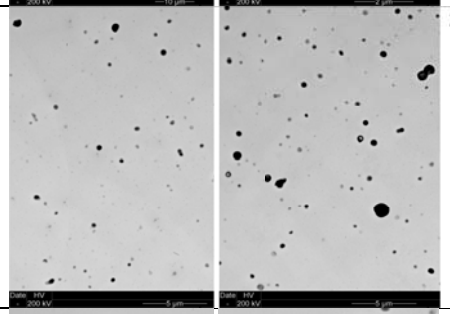
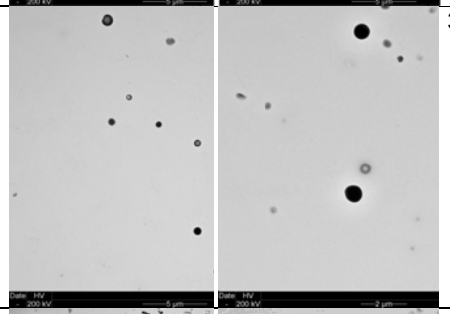
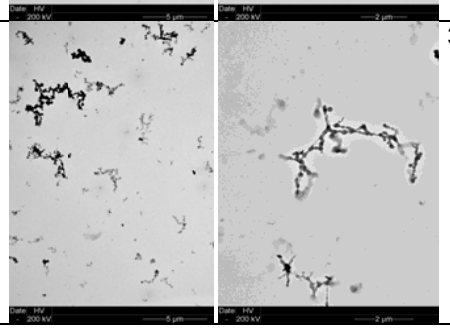
Thermal Precipitator 6

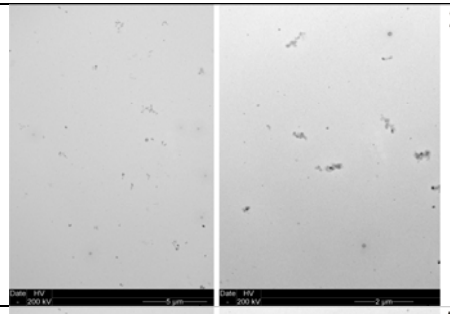
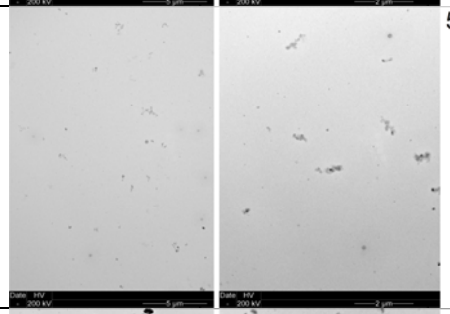
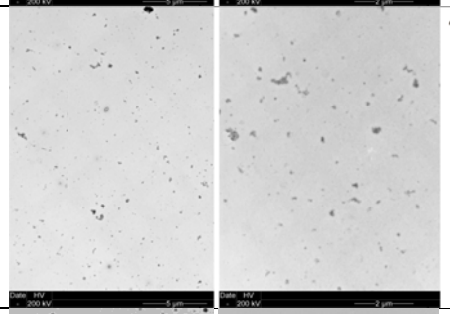
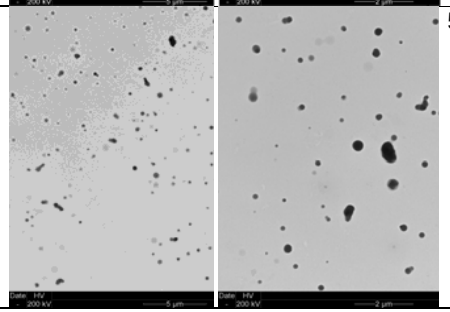
	Date viewed - comments	A	Date viewed - comments	B
1	3/19 - no particles	93 preage (SAMERFLT 46K)	2/13 - no particles	93 postage (SAMERFLT 46K)
2	3/8 - very tiny	94 preage (SAMERFLT 47K)	2/13 - Yes, few	94 postage (SAMERFLT 47K)
3	2/13 - yes, few	95 (SAMERFLT 48P)	2/13 - yes, various	96 (SAMERFLT 49L)
4	3/8 - yes, few	97 (SAMERFLT 50P)	2/13 - yes, few	100 (SAMERFLT 53T)
5	3/8 - yes, tiny	98 preage (SAMERFLT 51L)	2/13 - yes, few large & tiny spots	98 postage (SAMERFLT 51L)
6	3/8 - yes, small & large	101 preage (SAMERFLT 54K)	2/13 - yes, few	101 postage (SAMERFLT 54K)

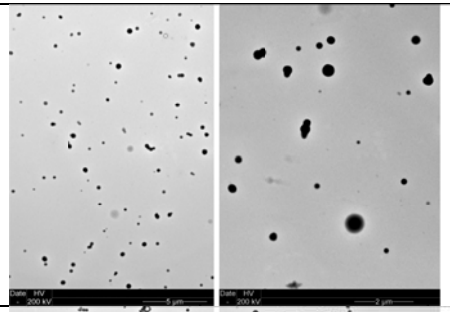
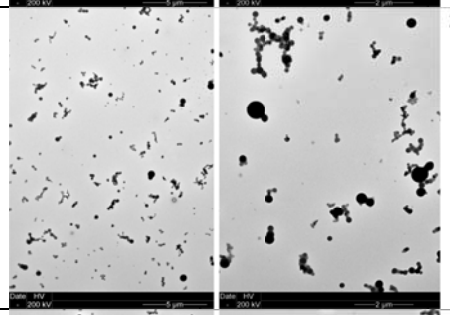
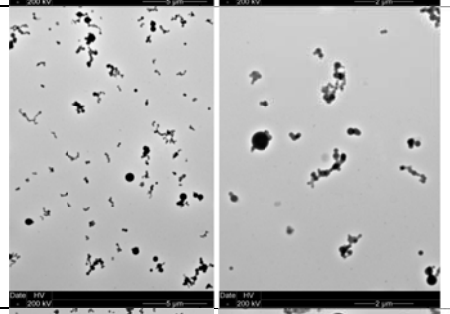
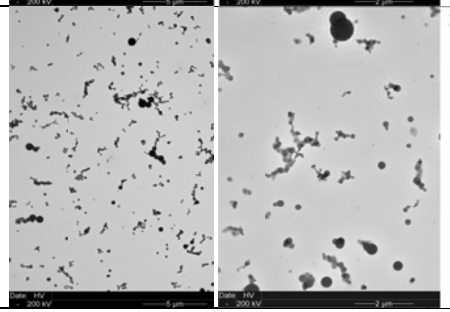
Aged?	Date TEM	Description	R_t/R_0	Morphology	Image 1	Image 2	M0 Best Diameter	M_1/M_0 (μm) AMD	Diameter of Average Mass (μm)	Hatch Choate sigma-g
Aged	8/24	Low Temp	1.304	No particles noted. No images			2,523,939	0.244	0.450	2.188
Unaged	8/24	Baseline High R_T/R_O	1.251	No aerosol particles noted – debris and random particles present .Particles <50nm in diameter visible at higher magnifications.			2,819,891	0.149	0.286	2.239
Unaged	9/1	Baseline High R_T/R_O	1.452	No aerosol particles noted – debris and random particles present; denser particles exhibit a primary particle size of <200nm.			3,654,789	65.14	0.148	2.079

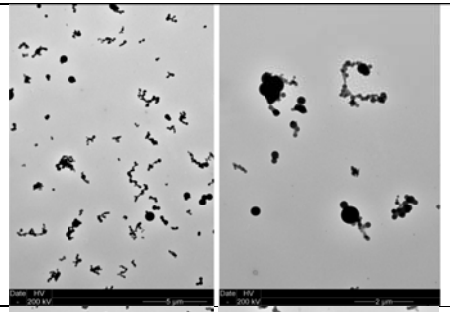
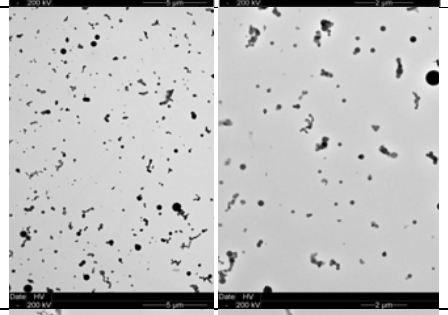
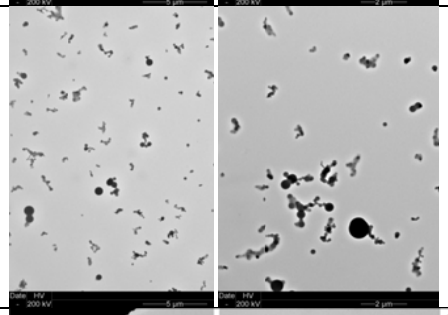
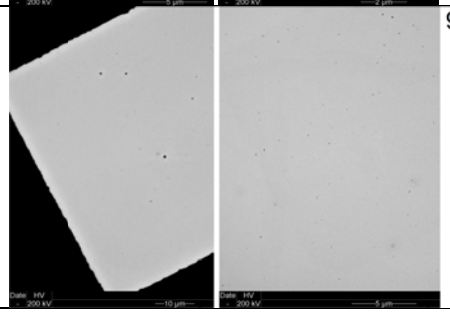
1	#51	Kapton	?	8/31	Baseline new formula	1.380	No particles noted – small population of chained particles similar to Teflon; primary particles in chains measure ~40-150nm.		1,825,597	0.056	0.097	2.091
1	#52	Pyrell	n/a	9/12	Baseline	1.220	No significant number of particles noted. Images exhibit a few particles downstream. Not enough to		3,110,971	0.208	0.405	2.262
1	#53	Silicone	Unaged	2/22	High Temp	1.350	No significant particles noted – some debris particles present plus the presence of background particles <50nm in diameter.		3,889,059	0.250	0.547	2.422
1	#67	Silicone	Aged	2/22	High temp aged STS det	1.350	Few particles noted – downstream – many noted upstream; ubiquitous particles <50nm noted.		1,352,021	0.385	0.668	2.100

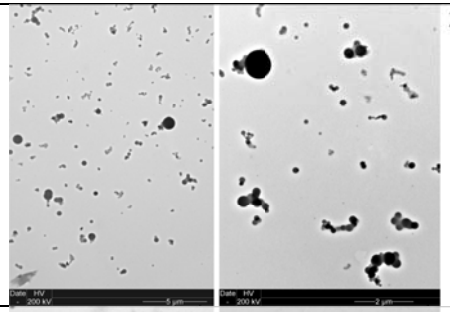
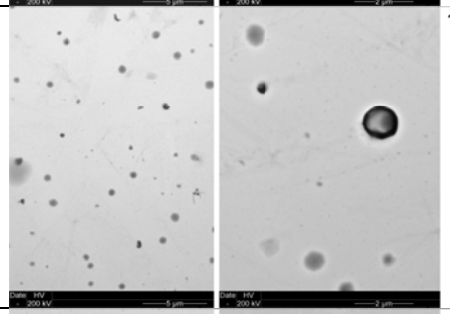
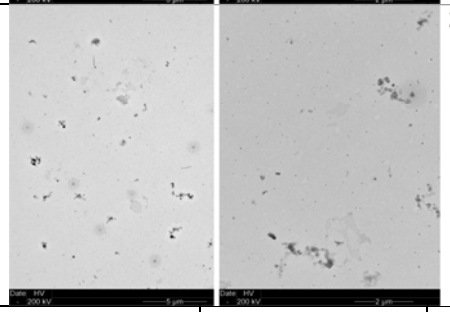
1	#68	Lamp wick	Aged	9/12	High temp aged STS det	1.280	No particles noted – exhibits small population of debris and chained particles similar to Teflon.		1,251,719	0.230	0.377	2.020
1	#69	Teflon	Aged	2/22	High temp aged STS det	1.450	Very few particles noted – some debris particles present.		840,336	0.446	0.0	
1	#70	Kapton	Aged	9/13	High temp aged STS det	1.480	No significant number of particles noted – small population of debris and chained clusters.		1,112,405	0.161	0.208	1.660
1	#71	Pyrell	Aged	2/22	High temp aged STS det	1.230	No significant particles – some debris.		1,352,317	0.321	0.434	1.731

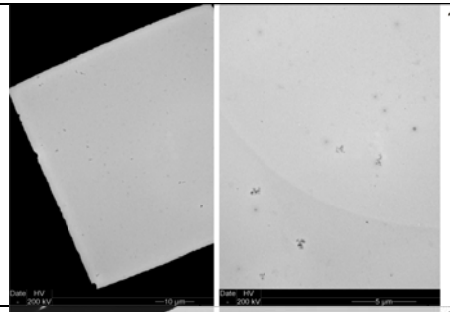
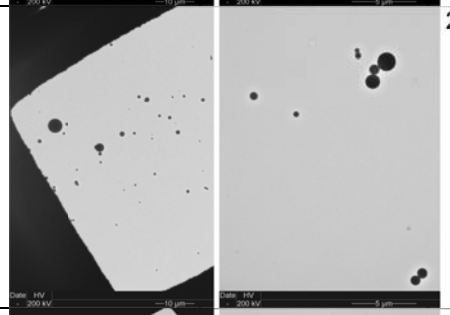
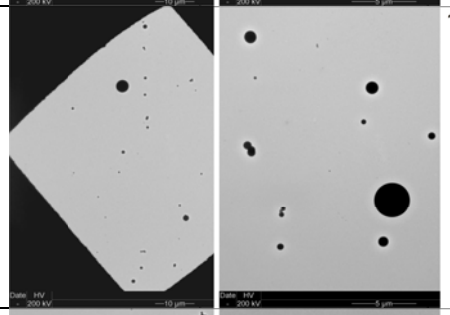
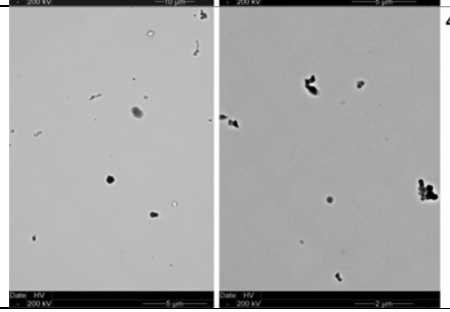
1	#72	Silicone	Unaged	2/22	Baseline STS det	1.327	No significant particles – some debris		2,987,870	0.182	0.419	2.491
2	#54	Lamp wick	Unaged	9/16 & 4/23	High R_T/R_O	1.260	Particle deposition starts upstream at grid opening y-3. Particles are large, spherical and usually occur singly; particles range in size from 200-1400nm.		3,197,762	0.110	0.253	2.493
2	#55	Lamp wick	?	9/28	High R_T/R_O test aging	1.253	Possibly a bit small smaller than #54; ranging ~240-1200nm (adjacent areas taken for image analysis).		3,373,889	0.116	0.275	2.528
2	#56	Teflon	Unaged	9/21	Baseline	1.445	Particle deposition – upstream at y-3; dense primary particles <50-200nm in diameter . Density differences could be due to age or the fact that Teflon is electron beam sensitive.		3,037,044	0.132	0.185	1.786

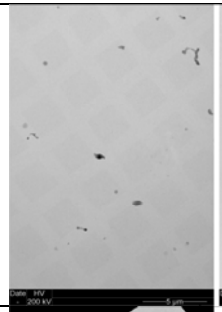
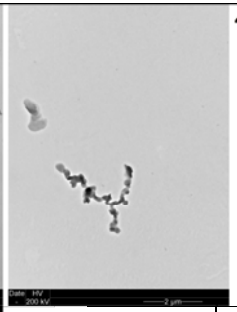
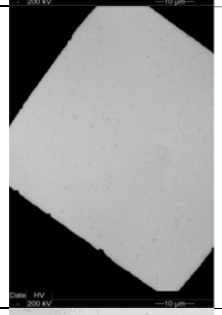
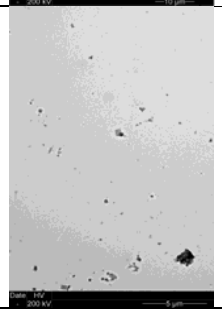
2	#58	Teflon	Aged	9/21	Baseline aging	1.445	Extremely few particles observed; faint chained clusters noted.		2,105,295	0.053	0.149	2.764
2	#58	Teflon	Aged	2/17	Baseline aging		Few particles observed – Teflon aggregates are very faint, not dense.		557,920	0.105	0.232	2.442
2	#60	Silicone		9/28 & 4/23	H Temp aging	1.350	Particles noted on second observation. Small and faint.		4,278,213	0.241	0.533	2.440
2	#62	Kapton	Unaged	9/16		1.480	Particle deposition – upstream at y-3; particles generally range in size ~90-430nm. Much smaller size than Lampwick.		5,850,382	0.148	0.251	20.66

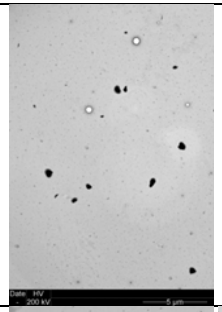
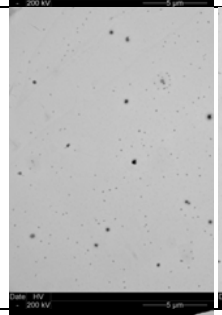
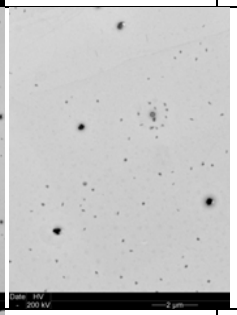
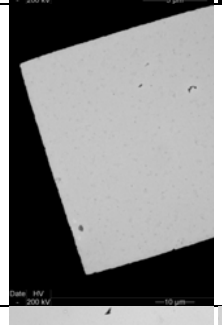
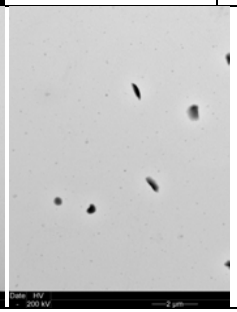
2	#62	Kapton	Aged	2/17	High Temp aging		Particle deposition upstream at y-4; spherical particles, most less than 1.0μm.		1,797,876	0.243	0.383	1.962
2	#63	Pyrell	Unaged	2/17	Baseline aging	1.220	Particle deposition upstream at y-4; heavy coverage with spherical particles and agglomerates. Primary particles generally measure >130nm – 1450nm.		3,862,679	0.207	0.392	2.222
2	#63	Pyrell	Aged	2/17	Baseline aging		Particle deposition upstream at y-4; again heavy coverage; particles appear a bit smaller than preage, <100nm-600nm.		1,414,788	0.359	0.539	1.892
2	#64	Pyrell	Unaged	9/21		1.230	Particle deposition – upstream at y-5; some chained Teflon-type particles; primary particles in chains range in size from ~90-270nm. Larger single particles often measure >700nm.		3,881,194	0.254	0.445	2.115

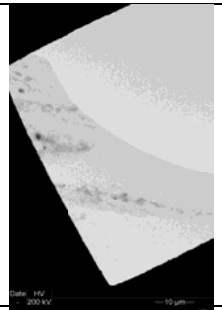
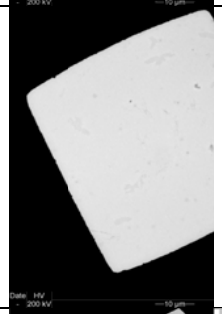
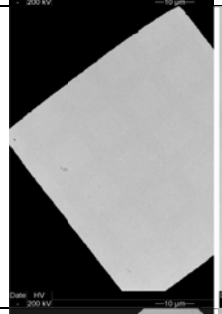
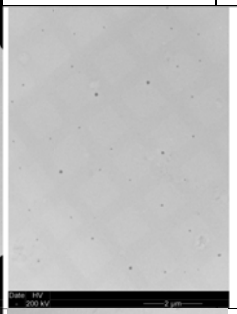
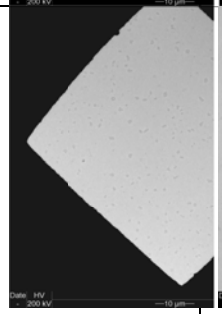
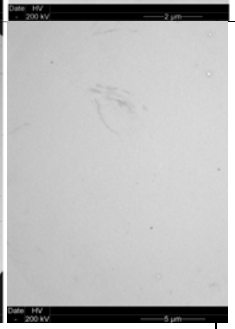
2	#64	Pyrell	Aged	2/22	High Temp aging		Particle deposition at y-4; particles are more highly aggregated with primaries measuring ~85-850nm.		1,417,571	0.450	0.616	1.750
3	#84	Pyrell	Unaged	10/5	Mid temp aging retry	1.225	Particle deposition – upstream at y-6; particles range from ~60-630nm. Particles agglomerated like Run #64 – adjacent areas taken for image analysis.		3,721,977	0.233	0.412	2.131
3	#84	Pyrell	Aged	10/5	Mid temp aging retry		Particle deposition – upstream at y-4-5; particles range in size ~60-1250nm. Particles highly agglomerated - adjacent areas taken for image analysis.		1,683,215	0.355	0.533	1.893
3	#85	Kapton		2/6	Mid temp aging run 75 repeat, low ptrack on peak one	1.430	Exhibits very tiny particles – not enough for image analysis particle size, exhibits a few denser particles ~100-200nm; also exhibits extraneous material.		954,553	0.112	0.174	1.936

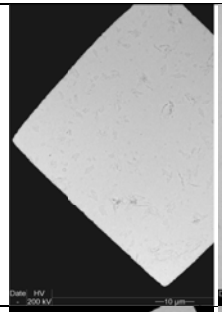
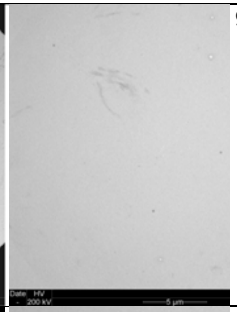
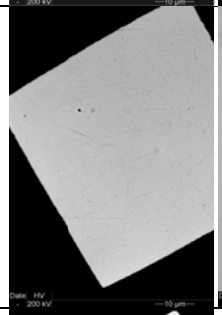
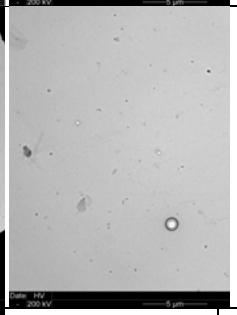
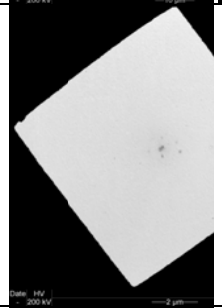
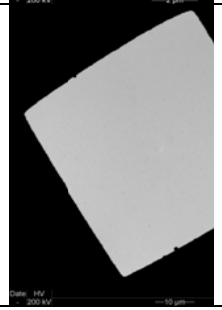
3	#87	Pyrell	Unaged	3/19	Baseline STS det	1.220	Heavy particle deposition – at y-5; small particles, range in size from <50nm – 200nm but could be up to~1000nm.			2,924,886	0.220	0.421	2.241
3	#106	Lamp wick		3/19	No flow run w/ aging	1.280	Deposition at y-4; large amount of background deposits, very faint particles ~200-500nm. Occasional extremely large particles usually are ~0.67-3.67μm.			105,517	0.442	0.600	1.736
3	#89	Teflon	Unaged	3/29	Baseline STS det	1.445	Small agglomerates at ~y-5 with tiny (<70nm) background particles. Small needle-like crystals noted.			2,774,308	0.055	0.151	2.733
3	#105	Teflon	Unaged	2/6	No flow run	1.450	Particle deposition upstream; not enough particles to measure. No images.			195,397	0.172	0.306	2.132

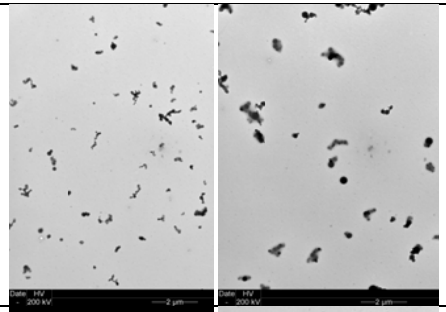
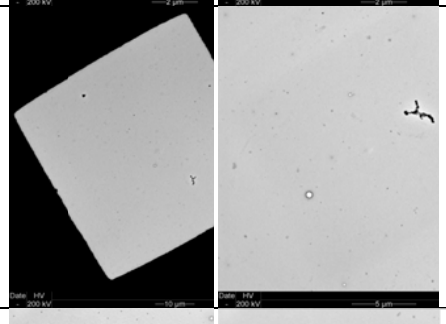
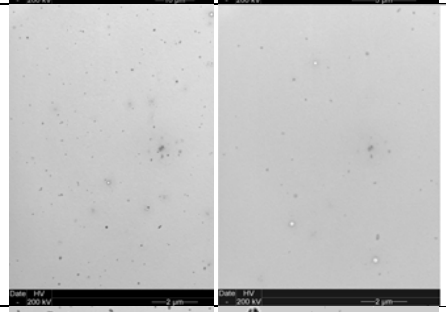
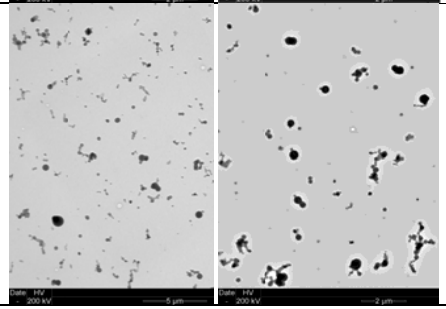
3	#105	Teflon	Aged	2/6	No flow run w/ aging		Not enough particles to measure; tiny agglomerates of chained particles with primary particles measuring ~<100nm.		146,578	0.162	0.324	2.298
3	#107	Pyrell		2/6	Low flow run w/ aging	1.230	Deposition at y-3, -4; larger particles measure ~200-		206,019	0.259	0.922	3.084
3	#107	Pyrell	Aged	2/22	No flow run w/ aging		Very few large spherical particles (~0.2-3.3 μm) present spread out over grid.		192,866	0.282	0.933	2.985
3	#108	Kapton	Unaged	3/29	Low flow run	1.480	Upstream ~y-7; couldn't observe at lower magnifications; particles very small. Small chains more visible at 10KX with primary particles ~90nm. Many particles ~200-600nm in diameter.		495,068	0.135	0.343	2.632

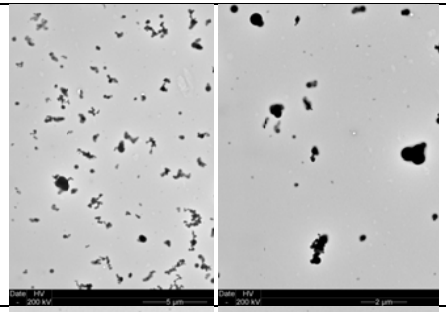
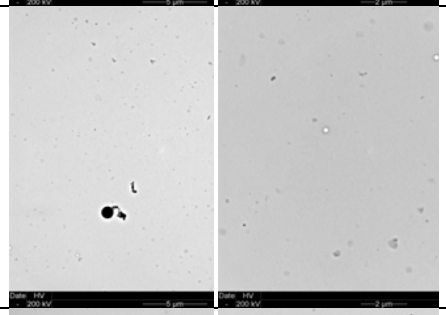
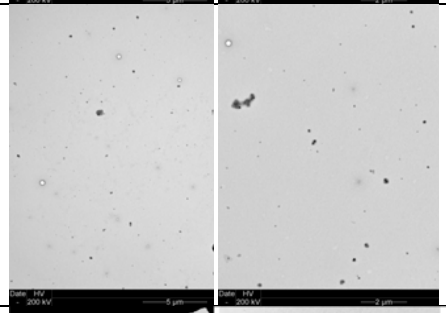
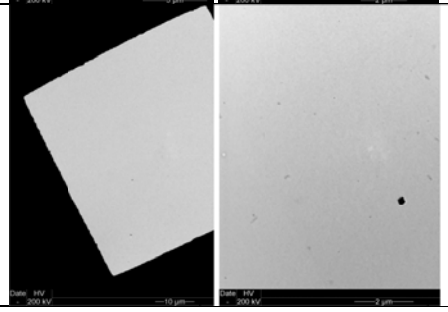
3	#108	Kapton	Aged	3/29	No flow run w/ aging		Deposited upstream at y-5; like preage sample.	 	451,723	0.145	0.351	2.561
4	#78	Lamp wick	Unaged	12/9	Baseline aging	1.245	No particles present. Carbon film is clean exhibiting little to no deposited material.		2,987,215	0.130	0.265	2.324
4	#78	Lamp wick	Aged	12/9	Baseline aging		Some 'junk' present – no particles.		1,158,632	0.224	0.355	1.975
4	#79	Teflon	Unaged	10/5	Baseline aging	1.445	Torn films plus a lot of debris.		2,596,154	0.163	0.193	1.511

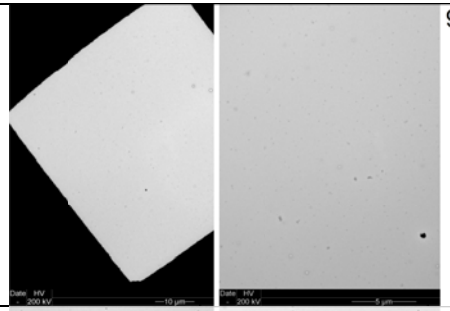
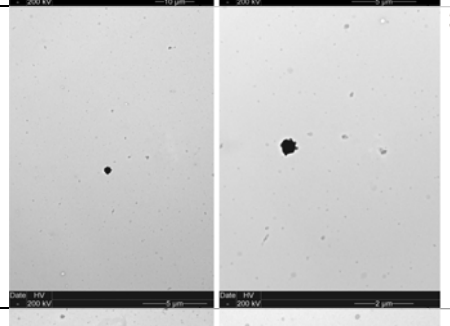
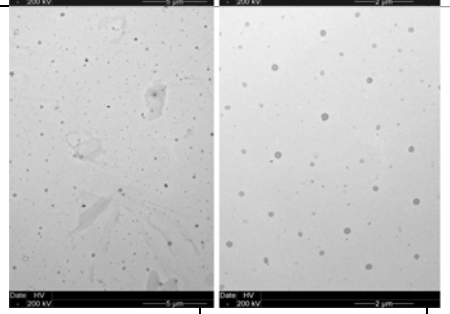
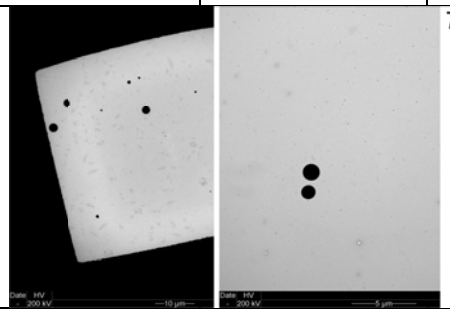
4	#79	Teflon	Aged	12/9	Baseline aging		Some debris on film – no particles. Non-aerosol deposit.			974,485	0.267	0.263	#
4	#80	Lamp wick	Unaged	12/9	High temp	1.280	No particles. Feathery or wispy deposit found in some areas along with other deposited material. Higher magnification of this material indicates a non-aerosol particle.			4,995,553	0.124	0.289	2.508
4	#80	Lamp wick	Aged	12/9	High Temp aging		No particles.			1,482,906	0.254	0.141	2.012
4	#81	Teflon	Unaged	10/5	High temp	1.450	No typical Teflon particles exhibited. Some alien-like junk present.			2,654,994	0.208	0.251	1.542

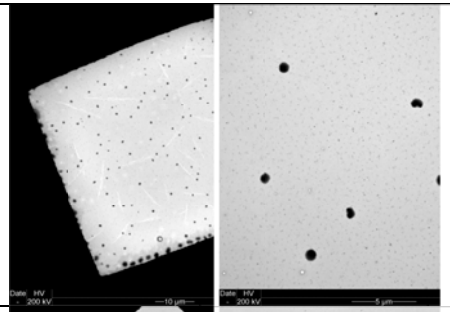
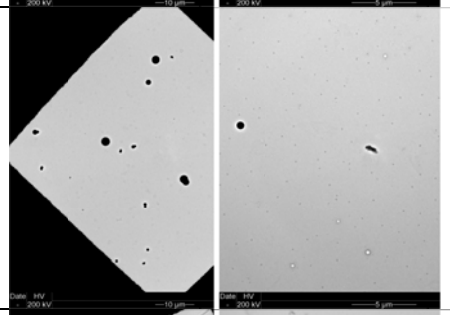
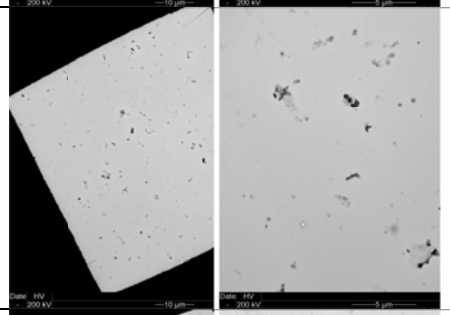
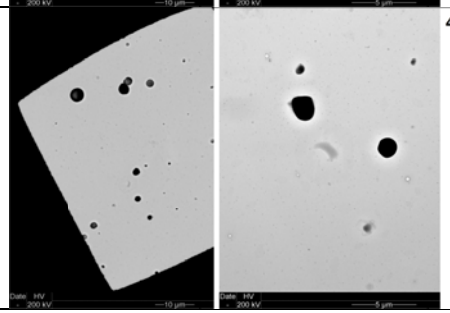
4	#81	Teflon	Aged	10/6	High Temp aging		Torn carbon film plus deposits.		920,600	0.385	0.348	#
4	#82	Lamp wick	Unaged	12/9	Mid temp	1.265	No particles.		3,463,686	0.139	0.290	2.357
4	#82	Lamp wick	Aged	12/9 & 4/23	Mid temp aging		No particles. Some ubiquitous tiny particles noted downstream at y+10.	 	1,343,413	0.230	0.385	2.051
4	#83	Teflon	Unaged	10/5	Mid temp aging	1.448	Very faint particles.	 	3,535,854	0.140	0.211	1.901

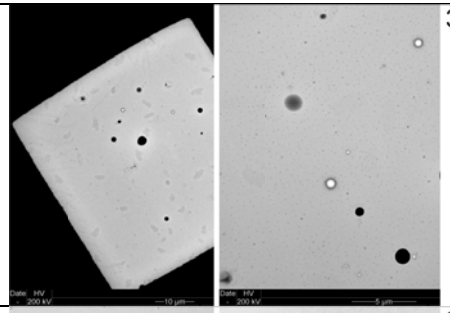
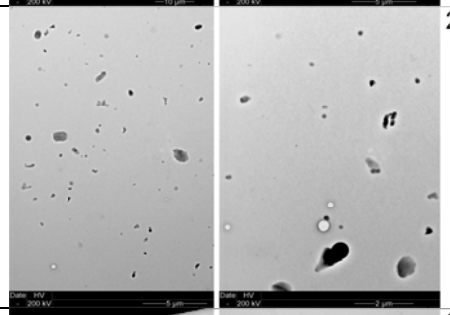
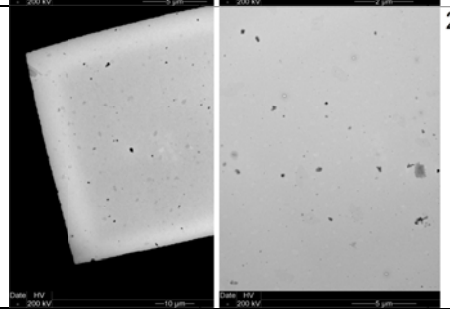
4	#83	Teflon	Aged	10/6	Mid temp aging		No particles – faint deposits.			909,243	0.337	0.325	#
5	#73	Kapton	Unaged	10/28	Baseline aging	1.380	No particles – some debris and deposits.			2,604,648	0.059	0.089	1.900
5	#73	Kapton	Aged	11/9	Baseline aging		Again, no particles – carbon film is extremely clean.			1,053,906	0.096	0.124	1.669
5	#74	Pyrell	Unaged	11/3	Baseline aging	1.220	No particles -			4,953,334	0.198	0.355	2.149

5	#74	Pyrell	Aged	11/9	Baseline aging		Many particles depositing upstream at -3y; particles ~100-400nm. Similar morphology to previously examined Pyrell (63P, 84P).		1,662,805	0.379	0.512	1.731
5	#75	Kapton	Unaged	10/28	Mid temp aging	1.430	No particles – some debris and deposits.					
5	#75	Kapton	Aged	11/9	Mid temp aging		Small population of extremely tiny particles, ~100-200nm; similar to #77K Kapton.		1,041,692	0.118	0.152	1.657
5	#76	Pyrell	Unaged	11/3	High temp aging	1.230	Lots of particles starting upstream at -7y. Size range of primary particles ~60-500nm with a few larger; exhibits small chained agglomerates. Morphology similar to #63 Pyrell.		3,275,642	0.354	0.515	1.843

5	#76	Pyrell	Aged	11/3	High Temp aging		Particles start upstream at -2y; particles generally between 60nm and 600nm. Slightly larger than preage samples.		1,632,140	0.445	0.643	1.835
5	#77	Kapton	Unaged	11/3	High temp aging	1.480	Very tiny particles, with a scattering of faint background particles measuring ~100nm. Some chained agglomerates; primary particles ~100-800nm.		3,460,328	0.088	0.107	1.541
5	#77	Kapton	Aged	11/3	High Temp aging		Higher population of extremely tiny (~100-200nm) particles with a few larger particles.		1,237,768	0.138	0.165	1.527
6	#93	Kapton	Unaged	3/19	Mid temp aging run 75 repeat	1.430	No particles except faint background particles ~90nm in diameter.		2,951,666	0.064	0.100	1.937

6	#93	Kapton	Aged	2/13	Mid temp aging 75 repeat		No particles (2 images) like preage.		990,234	0.105	0.155	1.864
6	#94	Kapton	Unaged	3/8	Mid temp aging run mid high	1.450	Extremely tiny particles, both upstream and downstream; background particles measure <50-75nm; extremely limited number of larger particles ~400nm.		3,018,896	0.092	0.116	1.626
6	#94	Kapton	Aged	2/13	Mid temp aging mid high		Small faint particles, most <200nm.		1,151,405	0.136	0.176	1.669
6	#95	Pyrell		2/13	No flow run w/ aging	1.220	A very few dense particles, most <2000nm.		71,388	0.140	0.683	3.521

6	#96	Lamp wick		2/13	No flow run w/ aging	1.245	Exhibits particles both upstream and downstream; typical dark spheres (~400-1200nm); aerosol deposits as a film in areas. Tiny particles are <100nm; also faint outlines of crystalline materials visible at lower magnifications.					
6	#97	Pyrell		3/8	No flow run w/ aging	1.230	Particle deposition upstream at y-4,-5; dense spherical particles measure ~400nm – 2.7 μm; faint background particles ~50-100nm.					
6	#100	Teflon		2/13	No flow run w/ aging	1.450	Typical Teflon aggregates with primary particles measuring ~100-300nm;					
6	#98	Lamp wick	Unaged	3/8	No flow run	1.280	Extremely large particles (up to ~7 μm) deposited upstream at y-2; very small (~25-50nm) background particles visible at higher magnifications.		438,729	0.168	0.403	2.546

6	#98	Lamp wick	Aged	2/13	No flow run w/ aging		Like preage with fewer large particles; exhibits significant amount of background material and very tiny particles.		329,084	0.186	0.431	2.503
6	#101	Kapton	Unaged	3/8	No flow run w/ aging	1.480	Particle deposition upstream at y-4, -5; typical small particles present (~200-1200nm) along with 'junk' or debris.		277,646	0.223	0.346	1.936
6	#101	Kapton	Aged	2/13	No flow run w/ aging		Exhibits a small population of smaller particles usually less than 1000nm.		253,110	0.234	0.351	1.893

Appendix F:

Particle Morphology and Size Results from the Smoke Aerosol Measurement Experiment-2

David L. Urban[1], Gary A. Ruff[2], Paul S. Greenberg[3], David Fischer[4], Marit Meyer[5]
NASA Glenn Research Center, Cleveland, Ohio 44135

George Mulholland[6]
The University of Maryland, College Park, Maryland 20742

Zeng-guang Yuan[7], Victoria Bryg[8]
National Center for Space Exploration Research, Cleveland, Ohio 44135
and
Thomas Cleary[9] and Jiann Yang[10]
National Institute of Standards and Technology, Gaithersburg, Maryland 20899

Results are presented from the Reflight of the Smoke Aerosol Measurement Experiment (SAME-2) which was conducted during Expedition 24 (July-September 2010). The reflight experiment built upon the results of the original flight during Expedition 15 by adding diagnostic measurements and expanding the test matrix. Five different materials representative of those found in spacecraft (Teflon, Kapton, cotton, silicone rubber and Pyrell) were heated to temperatures below the ignition point with conditions controlled to provide repeatable sample surface temperatures and air flow. The air flow past the sample during the heating period ranged from quiescent to 8 cm/s. The smoke was initially collected in an aging chamber to simulate the transport time from the smoke source to the detector. This effective transport time was varied by holding the smoke in the aging chamber for times ranging from 11 to 1800 s. Smoke particle samples were collected on Transmission Electron Microscope (TEM) grids for post-flight analysis. The TEM grids were analyzed to observe the particle morphology and size parameters. The diagnostics included a prototype two-moment smoke detector and three different measures of moments of the particle size distribution. These moment diagnostics were used to determine the particle number concentration (zeroth moment), the diameter concentration (first moment), and the mass concentration (third moment). These statistics were combined to determine the diameter of average mass and the count mean diameter and, by assuming a log-normal distribution, the geometric mean diameter and the geometric standard deviations can also be calculated. Overall the majority of the average smoke particle sizes were found to be in the 200 nm to 400 nm range with the quiescent cases producing some cases with substantially larger particles.

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[1] Branch Chief, Combustion and Reacting Systems Branch, 21000 Brookpark Road, MS 77-5, Senior Member.
[2] Advanced Capabilities Project Office, 21000 Brookpark Road, MS 77-7, Associate Fellow.
[3] Combustion and Reacting Systems Branch, 21000 Brookpark Road, MS 77-5.
[4] Combustion Branch, 21000 Brookpark Road, MS 77-5
[5] Combustion and Reacting Systems Branch, 21000 Brookpark Road, MS 77-5.
[6] Visiting Professor of Mechanical Engineering, Glenn L. Martin Hall.
[7] Staff Scientist, 21000 Brookpark Road, MS 77-5.
[8] Staff Scientist, 21000 Brookpark Road, MS 77-5.
[9] Engineered Fire Safety Group, 100 Bureau Drive, Stop 8664.
[10] National Fire Research Laboratory, 100 Bureau Drive, Stop 8662.

I. Introduction

Appropriate design of fire detection systems requires knowledge of both the expected signature of the events to be detected and the background levels of the measured parameters. Terrestrial fire detection systems have been developed based on extensive study of terrestrial fires.[1, 2] Unfortunately there is no corresponding data set for spacecraft fires and consequently the fire detectors in current spacecraft were developed based upon terrestrial designs. There are a number of factors that can be expected to affect the particle size distribution of the smoke from spacecraft fires. The absence of buoyant flow in low-gravity increases the residence time in microgravity fires and increases the transit time from the reaction zone to the detector.[3] Microgravity fires have been found to have radically different structure from their normal-gravity (a.k.a. 1-g) counterparts. The limited options available to respond to a spacecraft fire increase the importance of early detection. Finally the materials used in spacecraft are different from typical terrestrial applications where smoke properties were previously evaluated. All of these effects can be expected to change the smoke particle size distribution. The objective of this work was to make sufficient measurements of smoke from spacecraft fires to enable improved design of future detectors.

Smoke Background

Smoke is a general term that encompasses aerosol materials produced by a number of processes. In particular it can include unburned, recondensed, original polymer or pyrolysis products that can be either liquid or solid, hydrocarbon soot, condensed water vapor, and ash particles. Soot particles dominate the smoke particulate in established flaming fires while unburned pyrolysis products and recondensed polymer fragments are produced by smoldering and pyrolysis in the early stage of fire growth. Given the constrained space on any spacecraft, the target for the fire detection system is necessarily the early phase and not established flaming fires; consequently, the primary target for detection is the pyrolysis products and not the soot.

Prior spacecraft systems are summarized in more detail in papers by Friedman and Urban.[4, 5] In the Mercury, Gemini and Apollo missions, the crew quarters were limited and mission durations were short, consequently it was considered reasonable that the astronauts would rapidly detect any fire. The Skylab module, however, included approximately 30 UV-sensing fire detectors.[4] These devices were limited to line-of-sight and were reported to have difficulties with false alarms. The Space Shuttle Detectors were based upon ionization fire detector technology, the most advanced technology available at the time and used an inertial separator designed to eliminate particles larger than 1-2 micrometers. The International Space Station (ISS) smoke detectors use near-IR forward scattering, rendering them most sensitive to particles larger than a micrometer, outside of the range of sensitivity of the shuttle detector. As described by Friedman[4] there have been six overheat and failed component events in the NASA Orbiter fleet in addition to several similar incidents that have occurred on the ISS. None of these events spread into a real fire but as mission durations increase, the likelihood of failures increases. The experience on Mir in 1997 has shown that failure of oxygen generation systems can have significant consequences. As a result, improved understanding of spacecraft fire detection is critically needed.[6]

Previous work on smoke particles from low-gravity sources by Urban et al.[5] found that the particulate produced by low-gravity flames (soot or unburned fuel particles) tends to have larger size particles than in normal gravity. Results from the CSD (Comparative Soot Diagnostics) Experiment[5] which studied smoke properties in low-gravity from several spacecraft materials suggested that liquid smoke particles could achieve sizes larger than 1 μm while solid particulate remained in the sub-micrometer range. However, the CSD experiment did not produce sufficient data concerning the size of the liquid smoke particles to guide detector design. The combined impact of these limited results and theoretical predictions is that, as opposed to extrapolation from 1-g data, direct knowledge of low-g combustion particulate is needed for more confident design of smoke detectors for spacecraft.

II. SAME Experiment

To address the limited data from the CSD experiment concerning the likely size of spacecraft smoke particulate, another experiment was developed: the Smoke Aerosol Measurement Experiment (SAME). The SAME experiment sought to avoid the problems experienced by the CSD experiment by obtaining the particulate size statistics on-orbit with a reduced dependence upon sample return to Earth. This is a challenging endeavor because existing aerosol instrumentation is typically large, incompatible with spacecraft experiment constraints, and some systems require substantial sample return to Earth. As will be described below, an alternative approach was employed that used three discrete instruments to measure separate moments of the size distribution. When combined, these moments provide useful aggregate statistics of the size distribution. The measurements were made using smoke generated by overheated spacecraft materials in much the same manner as the CSD experiment however the sample temperature, flow field, and particle aging time were more rigorously controlled in the SAME experiment. The experiment flew

twice the first time in 2007 (SAME-1) and the second (SAME-2) in 2010. When discussion applies to both flights, this paper will refer to "SAME." If the discussion is specific to a particular flight, the flight number will be identified e.g. "SAME-2."

E. Moment Method

The approach used by the SAME experiment is termed the 'moment method' for convenience.[7] As will be described below, the approach consists of measuring three moments of the size distribution (zeroth, first and third) and using the properties of the log-normal distribution to estimate the geometric mean diameter and the standard deviation.

The average particle size and an estimate of the width of the size distribution will be estimated from various moments of the size distribution. The number distribution, $f_N(D)$, is defined as

$$f_N(D) = \frac{dN}{dD} \tag{1}$$

where dN is the number of particles per cm^3 with diameter between D and D + dD. The moments of interest consist of the number concentration, M_0, the first moment M_1, and the volume or mass concentration moment, M_3 and are defined as

$$M_i = \int D^i f_N(D) dD \quad i = 0,1,3 \tag{2}$$

When i=0, the zeroth moment of the distribution, M_0, equation (2) is simply the number of particles per unit volume. In the SAME experiment, this was measured using a condensation nuclei counter. The first moment, i=1, can also be thought of as the "diameter concentration" or integrated diameter per unit volume and is approximately proportional to the ionization detector moment (signal). For particles in the Mie scattering regime, particles sizes from 0.3λ to about 3λ (~0.2 μm to 2.0 μm for a red laser), the light scattering signal is approximately proportional to the third moment, i=3. From these moments, and a measurement of M_0 using a condensation nuclei counter, two mean diameters can be computed: the count (arithmetic) mean diameter $D_{0.5}$ or $\overline{D}$, which is equal to M_1/M_0 and the diameter of average mass $D_{1.5}$ or $D_{\overline{m}}$, which is equal to $(M_3/M_0)^{1/3}$. These calculations do not depend on the assumption of log-normal behavior. The basis for the subscript naming convention for $D_{0.5}$ and $D_{1.5}$ will be discussed later). The log-normal size distribution is widely used for describing the size distribution of aerosols including non-flaming smoke because for most aerosols; the bulk of the number concentration is associated with smaller particles.[8, 9] The number distribution $f_N(D)$ for the lognormal distribution is expressed as follows:

$$f_N(D) = \frac{N_t}{(2\pi)^{1/2} D \ln \sigma_g} \exp\left(- \frac{\left(\ln D - \ln D_g\right)^2}{2 \ln^2 \sigma_g} \right) \tag{3}$$

where N_t is the total number concentration of the aerosol (=M_0), and D_g and σ_g are the geometric mean diameter and geometric standard deviation defined by

$$\ln D_g = \int_0^\infty \ln D f_N(D) dD \Big/ \int_0^\infty f_N dD \tag{4}$$

$$\ln \sigma_g = \left[\int_0^\infty (\ln D - \ln D_g)^2 f_N(D) dD \Big/ \int_0^\infty f_N(D) dD \right]^{1/2} \tag{5}$$

For the log-normal distribution, one finds that the various diameter definitions given above are related to the geometric mean diameter, D_g, via the equation:[8 9]

$$D_p = D_g \exp(p \ln^2 \sigma_g) \tag{6}$$

For the count mean diameter, $D_{0.5}$, and the diameter of average mass, $D_{1.5}$, the corresponding values of p are 0.5 and 1.5. Fig. 1 shows a typical log-normal distribution for a D_g=1.0 and σ_g= 1.6. For this distribution, the corresponding values of $D_{0.5}$ and $D_{1.5}$ are1.17 μm and 1.39 μm, respectively. Using equation (6), one can relate σ_g to the ratio of $D_{1.5}$ and $D_{0.5}$ via the equation:

$$\sigma_g = \exp\left(\ln(D_{1.5} / D_{0.5})\right)^{1/2} \tag{7}$$

By combining these three moments it is possible to compute three mean diameters of the size distribution and the geometric standard deviation. Validation of this approach is discussed in Cleary, Weinert and Mulholland.[7] These statistics provide a strong basis for design of spacecraft smoke detectors.

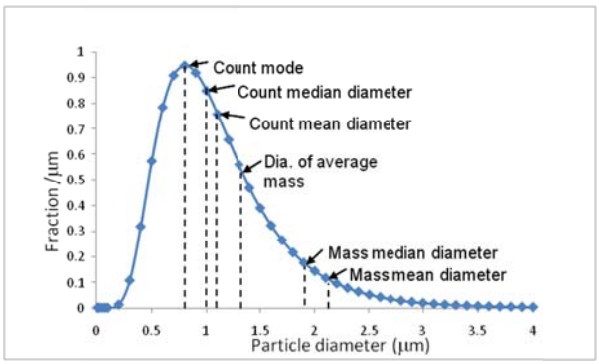


Figure 1: Diameters for a log-normal distribution with D_g=1.0 and σ_g= 1.6.

F. Instruments

These measurements were made using an assembly of three separate instruments. Two are industrial hygiene instruments manufactured by TSI and one is a modified residential smoke detector. More complete discussion of the SAME hardware is available in an earlier paper[10].

The zeroth-moment instrument is a condensation nuclei counter P-Trak™ (TSI Inc.). This device operates by passing the aerosol-laden particle stream through a region saturated with isopropanol vapor and then into a cooler region where the vapor condenses onto the particles increasing their diameter such that they can be readily counted by a light scattering device. This instrument is very robust and operates over a range of 0 to 10^5 particles/cm^3 and 20 nm to 1 μm diameter. Some dilution is required, since the smoke concentration ranges from about 0.5 × 10^6 to 5 × 10^6 particles / cm^3. The dilution was accomplished by controlled addition of filtered nitrogen to the aerosol sample being directed to the P-Trak. There was also a concern that the isopropanol condensate would not return to the wick in low-gravity.[5] To mitigate this issue, the condensing section of the device was modified with very small grooves to improve conductance of the condensate back to the wick. These changes were tested in a separate space experiment with good results indicating the modified device could be used successfully in low gravity.[5]

The first-moment instrument is the ionization chamber from a residential smoke detector. This device uses an alpha-particle emitter to generate ions in a region within a DC electric field. The drift of the ions in the electric field results in a current. The presence of aerosol particulate reduces the current as a result of the attachment of the ions to the particulate. The mobility of the charged aerosol is too small for it to be collected on the ionization chamber electrode. The required particle concentrations are on the order of 10^5 particles/cm^3 and no sample dilution was required.

The third-moment instrument is a light scattering device DustTrak™ (TSI Inc.). The device uses a 90 degree light scattering signal to quantify the aerosol mass density. For terrestrial dust particulate this signal correlates well with the mass concentration, however, as described in the calibration discussion below, additional compensation was employed to account for the range of materials and particle sizes that were seen in the SAME experiment. The device's operating range is from 0.001 mg/m^3 to 100 mg /m^3. These devices are equipped with an aerodynamic impactor at the inlet which captures particles larger than the selected size. The SAME-1 experiment included 2 DustTraks™ one with a 1 μm impactor and one with a 10 μm impactor. The difference in the signal from these two devices provided a measure of the fraction of the particulate that was larger than 1 μm. In some cases dilution was required owing to the high smoke concentration levels.

A schematic of the assembled hardware appears in Fig. 2. The system was installed in the Microgravity Science Glovebox (MSG), an ISS facility that provides many resources including: containment, power, data, video and uplink commanding. Smoke was generated by overheating a small sample of material in the smoke generation duct for approximately 60 seconds. During this interval, controlled flow was induced by a moving piston in the aging

chamber which drew the smoke into the chamber. The smoke was held in the chamber for a predetermined time, allowing the particles to coagulate. After a specified aging time, the smoke was then pushed by the piston into the diagnostics duct where the moment instruments made their measurements. Also installed in the diagnostics duct were space shuttle and ISS smoke detectors. As the smoke was monitored by the moment instruments, a sample of the smoke particles was deposited on Transmission Electron Microscope (TEM) grid via a Thermal Precipitator (Fig 3) which uses thermophoresis to deposit the particles on the grids. Inside the precipitator, the aerosol passed between a Transmission Electron Microscope grid and a heated wire. After the mission the grids were removed from the assembly and examined in a TEM to obtain an independent determination of the particle size and morphology.

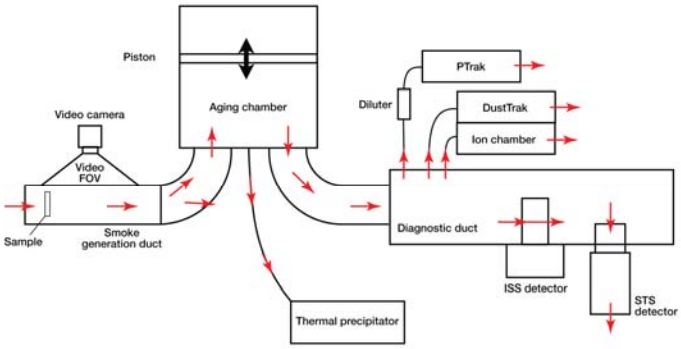


Figure 2. Schematic of the SAME hardware**.**



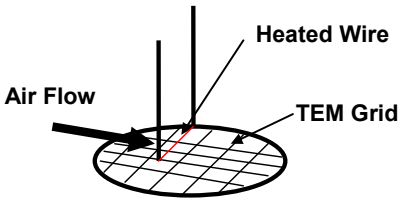


Figure 3: Photograph of a partially disassembled Thermal Precipitator Module with the inlet manifold exposed on the left, the hot wire leads in the red Vespel block in the middle and the outlet valves on the right. The drawing on the right displays the position of the hot wire above the TEM grid with the air flow passing over both. The heated wire drives aerosol particles onto

Calibration of the moment instruments was essential to properly interpret the flight data and was performed on the ground before the flight. Calibration was accomplished using two different aerosol generators one using mono-disperse particle generation using dioctyl phthalate (DOP) according to the approach by Mulholland and Liu[11] and the other using polystyrene spheres. The monodisperse droplet generator functioned by producing a spray of DOP diluted with isopropanol which is then evaporated and recondensed producing monodisperse droplets. The droplet size is controlled by the DOP dilution level. The generator will operate stably for tens of minutes. The aerosol from the generator was sampled simultaneously by the SAME instrument under test and a reference instrument. For the number count, the reference instrument was a TSI 3022A particle counter, for the mass concentration, a Tapered Element Oscillating Microbalance and for the first moment a TSI Electrical Aerosol Detector was used. The results for the P-Trak are shown in Fig. 4. As the number concentration increased, the effect of the particle diameter became more evident. Separate correlations were developed for each particle size and the closest correlation was used to analyze the flight data based on the initial estimates of the average particle size.

The first moment device, the ion chamber showed little effect of particle size as seen in Fig 5. Consequently a single correlation was used for all particle sizes. The third moment device (TSI- DustTrak®) is theoretically predicted to show non-monotonic behavior as particle size is increased and the response varies with the particle refractive index. This issue was addressed by calibrating the DustTrak for the smoke particulate from each material. Although the average particulate size is expected to change, the refractive properties are expected to remain the same. In the results reported here, the DustTrak® response was directly calibrated, for each smoke source, against

mass concentration measurements using a tapered element oscillating microbalance. Using estimated material densities, the diameter of average mass was computed for each sample.

For SAME-2, one of the DustTrak devices was replaced by a new device the MPASS (Multi-Parameter Aerosol Scattering Sensor) which is an optical scattering sensor, employing two detectors placed at fixed angles. The underlying design and performance have been reported previously.[12] The basic configuration is similar to the DustTrak, however, the MPASS has two angles of detection and it can measure two moment approximate moment quantities specifically the 3rd and 2nd moments. Further, the optical geometry is optimized using a Mie-scattering model for aerosols with light scattering properties typical of smoke: i) geometric mean diameters from 0.1 – 1.0 microns; ii) geometric standard deviations of 1.6 – 1.9; iii) the aerosols have lognormal size distributions and vi) a fixed, real value for refractive index of 1.6. This device was included as a technology demonstration of an aerosol sensor that measures two moments simultaneously. The underlying assumption is that by measuring two moments of the aerosol it is possible to correct the smoke detector signal for the particle size, unlike previous spacecraft smoke detectors which are limited to a single moment measurement[10]. This will enable the alarm to be less sensitive to larger particles that are typical of dust. The relative advantage of this approach was discussed in more detail in a prior paper[13].

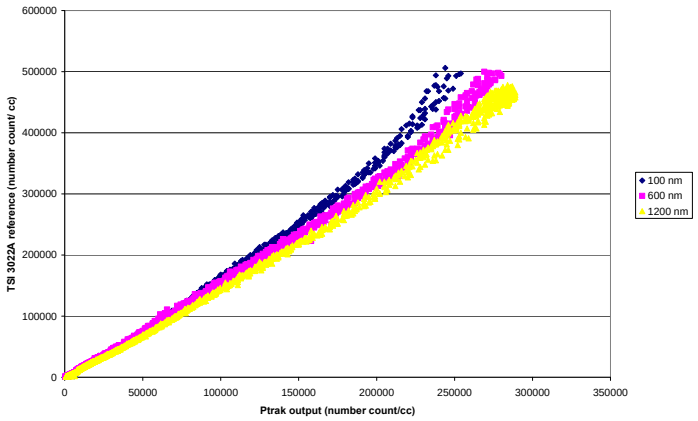


Figure 4: P-Trak Calibration results with Mono disperse DOP droplets.

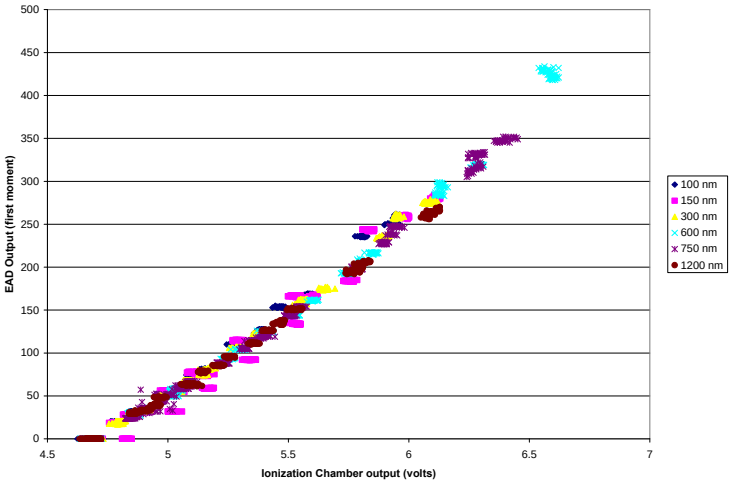


Figure 5: Ion Chamber Calibration results with Mono disperse DOP droplets

G. Experimental Results

Over 100 sample materials were tested. These were comprised of samples of 6 materials: Teflon™, Kapton™, Pyrell™, silicone rubber, cellulose (lamp wick), and dibutyl-phthalate deposited on a porous wick. The test conditions included multiple sample temperatures and air flow rates. The baseline air flow rate was 8 cm/s with runs conducted at rates down to 2 cm/s with a limited set conducted with no airflow while the sample was heated. The sample temperatures were based on the pyrolysis properties of the material and were thus material specific. The baseline temperature was selected to produce 1 to 2 mg of weight loss in 60 seconds and have reasonable signal on each of the moment instruments. To examine the effect of sample temperature, additional higher temperature conditions were also selected that would not saturate the moment instruments. The TEM girds recovered from the thermal precipitators on SAME-1 were unfortunately contaminated with extraneous particles whose origin is currently under investigation. For SAME-2 more rigorous assembly controls were followed and numerous good particle images were obtained. For typical runs, the particle concentrations were a few million particles / cc and the mass concentrations were 2 to 15 mg/m^3 however there were runs with fewer particles. For these conditions, the typical numbers of particles sampled by the instruments were in the billions although for the test with the lowest particle concentrations the sample sizes were of the order of several million particles.

1. Moment Instrument Results

Geometric mean, count mean, and average mass diameter results from the moment instruments for baseline runs are presented in Table 1. As described above, the count mean diameter and the diameter of average mass measurements are independent of the nature of the size distribution however, the calculation of the geometric mean diameter and the geometric standard deviation (σ_g) depend on the assumption of a log-normal distribution. This assumption cannot be directly tested with the available data since this would require a device that can separate the particles into size bins, However, there are two indications that the distribution can be approximated by a log-normal distribution. In the cases listed below, the count mean diameter is less than the diameter of average mass which is a necessary condition for a log-normal distribution (See Fig. 1). The second indication is that the geometric standard deviations are all physically reasonable (i.e. greater than 1 (see Eq. 3) and less than approximately 3.5 which is the practical limit on the magnitude of aerosol geometric standard deviation). The aging results show a substantial increase in the observed diameters. As expected the increase in the count mean diameter is greater than the corresponding increase in the diameter of average mass. This can be understood since coagulation consists of particle collisions joining two particles. For a polydisperse aerosol, these collisions usually involve the collision of one of the smallest (high mobility) particles with the largest (high cross-section) particles. This collision has little effect on the size of the large particle but it removes the smaller particles from the size distribution. This has a larger effect on the count mean diameter since it is linear in diameter while the diameter of average mass varies with the cube root of the diameter. Theoretical predictions suggest that this process will produce a near lognormal distribution and that the standard deviation will tend toward a self-preserving distribution with a value of 1.32.[14]

		Geometric Mean Diameter (Dg) (μm)	Count Mean Diameter (M_1/M_0) (μm)	Diameter of Average Mass (M_3/M_0) (μm)	σ_g
Kapton	Unaged	0.042	0.056	0.101	2.154
	Aged 720 s	0.089	0.109	0.161	1.872
Lampwick	Unaged	0.090	0.128	0.258	2.312
	Aged 720 s	0.229	0.276	0.398	1.834
Silicone	Unaged	0.128	0.196	0.465	2.530
	Aged 720 s	0.269	0.355	0.619	2.108
Teflon	Unaged	0.081	0.101	0.170	2.198
	Aged 720 s	0.070	0.105	0.232	2.442
Pyrell	Unaged	0.149	0.204	0.384	2.211
	Aged 720 s	0.293	0.359	0.539	1.892

Table 1: Diameter results for baseline runs for each material (Diameters are in micrometers).

Overall, the Teflon and Kapton particles were very small. The lamp wick and silicone results exhibited substantially larger diameters of average mass, particularly for the aged cases. As reported previously[10], the Dust Trak results for lamp wick and silicone demonstrated substantial portions of the particle distribution possessed aerodynamic diameters larger than 1 micrometer. The comparatively much smaller diameters for the Kapton and Teflon will make detection of this smoke challenging for light scattering devices, on the other hand the large sizes seen with the lamp wick and silicone would generate very large signals on a light scattering system. As discussed by Urban et al.[13], to have a system that would have the signal gain sensitive enough to trigger an alarm for the lower signal from small particles such as Teflon and Kapton without excessive nuisance alarms from large dust particles is difficult without some level of particle size discrimination. This suggests that detection of these particles against the background environment will require a detection system capable of measuring more than one moment of the particle size distribution.

The results for Teflon, Kapton and lampwick are plotted versus duct velocity in figures 6, 7 and 8. These figures show a strong trend of increased particle size at low flow rates. This indicates that smoke trapped in an avionics enclosure can be expected to be substantially larger than smoke released in ventilated areas.

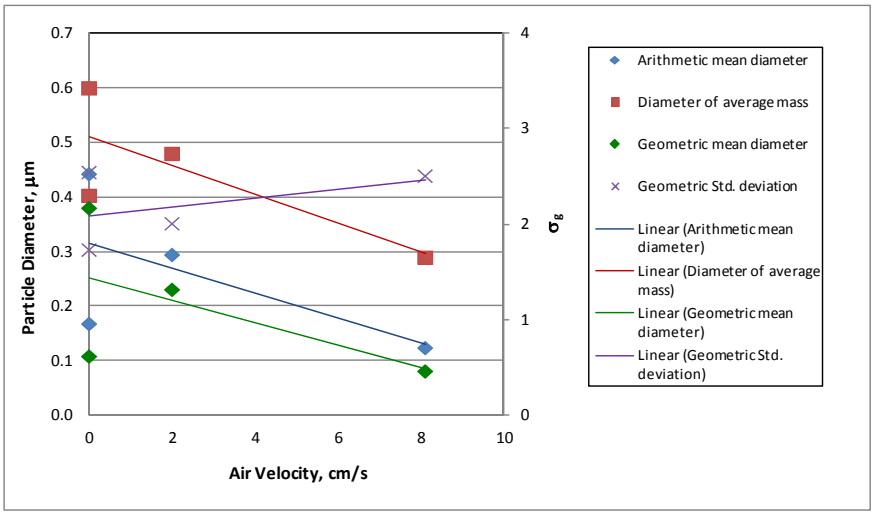


Figure 6: Particle diameters as a function of air velocity for lampwick. The lines are linear (least square) fits. The repeatability of the diameter measurements is estimated to be +/- 15%.

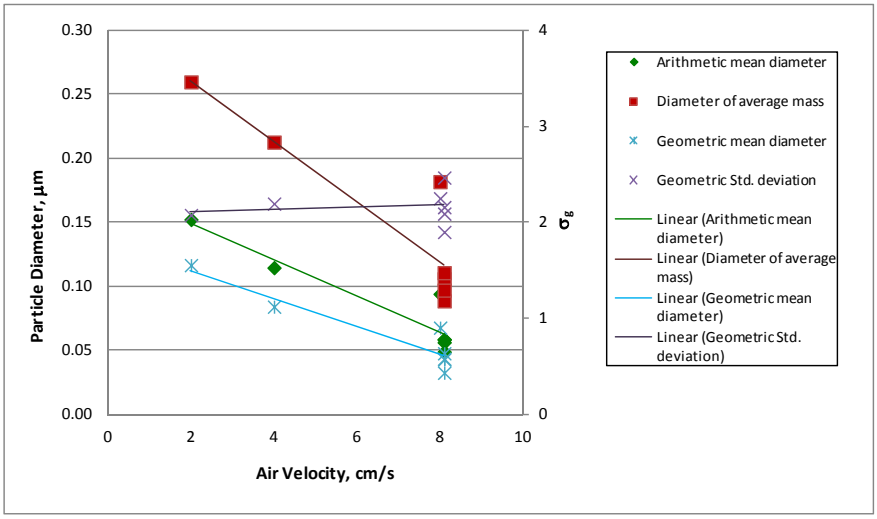


Figure 7: Particle diameters as a function of air velocity for Kapton. The lines are linear (least square) fits. The repeatability of the diameter measurements is estimated to be +/- 15%.

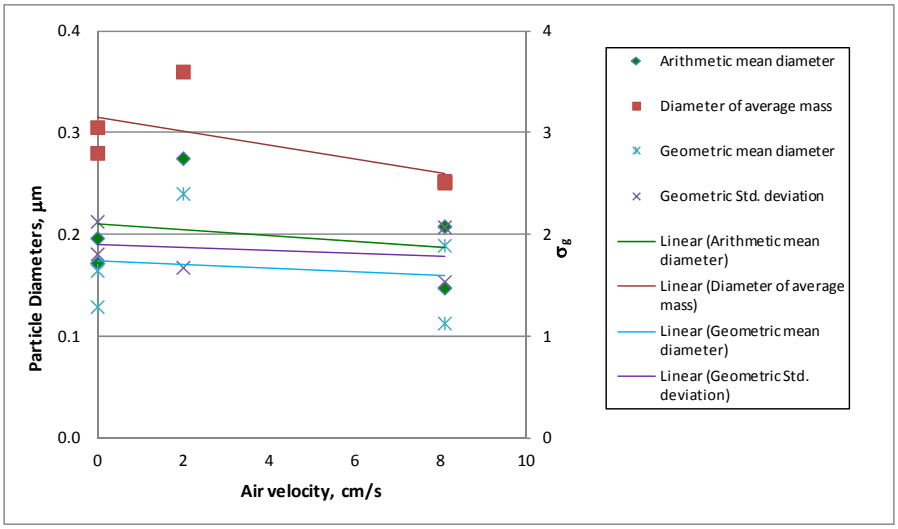


Figure 8: Particle diameters as a function of air velocity for Teflon. The lines are linear (least square) fits. The repeatability of the diameter measurements is estimated to be +/- 15%.

2. MPASS Results

The MPASS was included in SAME-2 to demonstrate the effectiveness of a two-moment smoke detector. Because of how they are used, a smoke detector is not calibrated for specific materials and instead has a single calibration for all conditions. The objective is to maintain a flat sensor response, regardless of the type of smoke or the particle size so the signal is proportional only to a selected moment (e.g. mass or number concentration). The approach used was to derive correlations to predict the smoke number concentration and mass concentration from the MPASS output. This correlation was made using the other SAME measurements (number and mass concentration) as a reference. A least-squares, third-order, correlation was computed on all test points with a signal to noise ratio greater than 2 independent of all other parameters (e.g. material, heating rate, and aging time).

The results are shown in figures 9 and 10, for prediction of the mass concentration and the number concentration, respectively. Ideally these response graphs should be horizontal lines. The normalized output is plotted along with the normalized correlation result as a function of geometric mean diameter. Ideally the normalized result should be horizontal lines. The uniformity of response (scatter) is expressed by the least squares variance relative to the mean (Δ) and is shown in the legend. Both figures show that the correlation decreases the relative variance by roughly a factor of at least 2 relative to that provided by a conventional single-output sensor. The resulting variance for the mass response is substantially lower than for the number concentration. This is consistent with the fact that the original optimization of the MPASS was to mass concentration not to number concentration. Thus with appropriate processing, the MPASS output can be proportional to either the number concentration or the mass concentration over a wide range of particle sizes and material compositions. This independence demonstrates that by combining 2 moments the MPASS is able to scale its response to the particle size being sampled giving a more accurate indication of the amount of smoke present than a single moment detector. The MPASS therefore has the ability to discern the difference between a strong signal from a high smoke concentration and a strong signal from a lower concentration of very large dust particles, a feature which can minimize false smoke detector alarms.

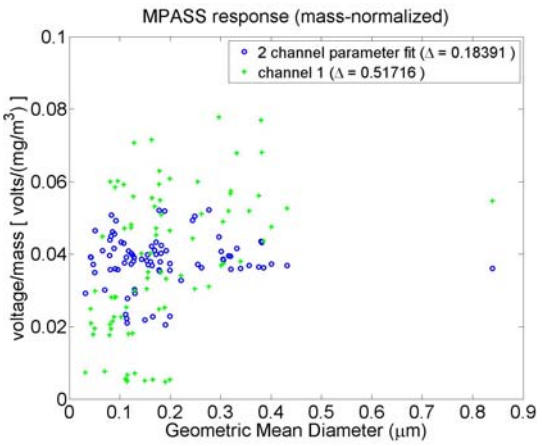


Figure 9: Mass normalized signal channel output voltage and correlated MPASS output voltage versus geometric mean diameter.

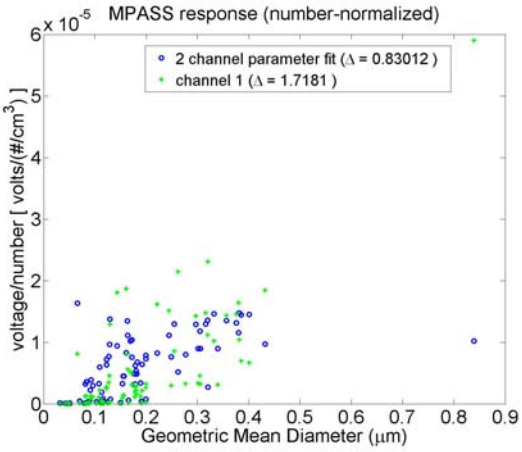
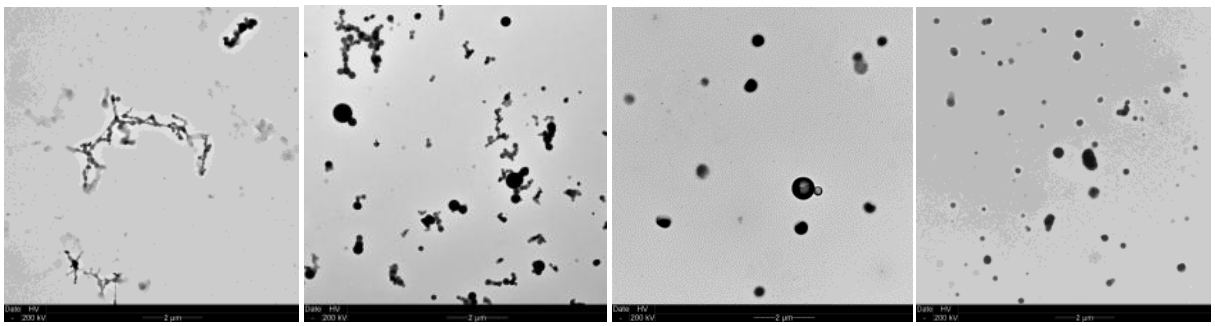


Figure 10: Count normalized output voltage and correlated MPASS output voltage versus geometric mean diameter.

3. TEM Results

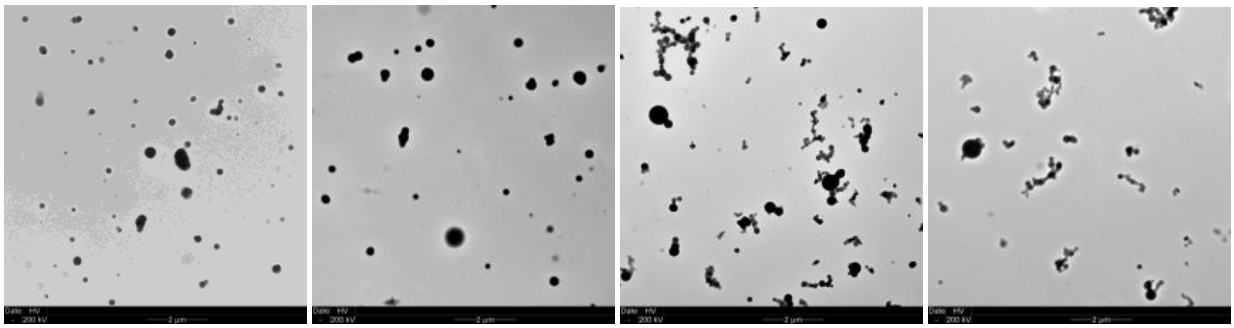
Overall, 72 particle samples were taken on Transmission Electron Microscope (TEM) grids using the thermal precipitator. These samples were imaged with a TEM to evaluate the particle morphology as a function of flow conditions, material and heating level. Overall, silicone rubber and DBP presented no particles that were visible via TEM, presumably because the particles were condensed liquid that evaporated before they could be imaged in the TEM. Figure 11 presents typical images for the major materials.



Length scale 2 μm ▬

Figure 11: TEM images for Teflon (Run 56), Pyrell (Run 63), Lampwick (Run 54) and Kapton

As seen in Table 1, aging did increase the average particle size, however this level of change is not readily visible in the TEM images, instead the most visible change is the disappearance of the smallest particles. (Fig 12) As described in the discussion of Table 1, this is consistent with expectations since the smaller particles have the highest mobility and consequently are readily incorporated into the larger particles. Particularly for the semi-liquid particles, this absorption of the smallest particles into larger particles is expected to have little or no apparent effect on the size of the larger particles because the diameter is the cube root of the volume. In the case of non-liquid particles, the smallest particles will nevertheless be unlikely to be visible via TEM when they are attached to larger particles.



Length scale 2 μm

Figure 12: TEM images showing the effect of aging for Kapton (Run 62) and Pyrell (Run 63). From Left to right, unaged Kapton, aged Kapton, unaged Pyrell, aged Pyrell. The notable change is the elimination of the smallest particles.

4. Impact of flow on particle morphology

As the samples are heated by the wire, chemical bonds are broken and gaseous products are released via pyrolysis. These products may include partially oxidized species resulting from the reaction with oxygen. As the gases move from the wire, the temperature and species concentration will decrease as a result of conduction/diffusion and convection. With the decrease in temperature, the super saturation ratio of the pyrolysis gas will pass a critical stage and nucleation of smoke particles will occur. These smoke nuclei grow rapidly from the condensation of the pyrolysis gases. The time scale for the nucleation and condensation is on the order of ms or less. Over a longer time scale, the growing smoke particles undergoing Brownian motion collide and stick together in a process known as coagulation. This process may continue for time scales of up to 10^3 s. If the particles are liquid, the colliding particles will coalesce and form larger droplets. This is the case for the lampwick, silicone rubber, and Kapton smokes (Fig. 11). If the particles are solid, they will retain their structure as they stick resulting in an agglomerate structure composed of many primary particles connected in a low density fractal-like structure. This is the case for the Teflon and Pyrell smokes (Fig. 11).

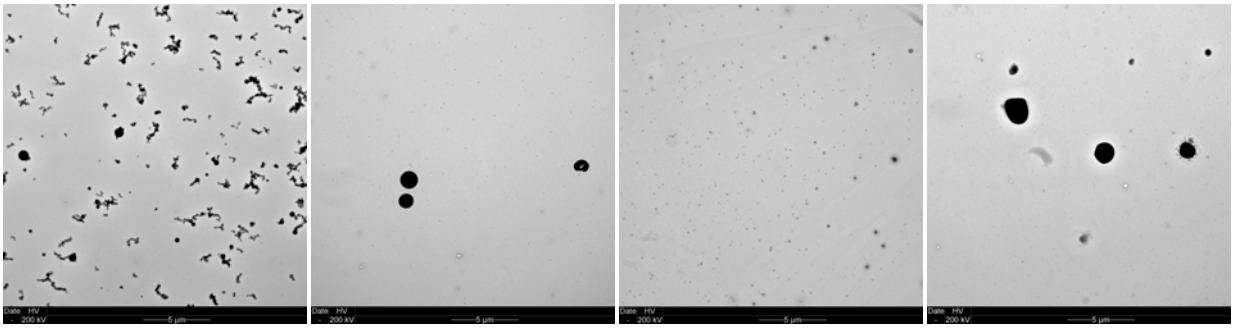
Key parameters affecting the smoke properties of the smoke produced here are the low-gravity condition, the convective flow past the heated wire, the wire temperature, and the aging time. As the wire temperature increases, more of the sample will be pyrolyzed and these gases will subsequently condense on the nuclei leading to a larger mass concentration of the aerosol and a larger diameter of average mass.

An increase in the convective flow for a fixed wire temperature is expected to result in an increase in the pyrolysis rate and thus an increase in the mass concentration of the aerosol. Some results follow this trend with silicone rubber, lampwick, and Pyrell increasing slightly with flow while Teflon and Kapton decrease. A possible explanation is that holding the wire temperature fixed does not necessarily mean that the sample surface temperature distribution is precisely the same at different air flow speeds. Although these are expected to be small variations, given the exponential dependence of the pyrolysis kinetics on temperature, changes in the heat transport environment could change the sample surface temperature between the wire coils.

A case of special interest is the case of zero-flow. The transport for the zero-flow case in μ-g is controlled by molecular diffusion and thermal expansion. Over a period of 60 seconds, the gases would diffuse a distance on the order of 4 cm. Assuming a cubical geometry this would correspond to the smoke being confined to a volume of about 64 cm^3 rather than being diluted in the 6000 cm^3 piston. If the same number of particles is generated as for the flow case, the initial concentration will be on the order of 50 times higher for the zero-flow case. This is expected to lead to more coagulation and a larger particle size. It is found for all of the materials the diameter of average mass is 30 % to 40 % larger for the zero-flow tests compared to the flow tests.

The zero flow condition was not in the original design of the SAME hardware but was examined for 11 tests by modifying the valve operation in the SAME hardware. This resulted in a very high concentration of particulate where the aging process was accelerated compared to that in the aging chamber at dilute conditions. An unfortunate side effect was that approximately half the smoke was lost from the system because it diffused out of the inlet into the MSG volume. As a result the total particle counts were lower than normal and the TEM grids had few particles. It is important to recognize that this condition is unique to low-gravity scenarios. If one performed the same experiment at 1-g, the buoyancy of the rising gases would greatly increase the mixing zone. So the larger particle diameters observed in these zero-flow conditions are important to consider in the development of improved fire detection equipment for space vehicles.

For the zero-flow case, the heat is transported primarily by conduction rather than convection. This is likely to result in a significant increase in the gas temperature relative to the case with 8 cm/s air flow. A striking effect of this temperature change is the change in the morphology of the Pyrell smoke from fractal-like clusters with up to 30 primary spheres for the case with a flow of 8 cm/s to spherical particles for the zero-flow condition (Fig 13).



Pyrell 8 cm/s air flow (Run 63) Pyrell no air flow (Run 95) Lampwick 8 cm/s air flow (Run 80) Lampwick no air flow (Run 80)

Length scale: 5 μm

Figure 13: TEM images showing the effect of flow for Pyrell and lampwick under 8 cm/s air

III. Conclusions

The results from the SAME experiment indicate that substantial differences in the size distributions of the smoke particles are evident for the five spacecraft materials that were tested. For the conditions of the SAME experiment, all samples produced significant numbers of sub-micron particulate that are readily detected using an ionization smoke detector. However, the aging results suggest that some materials such as silicone rubber can produce smoke that can reach sizes better detected by light scattering techniques. The zero flow conditions produced smoke particles that were 2 to 3 times larger than those produced at the nominal 8 cm/s flow rate. Since spacecraft fire conditions include an even wider array of materials than those tested here and a broader range of flow and temperature conditions, broad smoke aerosol size distributions can be expected from credible pre-fire overheat events. These results suggest that detection methods that can measure more than one moment of the size distribution may show more successful detection and false alarm rejection than single moment detectors. The results from the prototype 2-moment detector (MPASS) were very encouraging in this respect with the technique showing real potential as a means to respond to a broad particle range while avoiding excessive false alarms from nuisance sources. The TEM result demonstrated that fire particulate can be expected to have a broad range of morphologies influenced by heating rate, material and flow conditions.

IV. Acknowledgments

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Certain commercial entities, equipment, or materials may be identified in this document in order to describe an experimental procedure or concept adequately. Such identification is not intended to imply recommendation or endorsement by the National Aeronautics and Space Administration or the National Institute of Standards and Technology, nor is it intended to imply that the entities, materials, or equipment are necessarily the best available for the purpose.

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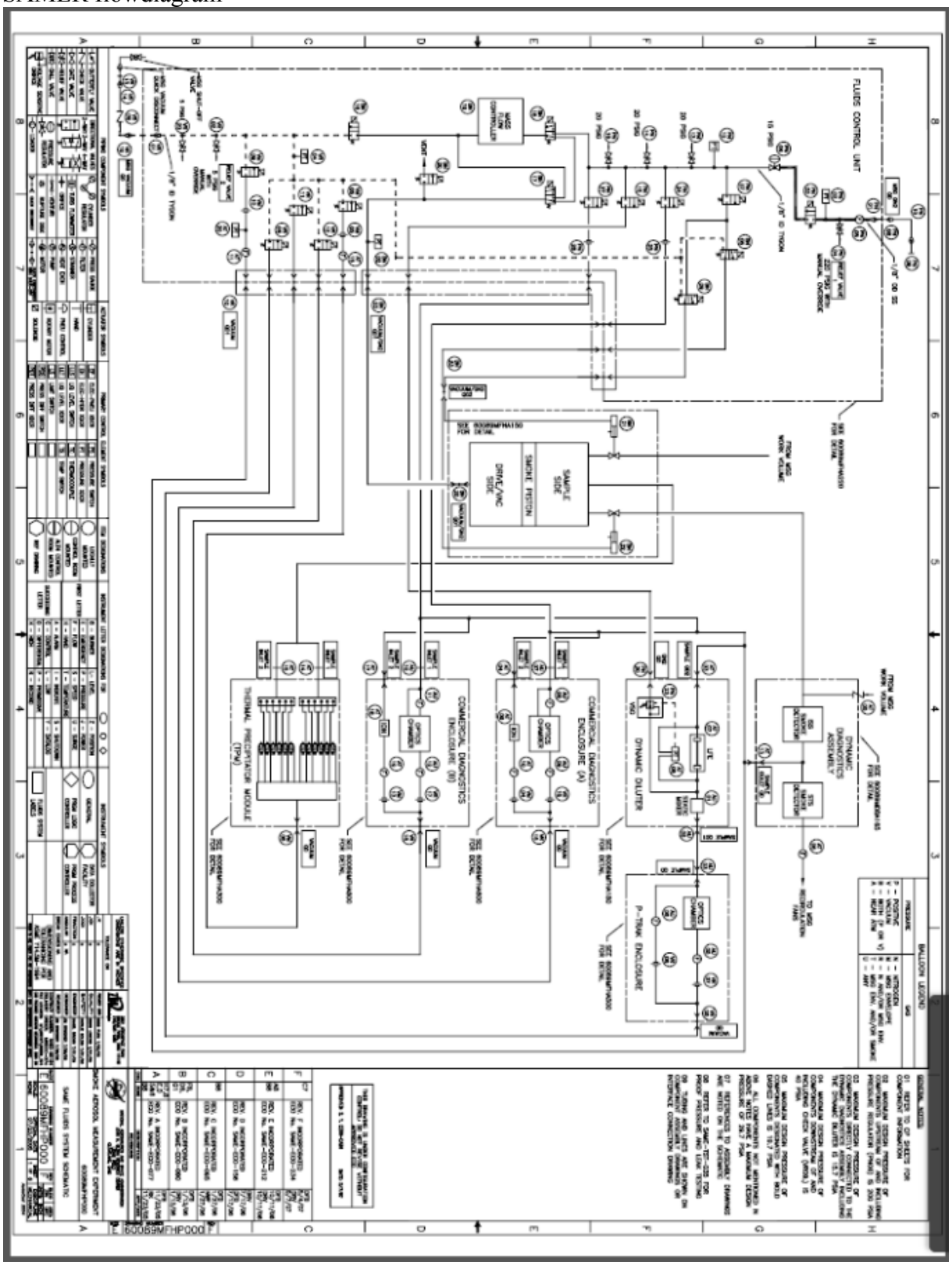
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SAMER flowdiagram



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Certain commercial entities, equipment, or materials may be identified in this document in order to describe an experimental procedure or concept adequately. Such identification is neither intended to imply recommendation or endorsement by the National Aeronautics and Space Administration or the National Institute of Standards and Technology, nor intended to imply that the entities, materials, or equipment are necessarily the best available for the purpose.

Publication list

Awards

SAE 2008 Arch T. Colwell Merit Award, , "Measurement of Smoke Particle Size under Low-Gravity Conditions,", David L. Urban, Gary Ruff, George Mulholland, Thomas Cleary, Jiann Yang, Zeng guang Yuan, 2008 SAE ICES, 08ICES-2089

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2. Pyrolysis Smoke Generated Under Low-Gravity Conditions, George W. Mulholland, Marit Meyer, David L. Urban, Gary A. Ruff, Zeng-guang Yuan, Victoria Bryg, Thomas Cleary, and Jiann Yang, in preparation for Aerosol Science and Technology.
3. Smoke Characterization and Feasibility of the Moment Method for Spacecraft Fire Detection, Marit Meyer, George W. Mulholland, Victoria Bryg, David L. Urban, Zeng-guang Yuan, Gary A. Ruff, Thomas Cleary, and Jiann Yang, in preparation for Aerosol Science and Technology.
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